IN-SPACe CAN-7USAT

INDIA STUDENT COMPETITION 2026

3rd Edition

PRELIMINARY DESIGN REVIEW (PDR) DOCUMENT

TEAM- VRITRA

INSTITUTE - LOVELY PROFESSIONAL UNIVERSITY

TEAM ID- 2026-INSPACe-CAN-7USAT-025

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1. CAN-7USAT MISSION OVERVIEW

We are team VRITRA with a team ID as 2026-INSPACe-CAN-7USAT-025 of Lovely Professional University. We are developing a CAN-7USAT for the IN-SPACe CAN-7USAT India Student Competition 2026 which includes designing, fabrication and flight of CANSAT. The CAN sized satellite will be carried to an altitude of 1000 m or ±100 m above the ground level (AGL) through a launch vehicle. It will be ejected from the rocket at the apogee and should have a two-stage recovery system while transmitting real time telemetry to a ground station.

Primary Mission Objective: This is the primary descent control system which is a hemispherical parachute connected to the outer body with no ejection mechanism and is deployed by the atmospheric drag. It will be activated directly after the ejection of CANSAT from the rocket having the descent rate of 20m/s with deviation of ±5m/s.

Secondary Mission Objective: This is the secondary descent control system with a secondary deployment of a RAM-air parachute at an altitude of 600 m AGL. This reduces the descent rate to 1-3 m/s at touchdown. It is also an optional objective which covers precision, controlled flight and targeted landing and is demonstrated by GPS guided RAM air parafoil which enables active recovery at the target landing point near launch pad.

This document presents the Comprehensive Design Review (CDR) of a CAN sized satellite designed by team VRITRA.

2. TEAM DESCRIPTION

2.1 Faculty Advisor and Mentor

NO.	NAME	ROLE	DESIGNATION
1	Mr. Mandeep Singh	Faculty Advisor	Assistant Professor and Assistant Dean
2	Mr. Ruhul Amin Choudhury	Mentor	Innovation Officer
3	Dr. Elaya Perumal	Mentor	Assistant Professor

Table 1: Mentor details

2.2 Team Members

NO.	NAME	ACADEMIC YEAR	BRANCH	ROLE
1	Numaan Aslma Khan	4th year	B. Tech – Aerospace Engineering	Structural & Mechanical Systems Design
2	Atharv Taralkar	3rd year	B. Tech –Electronics and Communication Engineering	Electrical power subsystem (EPS) design

3	Sujal Kalbhor	3rd year	B. Tech –Electronics and Communication Engineering	System and hardware integration
4	Vaishnavi Mahadik	3rd year	B. Tech –Electronics and Communication Engineering	Communication and data handling subsystem (CDH)
5	Rishav Raj	2nd year	B. Tech – Robotics & Automation	Fight software design
6	Aakanksha Rathore	2nd year	B. Tech – Robotics & AI (CSE)	Ground control subsystem (GCS)
7	Jasmeen Judge	3rd year	B. Tech – Aerospace Engineering	Descent control design
8	Ninad Nilesh Mestry	3rd year	B. Tech – Aerospace Engineering	Can-Sat integration and test

Table 2: Team Members details

Since the PDR stage, a new member is added in our team which is highlighted in the table. This was to fill the gap which was created due to the absence of previous member.

2.3 Project Milestone Chart

PHASE / ACTIVITY	START DATE	END DATE	RESPONSIBLE
PDR document preparation & submission	May 16, 2026	June 15, 2026	All members
Component Purchasing	Jun 15, 2026	Jul 5, 2026	Sujal Kalbhor, Atharv Taralkar
PCB design & fabrication	Ju1, 2026	Jul 30, 2026	Vaishnavi Mahadik, Atharv Taralkar
Mechanical fabrication (body, bays)	Jul 15, 2026	Aug 15, 2026	Ninad Mestry, Numaan Aslma Khan
Flight software development (Teensy FSM)	Jul 18, 2026	Jul 28, 2026	Rishav Raj, Aakanksha Rathore
Parafoil & servo integration	Jul 25, 2026	Aug 1, 2026	Rishav Raj, Numaan Aslma Khan
Ground station software development	Jul 20, 2026	Jul 25, 2026	Aakanksha Rathore, Jasmeen Judge
Subsystem-level testing	Jul 20, 2026	Aug 5, 2026	Atharv Taralkar, Ninad Mestry
CDR document preparation	Jul 28, 2026	Aug 8, 2026	Vaishnavi Mahadik, Jasmeen Judge

Full integration & functional test	Jul 25, 2026	Aug 30, 2026	Atharv Taralkar, Sujal Kalbhor
Flight Readiness Review (FRR)	Sep 1, 2026	Oct 15, 2026	Jasmeen Judge, Sujal Kalbhor
Launch & competition	Oct 25, 2026	Nov 30, 2026	All members

Table 3: Project milestone chart

3. LIST OF ACRONYMOUS

ACRONYM	EXPANSION
BMS	Battery Management System
CDH	Command & Data Handling
CFRP	Carbon Fibre Reinforced Polymer
CSV	Comma-Separated Values
DOF	Degrees of Freedom
EPS	Electrical Power Subsystem
FSM	Finite State Machine
GCS	Ground Control Station
GNSS	Global Navigation Satellite System
GPIO	General Purpose Input/Output
GPS	Global Positioning System
GS	Ground Station
I2C	Inter-Integrated Circuit (serial bus)
IMU	Inertial Measurement Unit
ISM	Industrial Scientific Medical (radio band)
LoRa	Long Range (radio modulation)
LOS	Line of Sight
MCU	Microcontroller Unit
NavIC	Navigation with Indian Constellation
PCB	Printed Circuit Board
PDR	Preliminary Design Review
PID	Proportional-Integral-Derivative (controller)
PM	Particulate Matter
PWM	Pulse-Width Modulation
RF	Radio Frequency

RTC	Real-Time Clock
SPI	Serial Peripheral Interface
UART	Universal Asynchronous Receiver/Transmitter
VCC	Voltage Common Collector (power supply)
CEP	Circular Error Probable

Table 4: List of acronymous

4. SYSTEM OVERVIEW

4.1 Brief summary of the mission

CAN-7USAT is the entry for team VRITRA in IN-SPACe CAN-7USAT India student Competition 2026 which focuses on building a CAN sized satellite with two stage guided descent and full real time telemetry.

PARAMETER	TARGET
Deployment altitude	1000 m (±100 m) AGL
Ejection shock survival	≥ 30G
Primary descent rate	20 m/s (±5 m/s)
Secondary activation altitude	600 m (±10 m) AGL
Secondary descent rate	1–3 m/s
Telemetry rate	≥1 Hz over LoRa
Stabilization	Mechanical gyroscopic wheel

Table 5: Mission requirement

4.1.1 Mission flow

1. Ejection and Stabilization: In this stage the CANSAT separates from the rocket at apogee by surviving the ejection shock and stabilizes itself using a mechanical gyroscopic wheel.

2. Primary Descent: After ejection from the rocket, the hemispherical shaped parachute is opened due to the atmospheric drag without any ejection mechanism making the descent rate to 20 m/s ±5 m/s.

3. Guided secondary descent: At 600 meters above the ground level, a GPS guided RAM air parafoil is deployed with the help of a servo-based ejection mechanism from the parachute bay. This design of parachute is made in such way that it cuts down the descent rate from 20 m/s to 1-3 m/s and is actively steering towards the launch site through the dual axis break line servo control.

4. Continuous telemetry: All 15 mandatory parameters are transferred with the help od LR1121 a LoRa at the rate of 1 Hz. It also has a backup microSD card for storing of data in case of packet or link loss.

Throughout the flight, the CAN-7USAT continuously acquires data from its onboard sensor stack and transmits all 15 mandatory telemetry parameters over a LoRa link at a minimum rate of 1 Hz, while simultaneously logging the complete dataset to an onboard microSD card as a backup in case of telemetry loss. The avionics architecture is built around a Teensy 4.1 primary flight computer running an 8-state finite state machine, supported by a secondary backup avionics board with an independent processing board to ensure telemetry continuity in the event of a primary processor failure.

Beyond the core mission, Team VRITRA also focusses on additional objectives of the mission which can help us in scoring additional points. The bonus capabilities for extra jury weightage are:

- Onboard video capture (descent + ejection)
- NavIC-enabled multi-GNSS positioning
- Environmental sensor suite — PM2.5/PM10, lightning, UV, ozone, CO_2
- Innovation payload: biochemical characterization of atmospheric bacterial isolates collected
- Custom Python ground station for live telemetry, command uplink, and mission logging

4.2 Any changes made from PDR stage

1. Onboard CANSAT telemetry antenna has been changed from Rubber duck omnidirectional antenna (AEACAC049018-S868) to Radio master T antenna of 865 MHz. This change was done due to dual band requirement of frequency during telemetry. Team VRITRA has its primary frequency as 867.0 MHz and the backup frequency as 86.75 MHz, which is important for telemetry transmission without any interference. T antenna from Radio Master is a dual antenna which makes it easy for us to switch the frequency whenever required.
2. Material of CANSAT has been changes which were Carbon fiber rods and glass fiber body. The body material which is selected is Acrylonitrile Butadiene Styrene (ABS) filament which is a tough, impact-resistant thermoplastic polymer. The plates are of aluminium and the rods are of glass fiber. This are changes made in material procurement which was done after the design testing on Ansys. According to the results the selected material gave better results as compared to the pervious in perspective of stress, etc.
3. The gyro wheel motor was changed from N20 to 2205 2300KV brushless DC motor. This was done after testing our RAM air parachute. To avoid spinning of CANSAT after parachute deployment we needed higher RPM motor than the N20.

The required test results are given below to justify about the changes made in CDR stage.

4.3 Overview of System Level Requirement

The CANSAT design is made by considering specific conditions and rules as given in the rulebook which covers mass, dimensions, structural survivability, descent performance, telemetry, and power performance.

Parameter	Required Value	Design Target	Margin

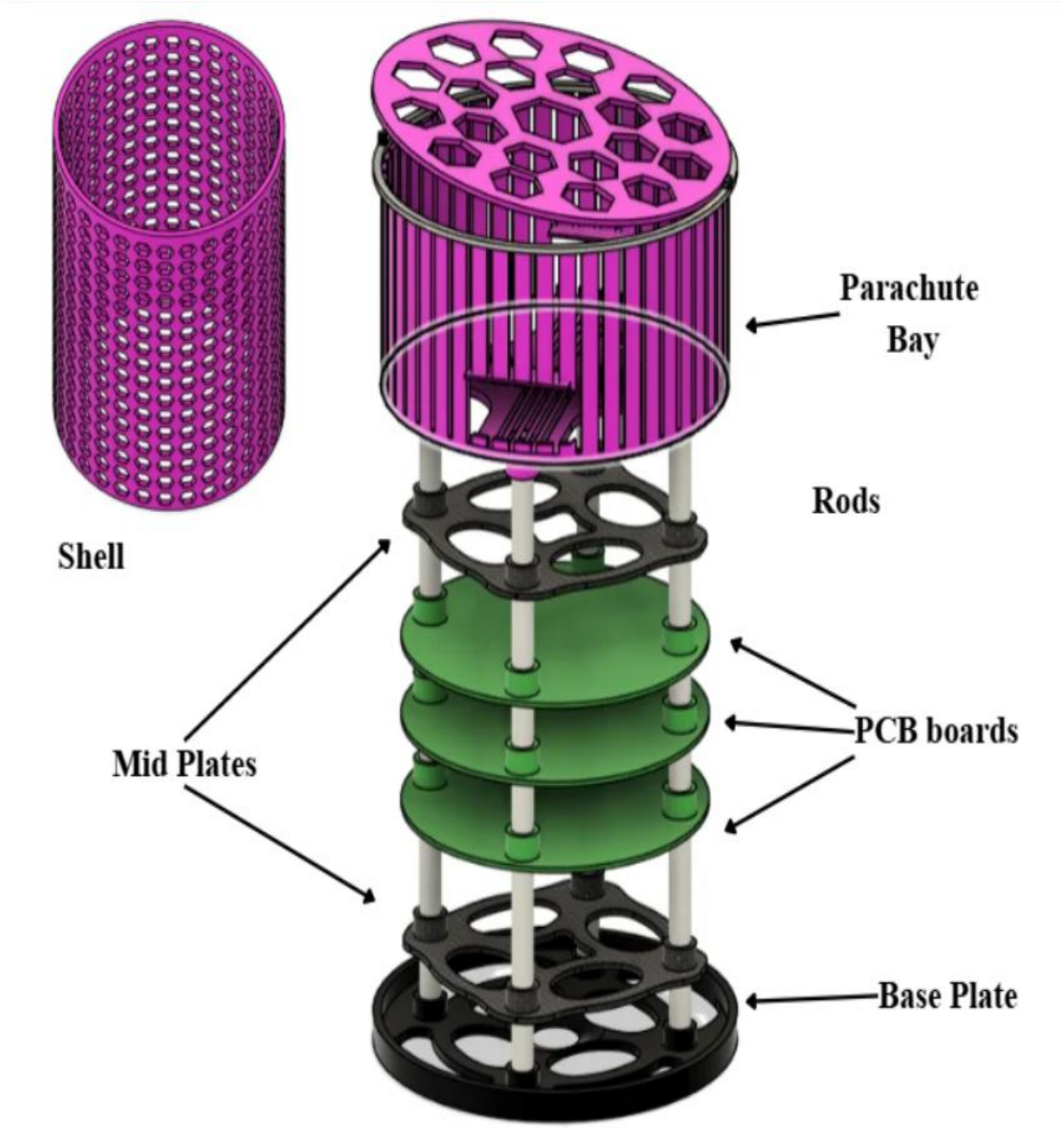
Total mass	≤ 1000 g ±50 g	~920 g ±65g	80 g
Diameter	≤ 150 mm	130 mm OD (fits rocket fairing)	20mm
Height	≤ 400 mm	232 mm	168
Launch acceleration survival	15G	Rated > 30G (Al standoffs + CFRP body)	2×
Shock survival	30G	LSM6DSO32 rated ±32g range; PCBs on M3 standoffs	—
Primary descent rate	20 m/s ±5 m/s	18–22 m/s	Within band
Secondary activation altitude	600 m ±10 m	BMP585, 5-sample average, ±3 m accuracy	±7 m
Secondary descent rate	1–3 m/s	1.5–2.5 m/s under parafoil	Within band
Telemetry rate	≥ 1 Hz	1 Hz (Teensy 4.1)	—
Battery endurance on pad	≥ 2 hours	~6 hours at avg. 750 mA draw	3×
Audio beacon SPL	≥ 92 dB	SFM20 beacon: ≥ 95 dB	3 dB

Table 6: System level requirements

This table has the design target with the required values in which each has a proper margin set so that it can tolerate manufacturing and flight variabilities. The key design considerations are:

- **Modular stacked design** — Base Plate, 3 Mid Plates, and a Parachute Bay connected by 4 continuous fiberglass rods, giving a rigid, serviceable structure (130 × 281 mm overall).
- **Payload bay** — 110 mm internal working diameter housing 3 stacked PCB layers for avionics, power, and mechanisms, within the 140 mm mechanical envelope constraint.
- **Parachute bay/shell** — hexagonal-lattice perforated cylinder (70 mm) for reduced mass and clean deployment-bag ejection.
- **Materials** — ABS for the shell/parachute bay (lightweight, printable), fiberglass rods for the primary load path (high strength-to-weight), and Aluminium 6061 plates for rigidity and mounting stability at joint interfaces.
- **Load path** — the fiberglass rods carry the full stack's structural loads through to the base plate, sized to handle the recovery system's peak opening shock (up to 11.1 G).

4.4 CANSAT physical layout



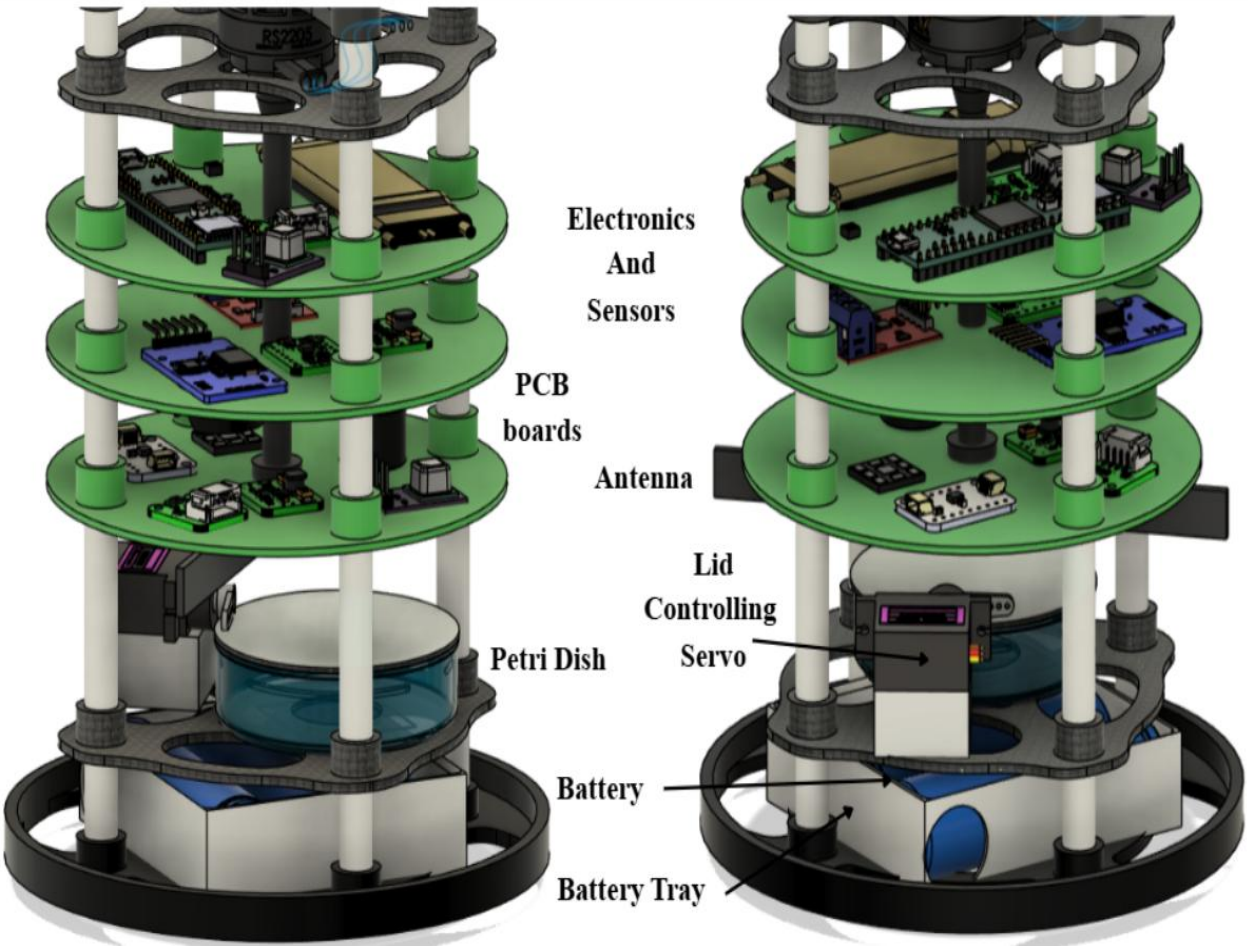

Electronics
And
Sensors
PCB
boards
Antenna
Lid
Controlling
Servo
Petri Dish
Battery
Battery Tray



Fig 1: CANSAT Figure Layout

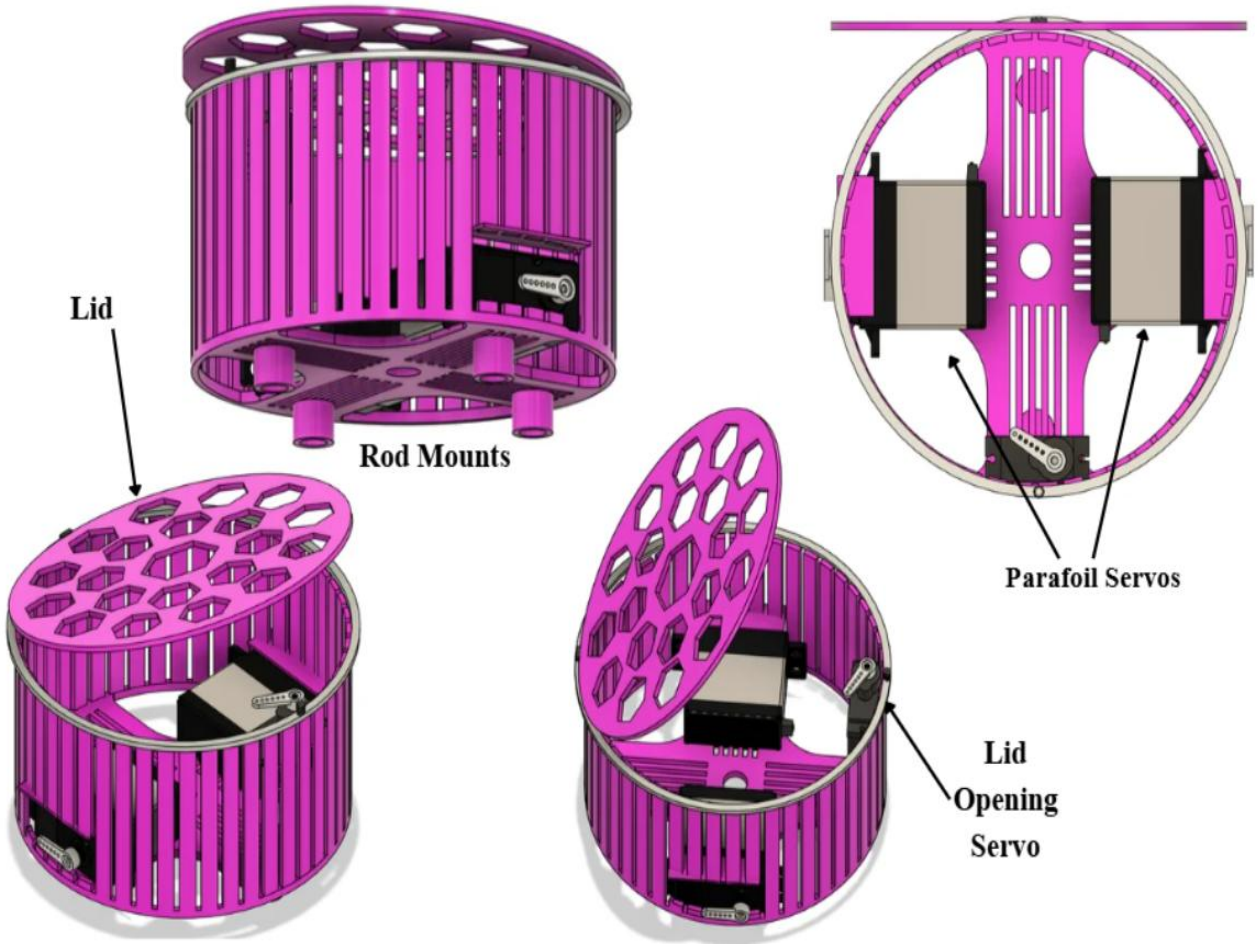


Fig 2: Parachute Pay design

This is the CANSAT physical layout in which all the parts are labelled properly. It consists of a cylindrical body having multiple subsystems inside it. Starting with the parachute bay which is designed for the Secondary descent GPS guided RAM air parafoil and the control mechanism to control the flow of parachute.

Next, we have designed a plate for the placement of gyro wheel and its motor to adjust properly. Followed by the gyro wheel we have electronics plates with all our telemetry components and backup board. The other mid plate after electronics is for battery holding purpose with a self-designed battery pack and can be easily replaced. Shell is designed to cover the CANSAT from external environment and to secure electronics as it has been instructed in the guidelines.

Below are some of the drawings of each part of our CANSAT with their dimensions:

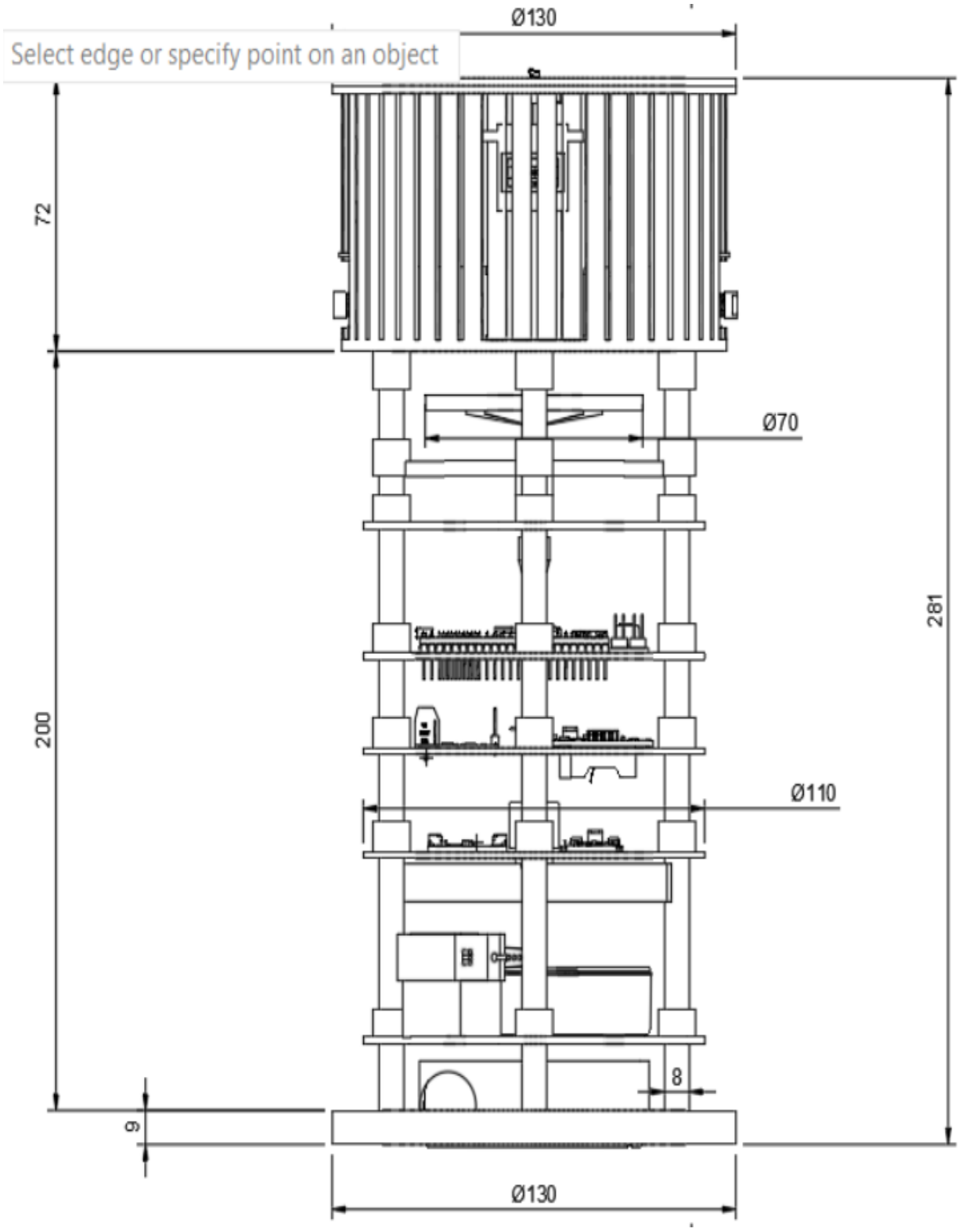


Fig 3: CANSAT Schematic design

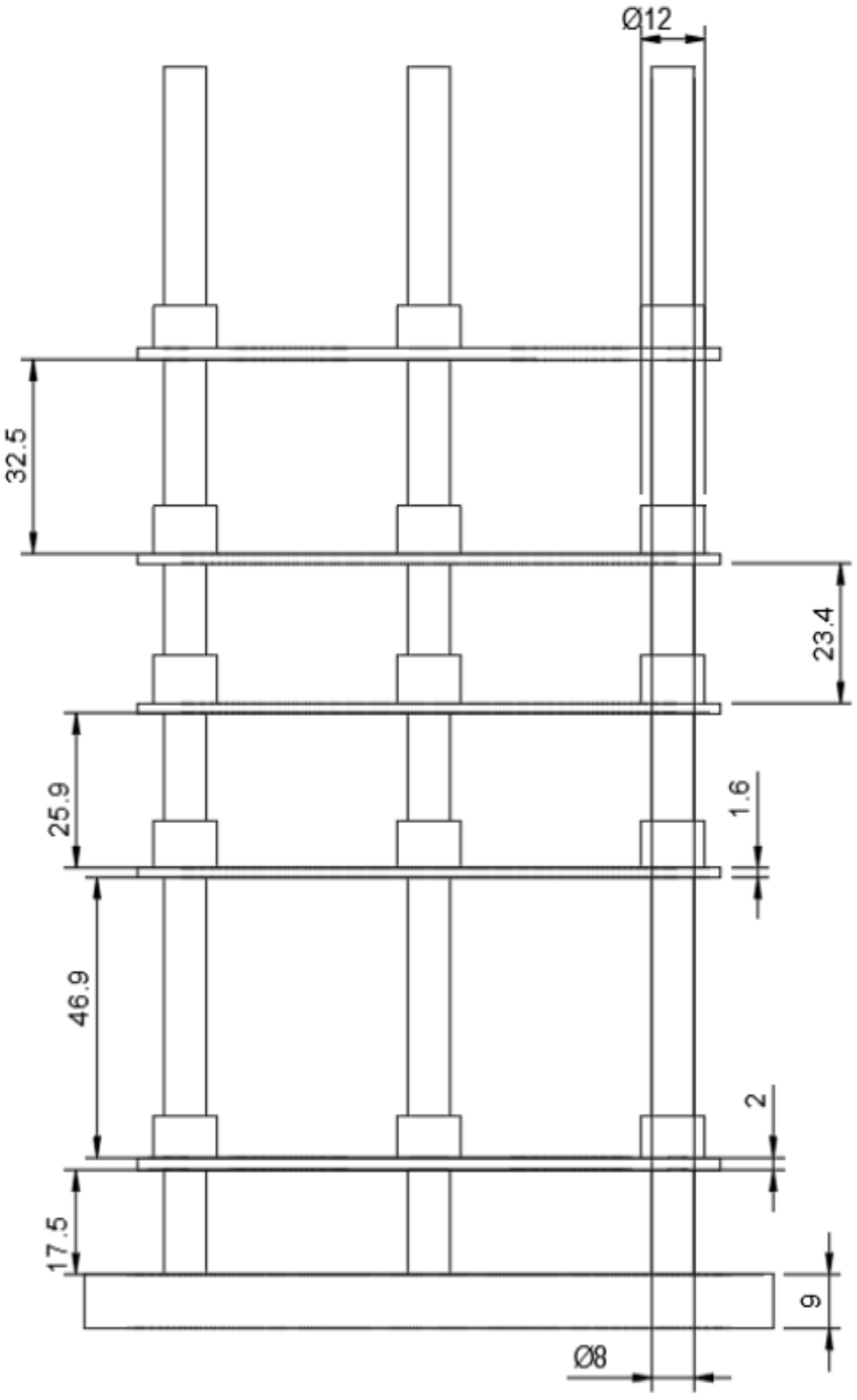


Fig 4: Rod Design

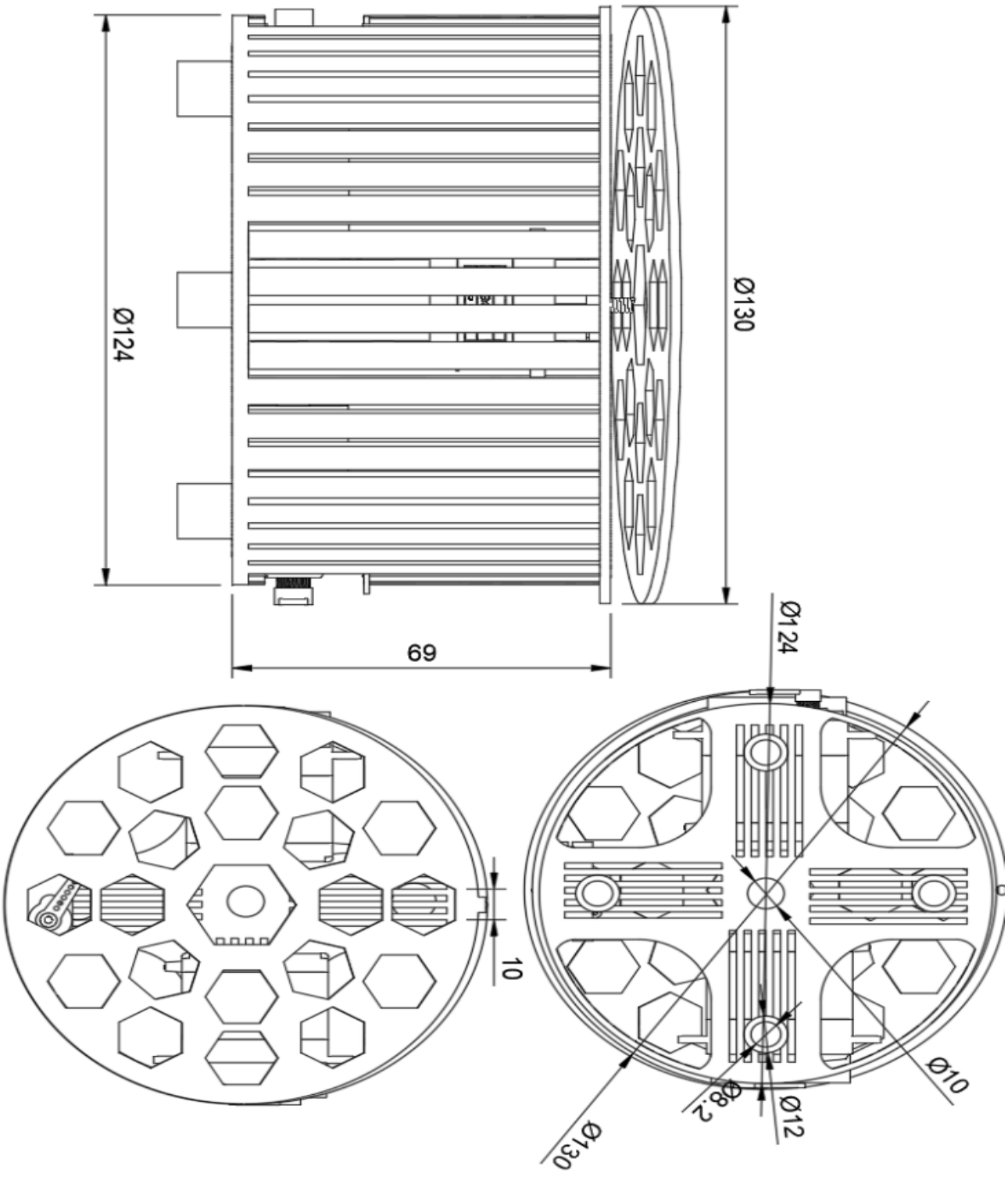


Fig 5: Parachute Bay Schematic Design

4.5 Entire CAN-7USAT Activities- Design Stage Through Launch, Recovery and Data Retrieval

The CANSAT mission consist of five different phases which goes from ground design and fabrication to post fight data recovery. Each phase carries its own verification checkpoints to ensure the system is flight-ready before proceeding to the next.

4.5.1 Design and Fabrication

Different subsystems such as the sensor subsystem, power subsystem, communication and data handling subsystem (CDH), etc have been designed as per the project milestone chart given above.

1. Mechanical subsystem

Reaction-Wheel Geometry and Moment of Inertia:

1.1 A Thin Ring (PRIMARY DESIGN)

The primary design assumes the reaction wheel is constructed as a thin ring (rim-weighted geometry).

Formula: $I = mr^2$

Where: m = wheel mass = 0.070 kg

r = wheel radius = 0.035 m

Calculation:

$$r^2 = (0.035)^2 = 0.001225\ m^2$$

$$I_{wheel, ring} = m \times r^2$$

$$I_{wheel, ring} = 0.070 \times 0.001225$$

$$I_{wheel, ring} = 0.00008575\ kg \cdot m^2$$

Final Result: $I_{wheel, ring} = 8.575000e\text{-}5\ kg \cdot m^2 = 85.75 \times 10^{-6}\ kg \cdot m^2$

This is the primary moment of inertia used throughout the calculation report.

1.2 Theoretical Motor No-Load RPM

Formula: $RPM = KV \times V$

Condition 1: 3S Nominal Voltage (11.1 V)

Nominal cell voltage: 3.7 V per cell

3 cells in series: $V = 3 \times 3.7 = 11.1$ V

Calculate theoretical no-load RPM:

$$RPM = KV \times V$$

$$RPM = 1590 \times 11.1$$

$$RPM = 17{,}649 \text{ RPM}$$

Theoretical RPM at 11.1 V = **17649 RPM**

Condition 2: 3S Fully Charged Voltage (12.6 V)

Full charge cell voltage: 4.2 V per cell

3 cells in series: $V = 3 \times 4.2 = 12.6$ V

Calculate theoretical no-load RPM:

$$RPM = KV \times V$$

$$RPM = 1590 \times 12.6$$

$$RPM = 20{,}034 \text{ RPM}$$

Theoretical RPM at 12.6 V = **20034 RPM**

1.3 RPM Margin Above Target Operating Speed

Target wheel speed: $N_{required} = 10000$ RPM

Condition 1: Margin at 11.1 V (3S Nominal)

Theoretical RPM = 17,649 RPM

Required RPM = 10,000 RPM

RPM margin = 17,649 − 10,000 = 7,649 RPM

Percentage margin = (7,649 / 10,000) × 100 = 76.49%

At 11.1 V: RPM margin = 7649 RPM (76.49% above target)

Target represents 56.66% of theoretical no-load speed.

Condition 2: Margin at 12.6 V (3S Fully Charged)

Theoretical RPM = 20,034 RPM

Required RPM = 10,000 RPM

RPM margin = 20,034 − 10,000 = 10,034 RPM

Percentage margin = (10,034 / 10,000) × 100 = 100.34%

At 12.6 V: RPM margin = 10034 RPM (100.34% above target)

Target represents 49.92% of theoretical no-load speed.

1.4 Angular Velocity Conversion

Formula: $\omega = 2\pi N / 60$

Condition 1: 5000 RPM

Step 1: N_rev/s = 5000 / 60 = 83.3333 rev/s

Step 2: $\omega = 2 \times 3.14159265 \times 83.3333 = 523.5988$ rad/s

Condition 2: 7500 RPM

Step 1: N_rev/s = 7500 / 60 = 125.0000 rev/s

Step 2: $\omega = 2 \times 3.14159265 \times 125.0000 = 785.3982$ rad/s

Condition 3: 10000 RPM

Step 1: N_rev/s = 10000 / 60 = 166.6667 rev/s

Step 2: $\omega = 2 \times 3.14159265 \times 166.6667 = 1047.1976$ rad/s

Condition 4: 12000 RPM

Step 1: N_rev/s = 12000 / 60 = 200.0000 rev/s

Step 2: $\omega = 2 \times 3.14159265 \times 200.0000 = 1256.6371$ rad/s

1.5 Moment of Inertia Summary

Primary design uses thin-ring geometry:

I_wheel, ring = 8.575000e-5 kg·m²

I_wheel, ring = 85.75×10^{-6} kg·m²

1.6 Angular Momentum

Formula: $L = I\omega$

Angular Momentum at 5000 RPM

$\omega = 523.5988$ rad/s

L = 8.575000e-5 × 523.5988 = 4.4899e-2 kg·m²/s

Angular Momentum at 7500 RPM

$$\omega = 785.3982 \text{ rad/s}$$

$$L = 8.575000e\text{-}5 \times 785.3982 = 6.7348e\text{-}2 \text{ kg·m}^2/\text{s}$$

Angular Momentum at 10000 RPM

$$\omega = 1047.1976 \text{ rad/s}$$

$$L = 8.575000e\text{-}5 \times 1047.1976 = 8.9797e\text{-}2 \text{ kg·m}^2/\text{s}$$

Angular Momentum at 12000 RPM

$$\omega = 1256.6371 \text{ rad/s}$$

$$L = 8.575000e\text{-}5 \times 1256.6371 = 1.0776e\text{-}1 \text{ kg·m}^2/\text{s}$$

1.7 Rotational Kinetic Energy

$$\text{Formula: } E = (1/2)I\omega^2$$

Rotational Energy at 5000 RPM

$$\omega = 523.5988 \text{ rad/s}$$

$$\omega^2 = 274155.677808 \text{ rad}^2/\text{s}^2$$

$$E = 0.5 \times 8.575000e\text{-}5 \times 274155.677808 = 11.754425 \text{ J}$$

Rotational Energy at 7500 RPM

$$\omega = 785.3982 \text{ rad/s}$$

$$\omega^2 = 616850.275068 \text{ rad}^2/\text{s}^2$$

$$E = 0.5 \times 8.575000e\text{-}5 \times 616850.275068 = 26.447456 \text{ J}$$

Rotational Energy at 10000 RPM

$$\omega = 1047.1976 \text{ rad/s}$$

$$\omega^2 = 1096622.711232 \text{ rad}^2/\text{s}^2$$

$$E = 0.5 \times 8.575000e\text{-}5 \times 1096622.711232 = 47.017699 \text{ J}$$

Rotational Energy at 12000 RPM

$$\omega = 1256.6371 \text{ rad/s}$$

$$\omega^2 = 1579136.704174 \text{ rad}^2/\text{s}^2$$

$$E = 0.5 \times 8.575000e\text{-}5 \times 1579136.704174 = 67.705486 \text{ J}$$

At 10,000 RPM, the wheel stores 47.0177 J of rotational energy.

1.8 Centripetal Acceleration

Formula: $a = \omega^2 r$

Centripetal Acceleration at 5000 RPM

$$a = 523.5988^2 \times 0.035 = 274155.677808 \times 0.035 = 9595.4487 \text{ m/s}^2$$

$$a_g = 9595.4487 / 9.81 = 978.1293 \text{ g}$$

Centripetal Acceleration at 7500 RPM

$$a = 785.3982^2 \times 0.035 = 616850.275068 \times 0.035 = 21589.7596 \text{ m/s}^2$$

$$a_g = 21589.7596 / 9.81 = 2200.7910 \text{ g}$$

Centripetal Acceleration at 10000 RPM

$$a = 1047.1976^2 \times 0.035 = 1096622.711232 \times 0.035 = 38381.7949 \text{ m/s}^2$$

$$a_g = 38381.7949 / 9.81 = 3912.5173 \text{ g}$$

Centripetal Acceleration at 12000 RPM

$$a = 1256.6371^2 \times 0.035 = 1579136.704174 \times 0.035 = 55269.7846 \text{ m/s}^2$$

$$a_g = 55269.7846 / 9.81 = 5634.0249 \text{ g}$$

1.9 Battery Theoretical Energy Storage

Battery Specification:

Type: 3S Li-ion

Nominal voltage: 11.1 V

Capacity: 5000 mAh = 5 Ah

Calculate theoretical stored energy:

Formula: $E_{battery} = V \times Ah$

$$E_{battery} = 11.1 \times 5 = 55.5 \text{ Wh}$$

Convert to joules:

$$1 \text{ Wh} = 3600 \text{ J}$$

$$E_{battery} = 55.5 \times 3600 = 199{,}800 \text{ J}$$

Final result:

Battery theoretical stored energy = 55.5 Wh = 199,800 J

This represents total electrical energy available from a fully charged pack. Actual deliverable energy depends on discharge rate, temperature, cycle life, and battery C-rating.

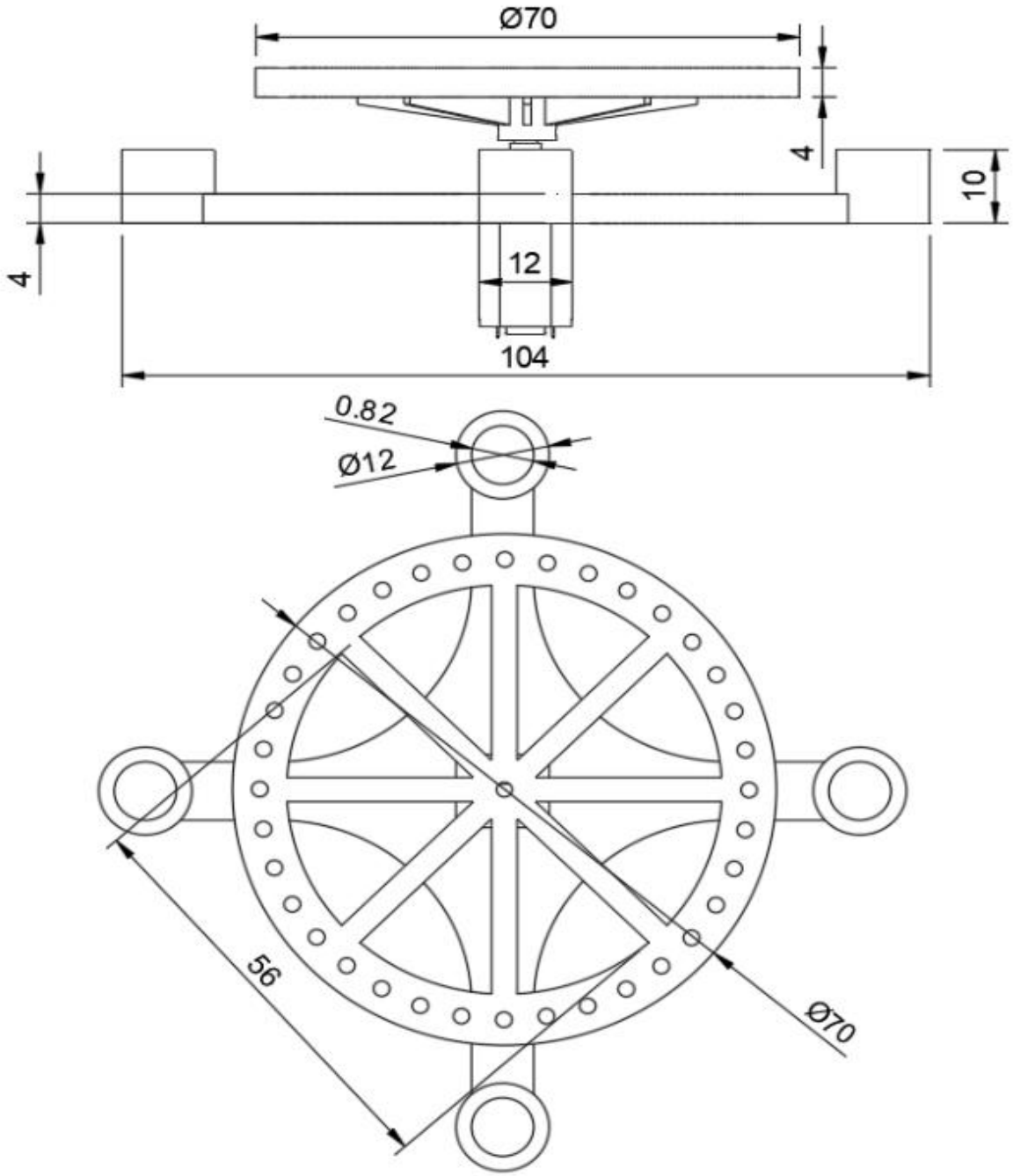
1.10 Reaction-Wheel / CANSAT Moment of Inertia Ratio

Calculate: I_wheel / I_cansat = 8.575000e-5 / 7.969063e-3 = 1.076036e-2

Inertia ratio = 1.076036e-2 = 1.0760%

The reaction wheel's moment of inertia is approximately 1.076% of the CANSAT moment of inertia. This small ratio indicates that significant wheel torque is needed to produce CANSAT Altitude changes, typical for CubeSat-class satellites with limited actuators.

Gyro wheel design:


Ø70
4
10
4
12
104
0.82
Ø12
56
Ø70

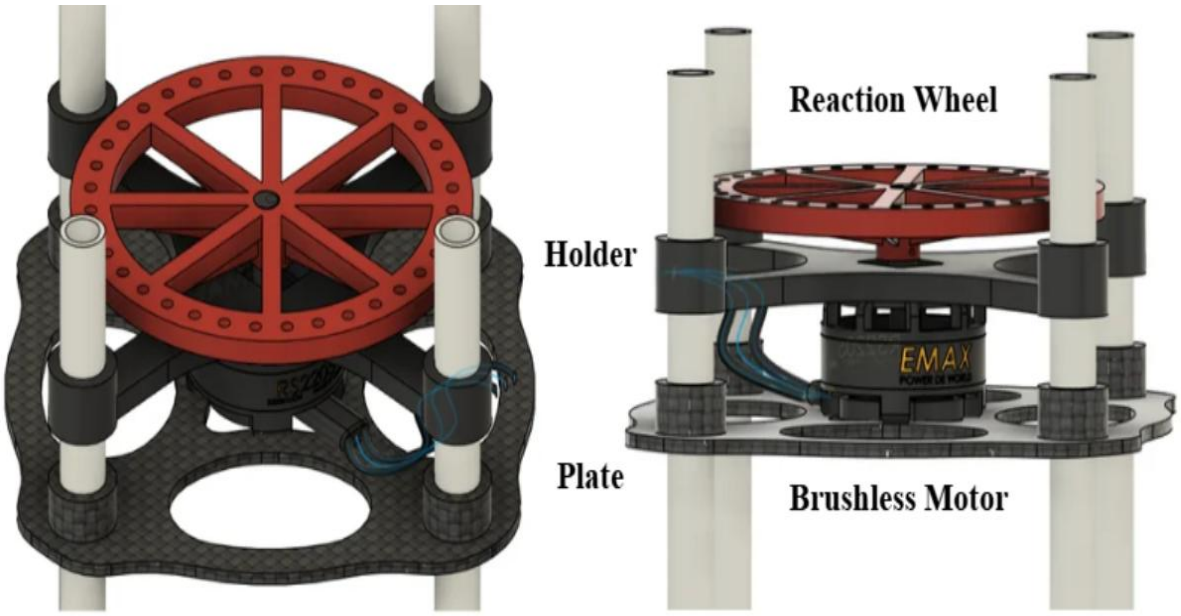


Fig 6: Gyro Wheel Schematic design

2. Command and Data Handing (CDH) Subsystem

- Teensy 4.1 was selected as the primary flight computer after comparing multiple microprocessors. The optional microprocessors which were compared a
- re Raspberry pi 0, ESP 32 P4, STM 32, Teensy 4.0, etc. Teensy 4.1 was choose because of its specification which were good as compared to other such as processor speed, memory, pin count, footprint.
- 8-state flight software FSM architected and coded in C++ (Arduino/Teensyduino framework), running a 10 Hz sensor/control loop with 1 Hz telemetry sub-rate.
- XIAO ESP32S3 camera module is added as an independent recording subsystem (isolated from flight-critical control, so a camera failure cannot affect mission success).
- DS3231M RTC and onboard EEPROM is included for mission-time and flight-state persistence across any processor reset.
- SanDisk Ultra MicroSDHC selected for onboard data logging after comparing read/write speed and cost against SanDisk Extreme Pro and Kingston alternatives.

3. Sensor Subsystem

- All sensors are selected through comparative trade studies based on the accuracy, power draw, interface compatibility, etc for each measurement type:

  - BMP585 - barometric pressure/altitude ($\pm$0.5 Pa accuracy).

- LSM6DSO32 -6-DOF IMU for Secondary parachute control (±32g, ±4000 dps range).
- ICM-20948 -9-DOF IMU/magnetometer for acceleration and spin-rate sensing.
- L89HA GNSS - multi-constellation GPS with NavIC support, dual-band L1+L5 for <1.8 m CEP accuracy.
- INA226 - real-time bus voltage/current/power monitoring.
- Bonus payload sensors -BMV080 (PM), SEN0290 (lightning), ML8511 (UV), MQ131 (ozone), SCD41 (CO_2).

- Telemetry packet format (15 mandatory and other bonus fields) defined and mapped to source sensors, with resolution/units specified per Interface Control Document.
- Sensor interfacing finalized across I2C, SPI, and UART buses, with bus-conflict mitigation planned (e.g., BMP585 moved to SPI or watchdog-recovery routine if I2C contention arises with the RTC).

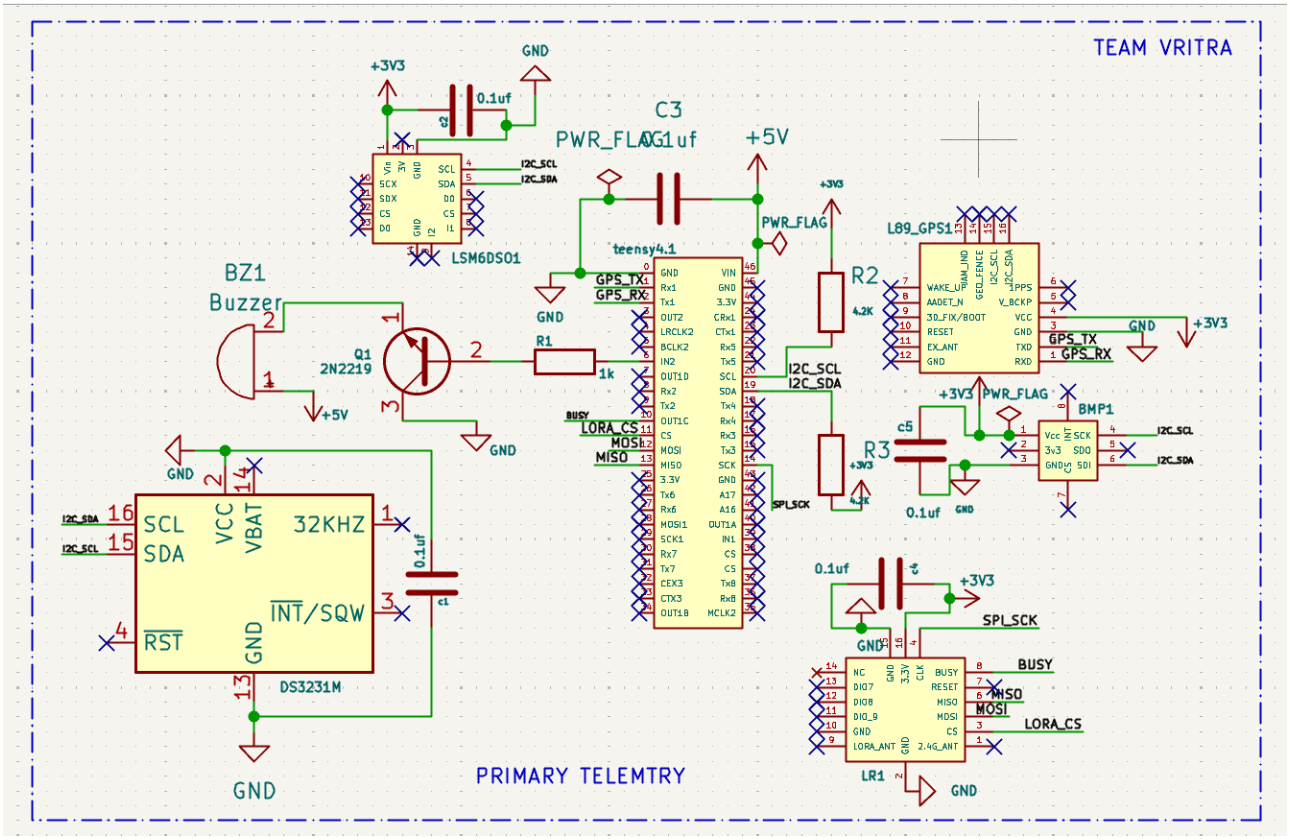


Fig 7: Primary telemetry Schematic

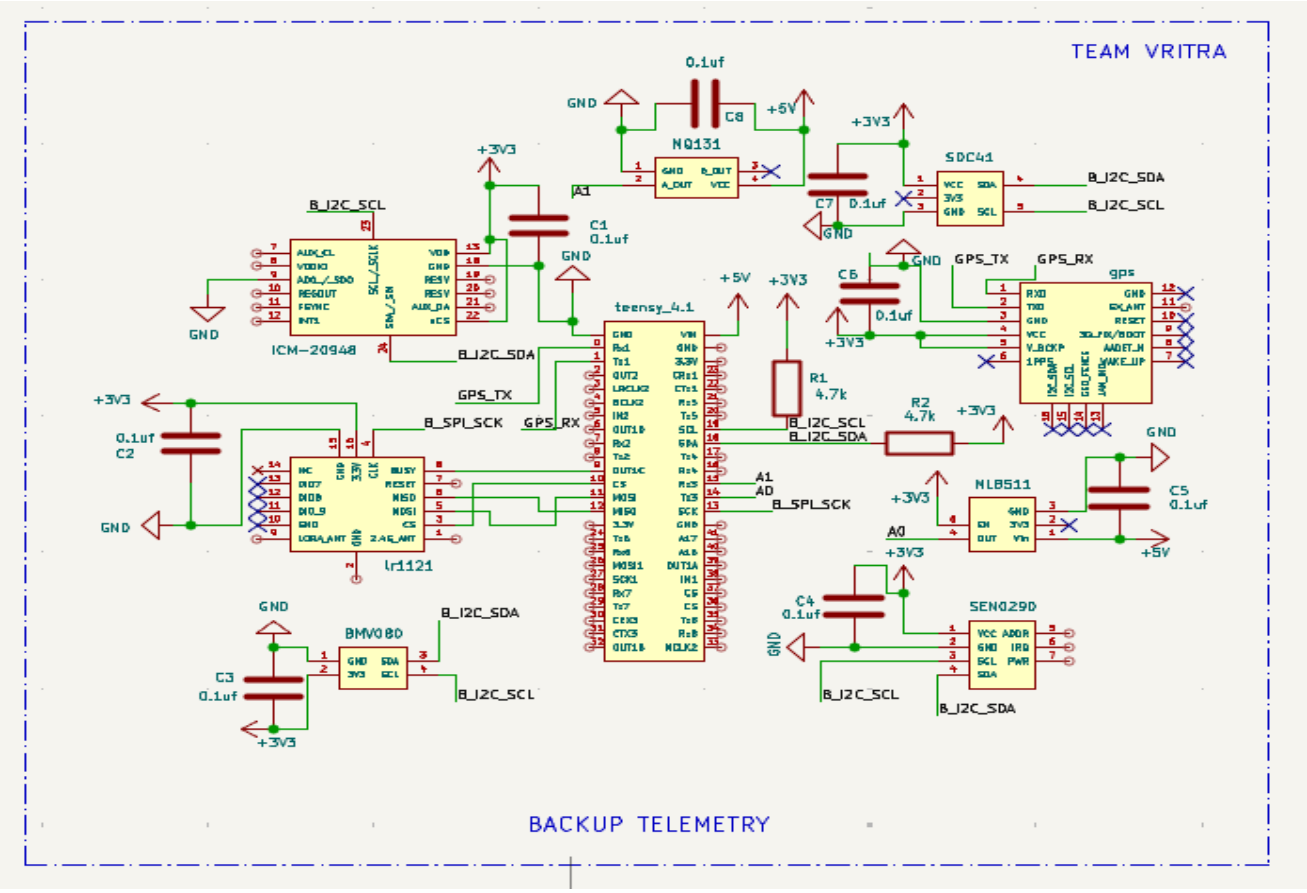


Fig 8: Backup Telemetry Schematic

This is the schematic design for all the important telemetry required from the CANSAT. All sensors are attached with a coupling capacitor to avoid current spikes in the circuit. Teensy 4.1 is the main processor of the CANSAT through which all other sensors are connected.

This is the Schematic design of CANSAT's backup system and also about the additional environmental sensors used for bonus jury weightage. Backup system has a different processor which is Teensy 4.1. This will help us in getting the data even when the main processor shuts down.

4. Communication & Data Handling Subsystem

- LR1121 multi-band transceiver is selected over SX1262/SX1280 for its dual Sub-GHz + 2.4 GHz flexibility and lower power draw.
- Configured as LoRa at 867 MHz (India ISM band) with team-specific NETID/PANID (0x98) to prevent cross-team packet collision.
- Bandit 915/868 MHz T antenna is selected for onboard transmission based on gain, frequency match, and connector compatibility.
- Radio interference mitigation plan defined: orthogonal spreading factor (SF10) selection, directional ground antenna gain, and packet-counter-based loss detection.
- Secondary/backup LR1121 link designed on the backup avionics board for failover telemetry continuity.

5. Electrical Power Subsystem (EPS)

- 3S Li-ion battery pack (Molicel INR-21700-P50B, 5000 mAh) is selected and sized against the full mission power budget.
- 3S 40A BMS is used for overcharge/over-discharge/overcurrent/short-circuit protection and cell balancing.
- Dual buck converter architecture is designed by using 7Semi AP64500 (5V/5A rail) and AP62301 (3.3V/2A rail), sized against actuator and sensor current draw respectively.
- INA226 power monitor and status LED incorporated for real-time health telemetry and visual power-on confirmation.
- Power budget is calculated as per mission phase such as standby, pre-launch, ascent, descent, peak, etc to operate for 2-hour pad-endurance requirement and the design achieves roughly 6 hours of margin.

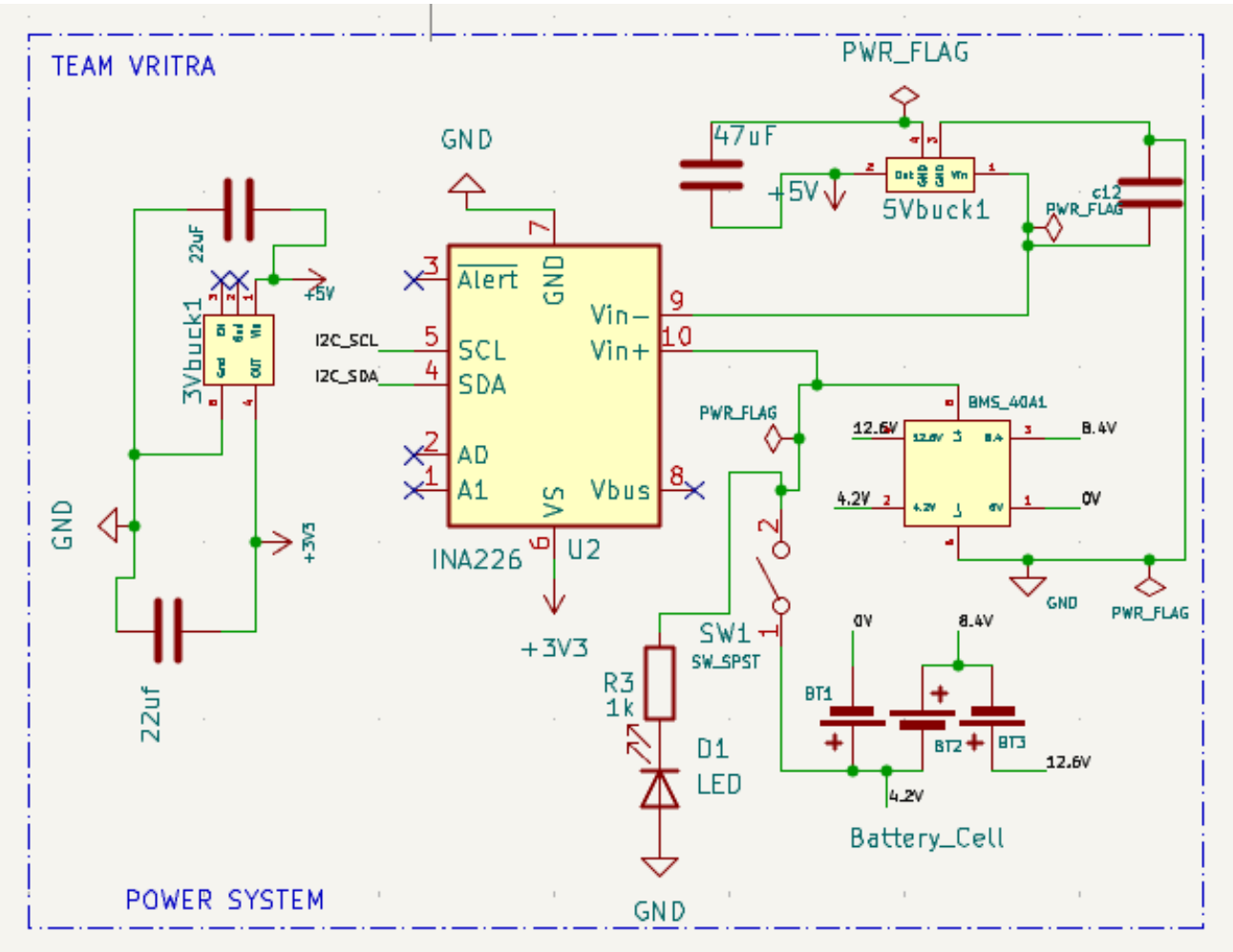


Fig 9: Power System

This schematic is about the power subsystem which is required for complete operation of CANSAT. It consists of 3 molicel 21700 lithium-ion batteries connected in series. This power schematic also consists of BMS and INA226. It has 5V and 3.3 V buck converters for power rails as required. A mandatory switch and a LED is connected through the power supply for the purpose of indication.

6. Descent & Recovery Subsystem

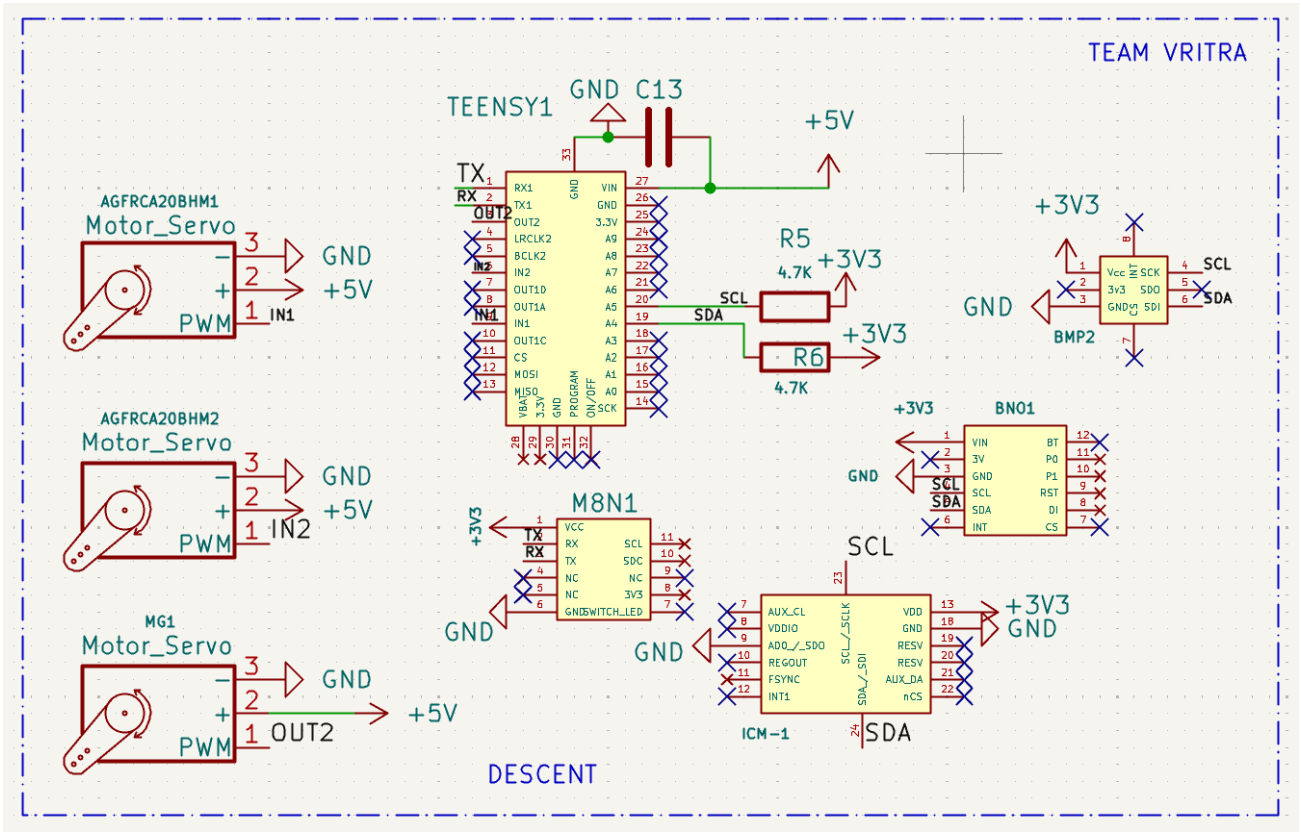


Fig 10: Descent Schematic

This schematic is about the descent subsystem. Team VRITRA's secondary descent subsystem is about GPS guided RAM air parafoil which includes a different processor independent from any other which is used for telemetry. Teensy 4.0 has been selected for Descent control, so that any error in telemetry side will not affect the descent mechanism of parachute.

Primary Parachute Design:

A flat circular vented canopy is selected for primary parachute on the basis of:

- Fastest and most reliable deployment (critical for primary)
- No preferred orientation required
- Established drag coefficient data

Drag coefficient referenced to constructed (nominal) area = 0.75

At equilibrium descent, aerodynamic drag equals gravitational force:

$$\frac{1}{2}\rho 1 V_{s,P1}^2 S_{P1} C_{D,P1} = mg$$

Solving for plan area S_{P1}:

$$S_{P1} = \frac{2mg}{\rho 1 V_{s,P1}^2 C_{D,P1}}$$

Substituting values ($V_{s,P1}$= 15 m/s):

$$S_{P1} = \frac{2 \times 1.000 \times 9.810}{1.112 \times 15^2 \times 0.75} = \frac{19.620}{187.65} = 0.1045\ m^2$$

Canopy diameter from plan area:

$$D_{P1} = \sqrt{\frac{4S_{P1}, design}{3.14}} = 0.383\ m$$

Central vent diameter:

$$d_v = 0.12 \times D_{P1} = 0.12 \times 380 = 45.6\ \text{mm}$$

Shroud Line length:

$$L_s = 1.5 \times D_{P1} = 1.5 \times 380 = 570\ \text{mm}$$

Parameters	Symbols	Values
Canopy type	-	Flat circular, vented
Plan area	S_{P1}	0.1134 m^2
Canopy diameter	D_{P1}	380 mm
Central vent diameter	d_v	45 mm
Drag coefficient	$C_{d,P1}$	0.75
Equilibrium sink rate	$V_{s,P1}$	14.4 m/s at 1000 m AGL
Number of shroud lines	n_s	8
Shroud line length	L_s	570 mm
Line spacing on canopy edge	Δs	149 mm
Line rated strength	-	133 N (30 lb)

Table 7: Primary parachute parameters

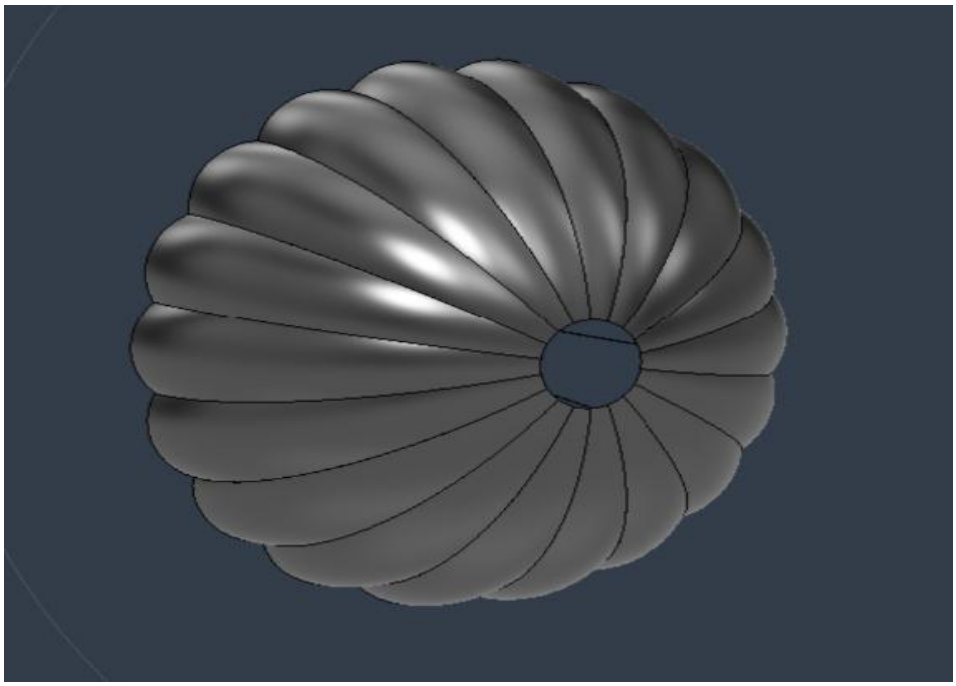


Fig 11 : Primary Parachute

Secondary Parachute design:

A ram-air parafoil is selected for P2 because:

1. It can be steered — two brake lines controlled by servos.

2. Its low equilibrium sink rate satisfies R-02.

3. Forward glide capability allows landing point adjustment.

A rectangular planform with low aspect ratio (AR = 2.30) is chosen for:

- Maximum collapse resistance (critical for autonomous control)
- Stable deployment without feedback
- Forgiving response to asymmetric brake inputs

Basic concept behind the RAM-air parachute idea is:

- Autonomous multi-stage descent controller for mission execution.
- Sensor-fusion-based altitude estimation using barometer and IMU data.
- GPS-guided navigation with wind compensation for target-oriented descent.
- Controlled flare and braking mechanism for safe and accurate landing.
- Real-time telemetry, onboard logging, and fault-recovery capabilities for mission reliability.

Minimum Canopy Area Derivation:

$$S_{P2} = \frac{2mg}{\rho 2 V_s^2 C_L (1 + GR^2)}$$

$$= \frac{19.620}{1.155 \times 6.250 \times 0.90 \times 7.250}$$

$$= 0.415 m^2$$

Performance verification:

$$V_{total} = \sqrt{\frac{2mg}{\rho_2 S_{p2} C_L}}$$

$$= \sqrt{\frac{2 \times 1.000 \times 9.810}{1.155 \times 0.40 \times 0.90}}$$

$$= \sqrt{\frac{19.620}{0.4158}} = \sqrt{47.19} = 6.87\ m/s$$

Forward speed:

$$V_f = GR \times V_s = 2.50 \times 2.55 = \mathbf{6.37}\ \text{m/s}$$

Canopy Loading:

$$\frac{W}{S_{P2}} = \frac{mg}{S_{P2}} = \frac{1.000 \times 9.810}{0.40}$$

$$= 24.5 \frac{N}{m^2} = 2.50\ kg/m^2$$

Profile Feature	Ratio (% chord)	Formula	Dimension
Max thickness	14%	0.14 × 420	58.8 mm ≈ 59 mm
Position of max thickness	32% from LE	0.32 × 420	134 mm from LE
Max camber (upper surface)	9%	0.09 × 420	37.8 mm ≈ 38 mm
Position of max camber	35% from LE	0.35 × 420	147 mm from LE
Leading edge radius	2.5%	0.025 × 420	10.5 mm ≈ 11 mm
Trailing edge thickness	1.5%	0.015 × 420	6.3 mm ≈ 7 mm

Table 8: Secondary parachute summary

Parameter	Symbol	Value

Canopy type	-	Ram-air parafoil, rectangular, 7 cells
Plan area	S_{P2}	0.400 m2
Span	b	960 mm
Mean chord	C	420 mm
Aspect ratio	AR	2.30
Number of cells	n_{cell}	7
Number of ribs	n_{rib}	8
Cell width	w_{cell}	137 mm
Max profile thickness	t_{max}	134mm
Max camber	f_{max}	147 mm
Entry mouth height per cell	h_{inlet}	30mm
Lift coefficient	C_L	0.90
Glide ratio	GR	2.50
Equilibrium sink rate	V_s	2.55 m/s at 600 m AGL
Forward speed	V_f	6.37 m/s
Total trim airspeed	V_{total}	6.87 m/s
Canopy loading	W/S	24.5 N/m2
Main line length	L_{line}	870 mm
Brake line length	L_{brake}	1050 mm
Riser length	L_{riser}	90mm
Total suspension length	L_{total}	960 mm
Total lines	-	16 (14 main + 2 brake)
Brake drag force	-	0.661 N per side

Table 9: Secondary parachute parameters

Parachute material:

Selected material	Selection reason
 **Ripstop Nylon 40D**	• Lightweight and compact for CanSat parachutes • High tear resistance due to ripstop weave • Good strength-to-weight ratio • Water-resistant and weather-resistant • Easy to fold and deploy

Parachute threads:

Selected thread	Selection reason
 **Nylon Thread**	• High tensile strength for parachute suspension lines • Lightweight and flexible • Good resistance to wear and abrasion • Easy to stitch and handle during assembly • Cost-effective and readily available

7. Ground Station

- Ground hardware architecture is defined as laptop, LR1121 receiver via ESP32-WROOM bridge, Radio master omnidirectional T antenna 865 MHz.
- Ground software developed in Python (PyQt5 + Matplotlib) for real-time telemetry display, live plotting of all 15 fields and additional telemetry data, CSV logging, command uplink (TX ON/OFF, calibrate), and GPS track visualization.

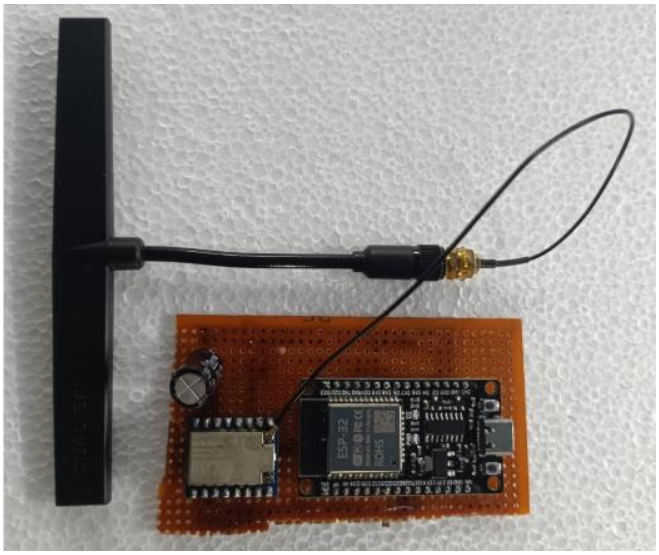


Fig 12: Ground Station

4.6 Material Procurement details

All components of team VRITRA are sources from combination of different domestic electronics suppliers and international distributers. The main preference is given to the Indian manufactured parts where precision is better as compared to other international components because according to the guidelines of competition, Make-in-India components are given additional points.

4.6.1 The procurement strategy

- Primary domestic suppliers: Robu.in, Amazon India and local Indian electronics vendors for sensors, connectors, and mechanical hardware.
- International/specialty suppliers: Digikey for components not readily available domestically (e.g., specific IMU breakout variants, precision servos).
- Direct manufacturer sourcing: A significant share of the sensor and power-regulation stack such as BMP585, ICM-20948, BMV080, AP64500, AP62301, and others, is sourced directly from **7Semi Technologies**, an Indian company based in Navi Mumbai specializing in embedded sensors and wireless communication modules — these components are highlighted in the BOM as Make-in-India contributions.
- Structural materials: Glass fiber rods, plates, and sheet stock procured separately for the shell, base plate, and internal structural elements.
- Procurement timeline: Component purchasing is scheduled as a dedicated milestone immediately after PDR submission, ahead of PCB fabrication and mechanical fabrication, to avoid schedule slip on integration and testing.

4.6.2 Budget Compliance

COMPONENT	PURPOSE	QTY.	SOURCE	TOTAL COST (₹)
BMP585	Pressure & Altitude Sensor	3	7 Semi	1,675.20
LSM6DSO32	6-DOF IMU + Gyroscope	1	Digikey	1705.00
ICM-20948	9-DOF IMU + Magnetometer	2	7 Semi	1,536.00
SCD41	CO_2, Temperature & Humidity Sensor	1	7 Semi	6,224.00
SEN0290 (AS3935)	Lightning Detection Sensor	1	Digikey	2,199.00
ML8511	UV Sensor	1	Robu.in	2,099.00
MQ131	Ozone (O_3) Sensor	1	Robu.in	1,449.00
BMV080	PM2.5 Air Quality Sensor	1	7 Semi	1,092.00
TPL5010	Watchdog Timer	1	Digikey	4067.00
W25Q64 SPI Flash	Data Backup Memory	1	7 Semi	111.20
INA226	Voltage & Current Monitor	2	Robu.in	219
SN74LV4052A	Analog Multiplexer	1	Digikey	290.00

DS3231N RTC	Real-Time Clock	1	Think Robotics	299.00
Teensy 4.1	Primary & Backup Flight Controller	2	Robu.in	7998.00
ESP32-S3	Ground Station Processor	1	Robu.in	2,709.00
XIAO ESP32-S3	Camera Module	2	Robocraze	2,011.00
LR1121 LoRa	Telemetry Transceiver	2	Think Robotics	2450
T antenna 865 MHz	Onboard and ground station omnidirectional antenna	2	Radio master	3700
L89 GPS	NavIC/GPS/GLONASS GNSS	2	7 Semi	1,598.40
NavIC GPS Antenna	Active GNSS Patch Antenna	2	Edge hex	1,500.00
AGFRC A20BHM	Parafoil Line Control	2		9000
MG90S Servo	Secondary Deployment Mechanism	2	Robu.in	444.00
EMAX2205 2300kv	Gyroscope Wheel Motor	1	Robu.in	750
Bi directional ESC 40A	Motor Driver	1	7 Semi	1167
3S 40A Li-ion BMS	Battery Protection Circuit	1	Amazon India	817.00
DFR1015	Multi-Output Buck Converter	2	Digikey	1,000.00
AP64500 5V 5A Buck	Servo Power Supply	3	7 Semi	1,675.20
AP62301 3.3V 2A Buck	Sensor Power Supply	3	7 Semi	235.20
Teensy 4.0	Secondary descent MCU	1	Robu.in	2,549.00
NEO M9N	GPS module	1		2,000.00
XT30 Connector	Battery Connection	5	7 Semi	245.00
Capacitors	470 µF & 0.1 µF	Set	Amazon India	100.00
Resistors	Various Values	Set	Amazon India	125.00
Glass Fibre Rod	Structural Component	4		1700.00
SD Cards	Camera & MCU Storage	2	Amazon India	2503.00
2104-SM400204-1-ND	Buzzer	1	7 Semi	930.00
MAX98357A	Audio Amplifier	1		730.00
Parachute Material	Recovery System	-		12000
PCB Custom Fabrication	Main & Secondary PCB	2		5000
LEDs	Status Indicators	2	Amazon India	10.00
Power Switch	Main Power Control	1	7 Semi	252.00

Molicel 21700 power cells	Battery pack	6	Robu.in	3500
Aluminium	Structure	-	Locally Sourced	2000
ABS	Outer Body	-		1200
Contingency Reserve	-	-		6,000
			Total	**99212**

Table 10: Budget sheet

Make-in-India components from 7 Semi are highlighted in the budget sheet. Source of the components are mentioned with it. 7 Semi which is an Indian Manufacturing company, has helped team VRITRA by providing 20 per cent of discount during our CANSAT preparation.

4.6.3 Mass budget

COMPONENT/PART	PURPOSE	WEIGHT (g)
BMP585	Pressure and altitude sensing	3 (Quantity-2)
LSM6DSO32 (Adafruit)	6-DOF IMU and gyroscope	2
ICM-20948 7Semi	9-DOF IMU and magnetometer	8 (Quantity-2)
SCD41	CO_2, temperature, and humidity sensing	8
SEN0290 DFRobot AS3935	Lightning detection	8
ML8511	UV radiation sensing	2
MQ131	Ozone (O_3) gas sensing	8
BMV080 7Semi	Fanless laser PM sensor	1
TPL5010	Watchdog timer	2
W25Q64 SPI Flash	In-flight data backup	1.5
INA226	Voltage and current monitoring	5
SN74LV4052A	Multiplexer switch	2
DS3231N RTC	Real-time clock	3
Teensy 4.1	Primary and backup flight controller MCU	12 (Quantity-2)
XIAO ESP32-S3 Sense	Camera module	5.6
Teensy 4.0	Secondary ejection control MCU	10
LR1121 LoRa Transceiver	Telemetry communication	2
Bandwidth T antenna 865 MHz	CanSat antenna	22
L89 GPS	NavIC, GPS, and GLONASS GNSS module	6.7
NavIC GPS Antenna	Active GNSS patch antenna	28
AGFRC A20BHM	Paraglider line control	18 (Quantity-2)
MG90S Servo	Secondary deployment/ Bio gel lid	12.5 (Quantity-2)
2205 2300KV BLDC Motor	Gyroscope wheel motor	30

40A ESC	Motor Motor control	30
AP64500 5V 5A Buck	Servo power regulator	0.6
AP62301 3.3V 2A Buck	Sensor power regulator	0.4
AMS1117-1.8 LDO	1.8V voltage regulation	0.5
XT30 Connector	Battery connection	3
Capacitors (470µF + 0.1µF)	Power filtering and stabilization	—
Resistors (0.1Ω, 4.7kΩ, 330Ω, 10kΩ)	Signal conditioning and protection	—
Diodes	Protection and switching	—
Molicell 21700 5000mAh	Lithium-ion battery	72 (Quantity-3)
M9N GPS module	Secondary mechanism GPS	12
Rods	Structural weight	35.1
Base plate	Structural weight	77.11
Descent parachute bay	Structural weight	62.4
Bay lid	Structural weight	15.5
Outer shell	Structural weight	
Petri plate	Innovation idea	10
Primary Descent	Parachute	33
Secondary Descent	RAM-air Parachute	105
	Total	**897+shell**

Table 11: Mass budget

4.6.4 Procurement Risk Mitigation

Critical long-lead items such as Teensy 4.1, LR1121, GNSS modules and other important telemetry sensors are prioritized for early ordering to prevent delays to PCB design and integration testing. Dual-sourcing or set orders considered for common passive components such as capacitors, resistors, connectors, etc to avoid single-supplier bottlenecks. A contingency budget of ₹6,000 is reserved specifically to overcome price fluctuations or replacement needs identified during subsystem testing.

4.7 System and sub-system level Analysis and Test Results

This section presents the analytical basis for CAN-7USAT's design margins across each subsystem, paired with the test plan defined to verify those margins on hardware. As of CDR stage the analytical work is complete and hardware test execution is scheduled as per the project milestone chart. Subsystem level testing is done according to the requirement for CDR stage and the rest results which needs upgradation will be shown in FRR.

4.7.1 Mechanical Subsystem (Structural Analysis)

4.7.1.1 Base Plate

Software Used: **Ansys**

Material Selection: Aluminium Alloy

Boundary Conditions: Fixed support applied at rod mounting bosses

Acceleration application: 30G in longitudinal axis

Mesh Setup

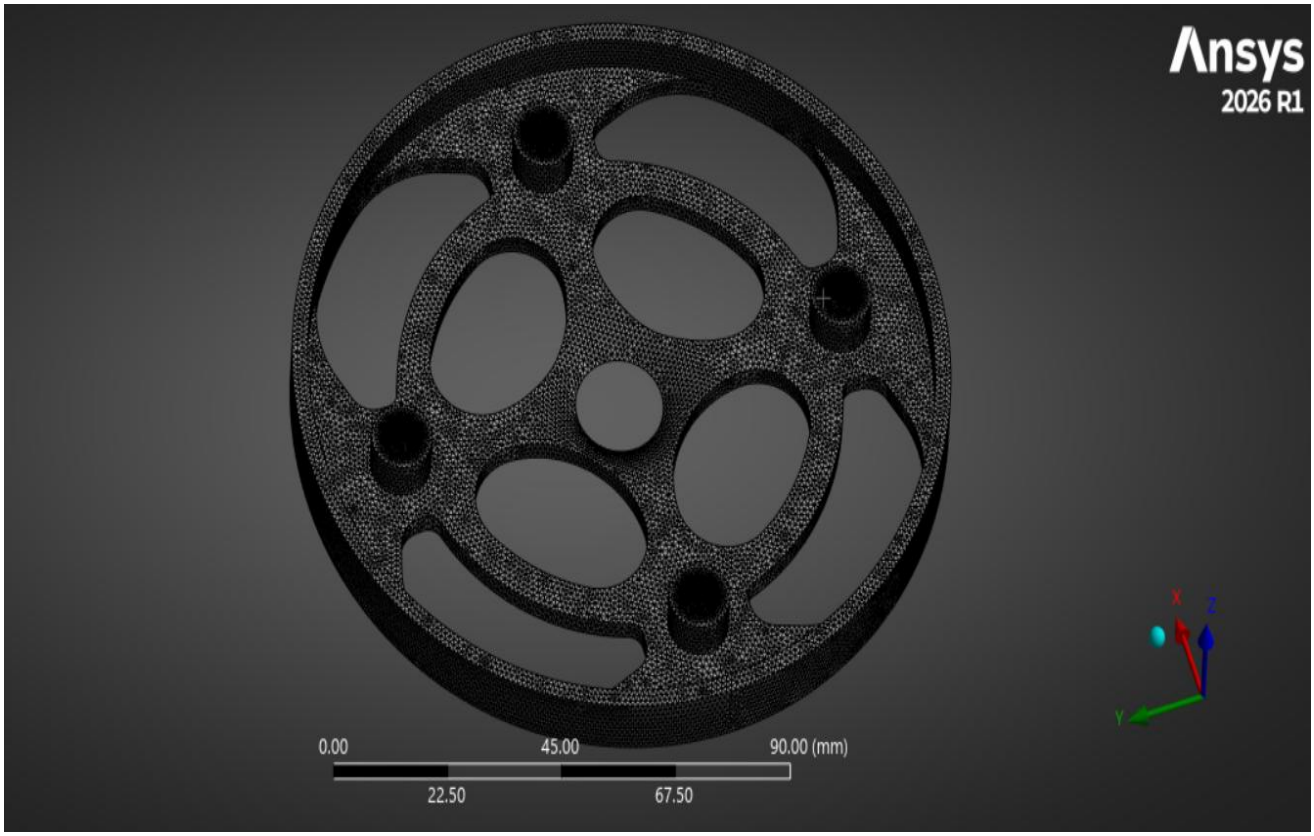


Fig 13: Meshing of Base Plate

Type: **Tetrahedral meshing**.

Refinement: Finer mesh applied to critical areas, including the holes and thrust pad.

Parameters Simulated: Von Mises Stress, Total Deformation

Results:

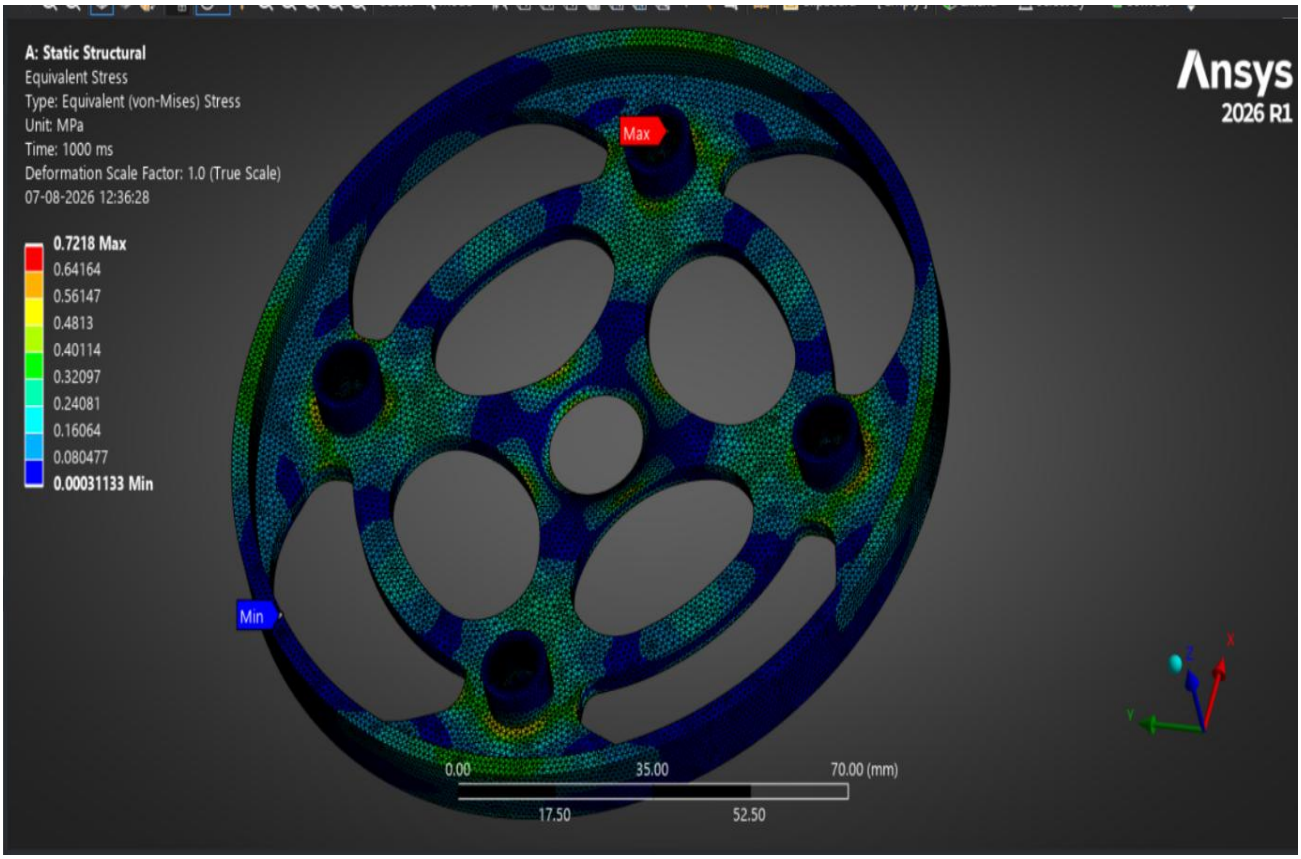


Fig 14: Stress Analysis of Base Plate

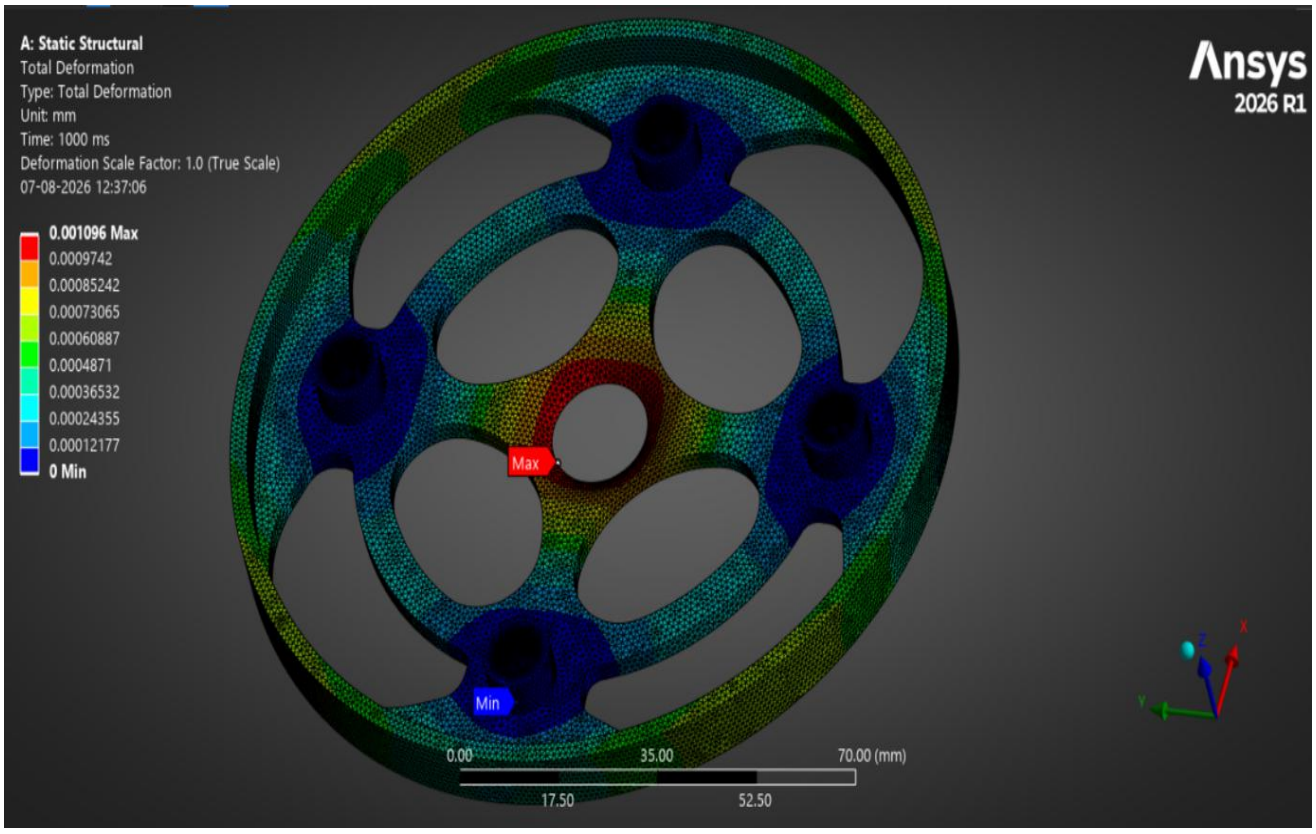


Fig 15: Total Deformation Analysis of Base Plate

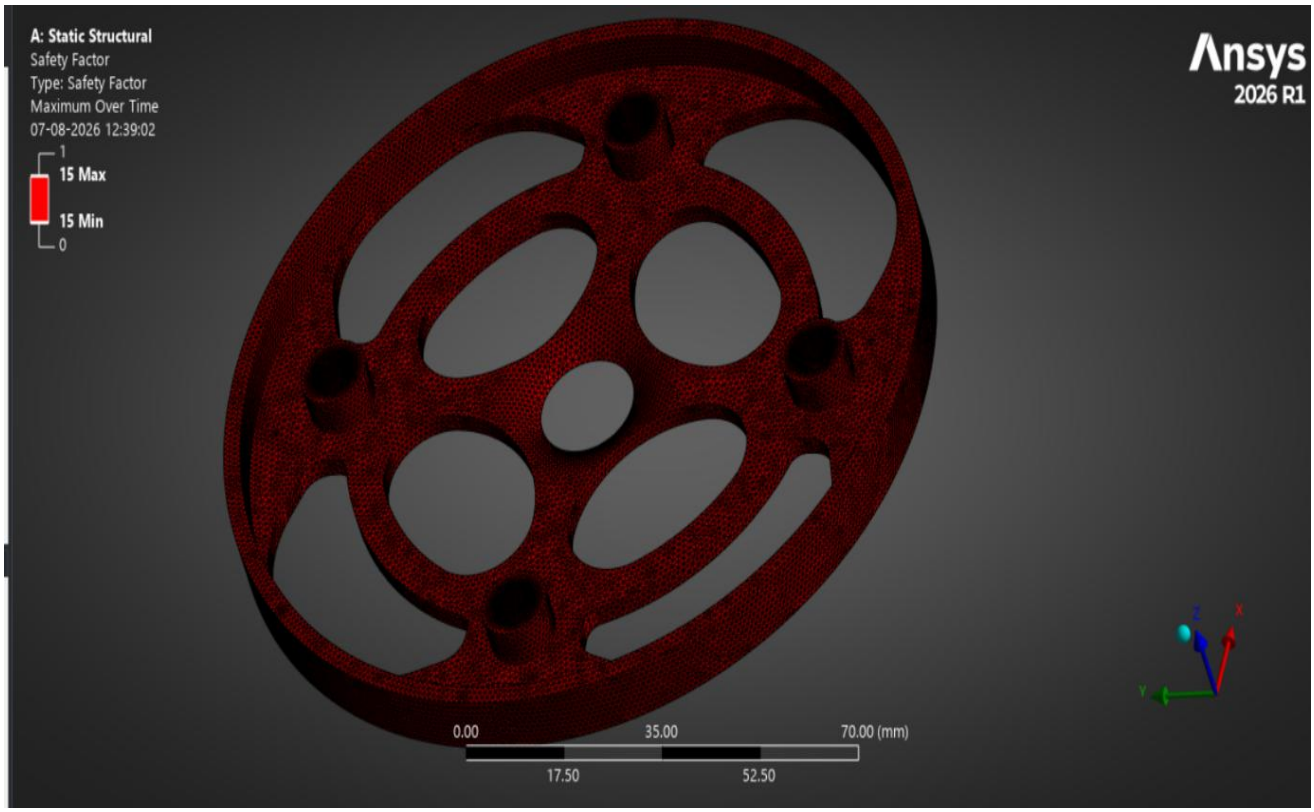


Fig 16 : FOS Analysis of Base Plate

Conclusion

After simulating, the results prove that the structure of the base plate is sound at the applied loads. The stress falls under the acceptable range of selected material's strength and the deformation value is close to zero. Thus, the component can be considered safe for all the required operations.

Further mass was reduced by removing non critical material while preserving the load path. Topology optimization was used to identify these regions. The optimized design achieves a lower structural mass while satisfying all structural requirements.

4.7.1.2 Middle Plate

Software Used: **Ansys**

Material Selection: Aluminium Alloy

Boundary Conditions: Fixed support applied to the inner faces of rod mounting bosses

Acceleration application: 30G in longitudinal axis

Mesh Setup

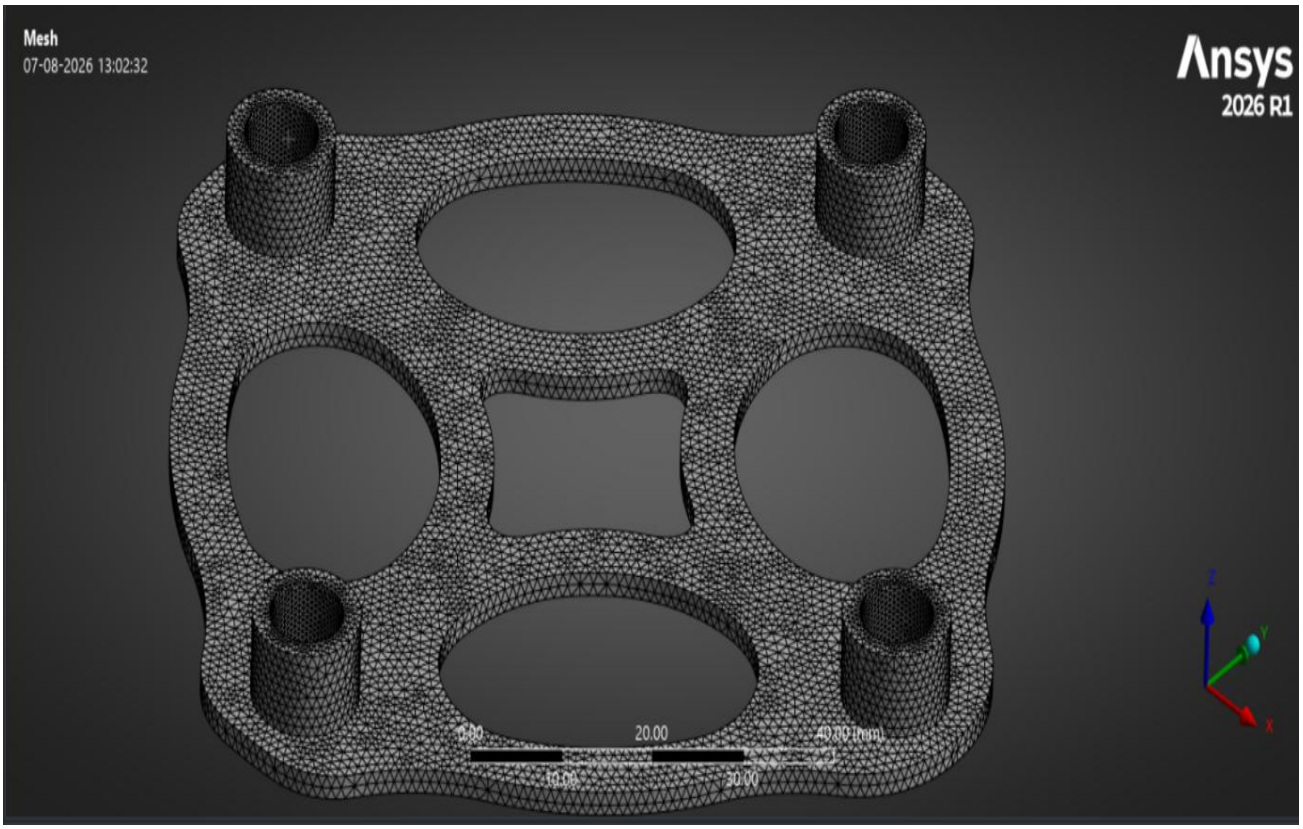


Fig 17: Meshing of Base Plate

Type: **Tetrahedral meshing.**

Refinement: Finer mesh applied to critical areas, including the holes and thrust pad.

Parameters Simulated: Von Mises Stress, Total Deformation

Results

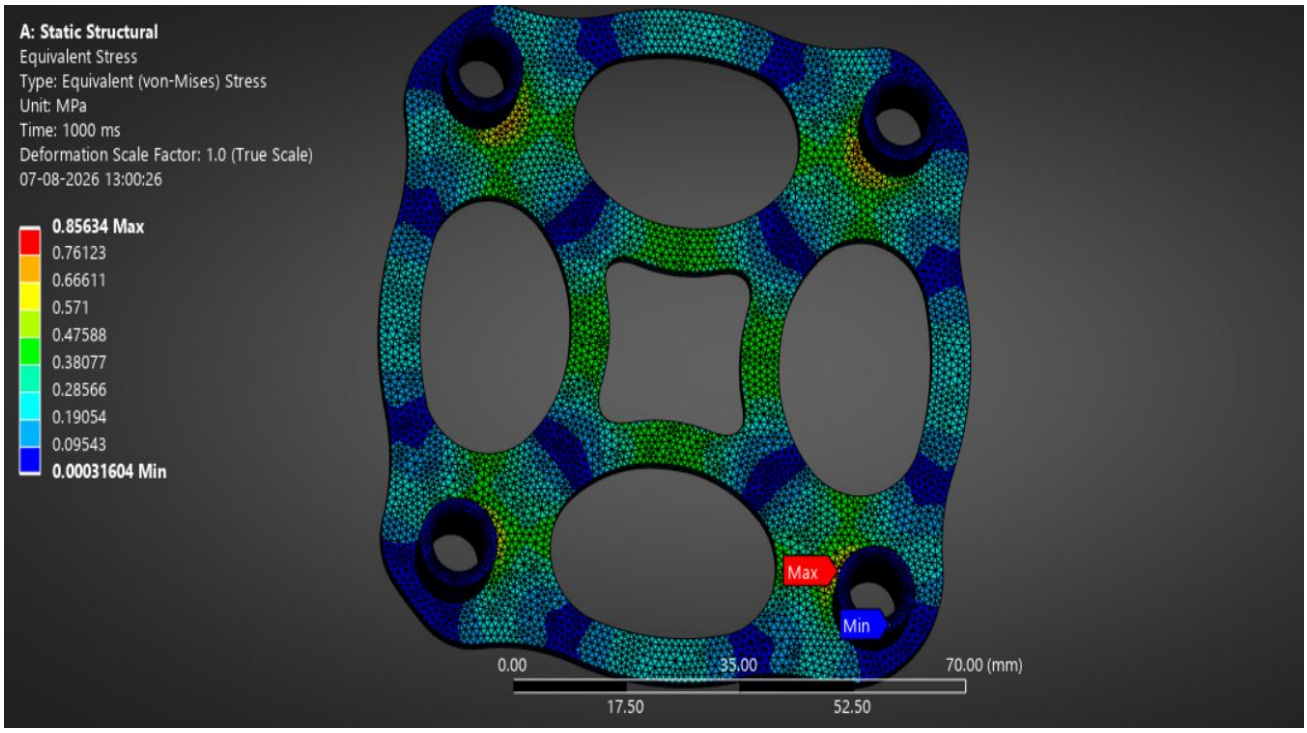


Fig18: Stress Analysis of Middle Plate

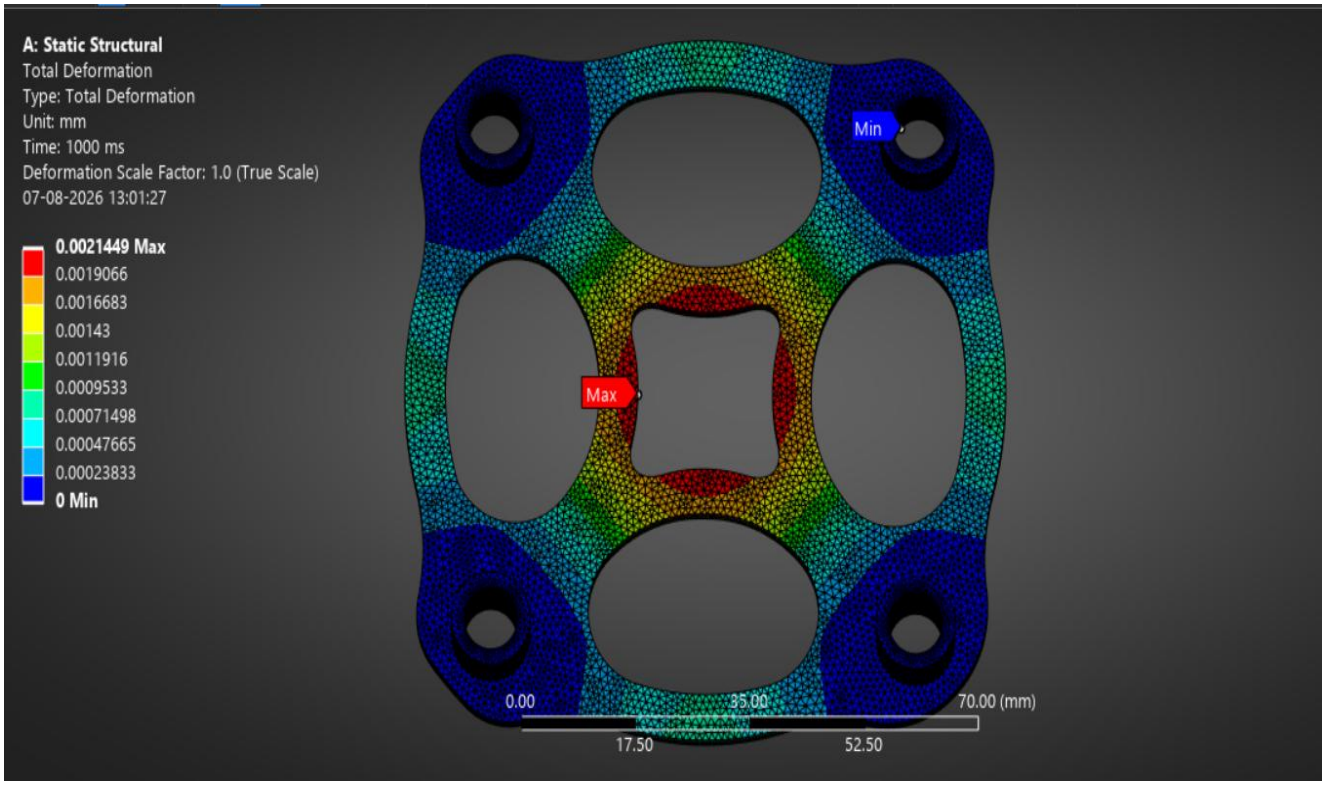


Fig 19: Total Deformation Analysis of Middle Plate

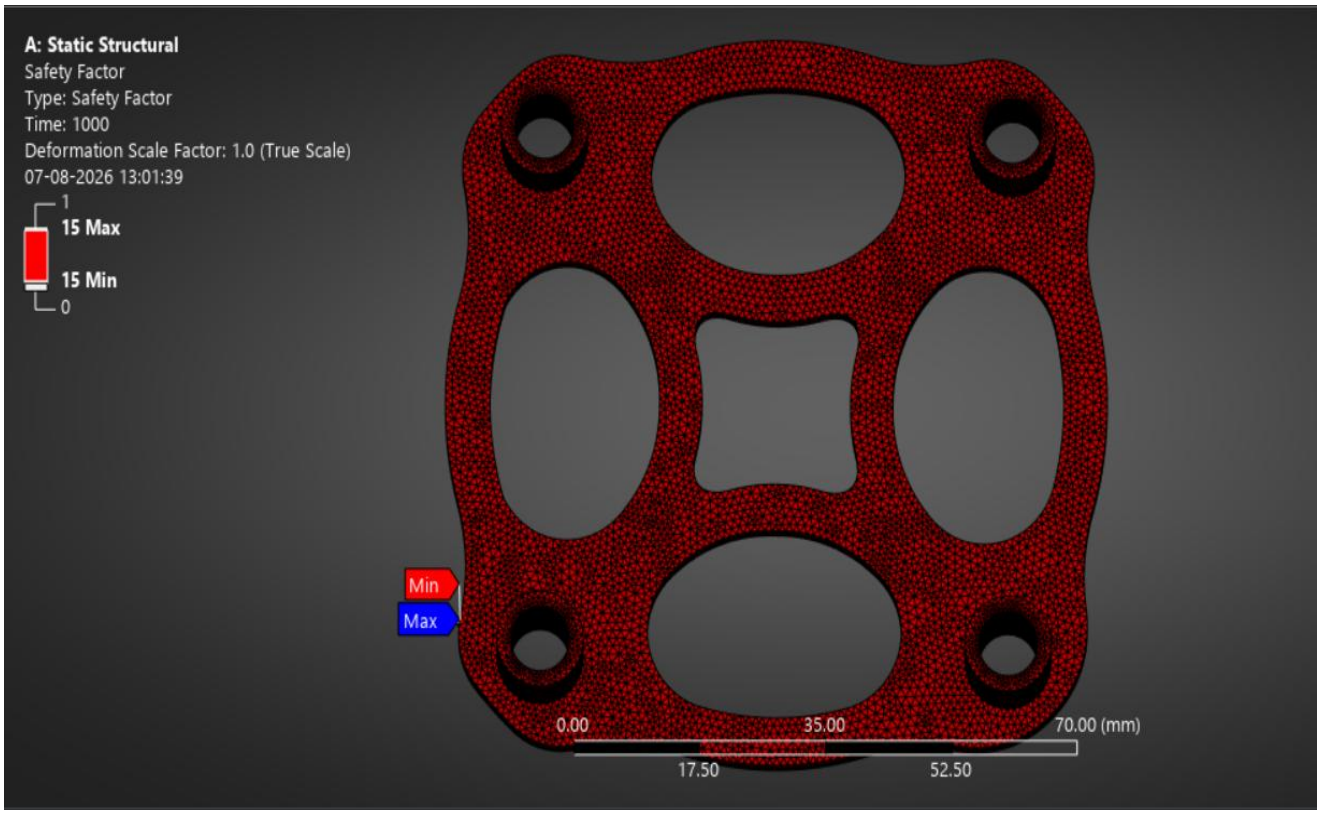


Fig 20: FOS Analysis of Middle Plate

Conclusion

The results prove that the plate has enough structural stability to withstand applied loads. The level of stress induced is far below the maximum acceptable strength of the material and the deformation in the structure is miniscule.

Topology optimization was done to remove materials from areas not critical for load paths. San enhanced strength to weight ratio is maintained without compromising on structural performance.

4.7.1.3 CANSAT Assembly

Software Used: **Ansys**

Material Selection: Aluminium Alloy for base and middle plates

- Fiberglass for rods
- FR 4 for PCB plates

Boundary Conditions: Fixed support applied at rod mounting bosses at the base plate.

Acceleration application: 30G in longitudinal axis

Component mass representation: The masses of integrated electronic components and other onboard hardware were added as distributed masses in respective places to accurately simulate the real loading conditions.

Mesh Setup

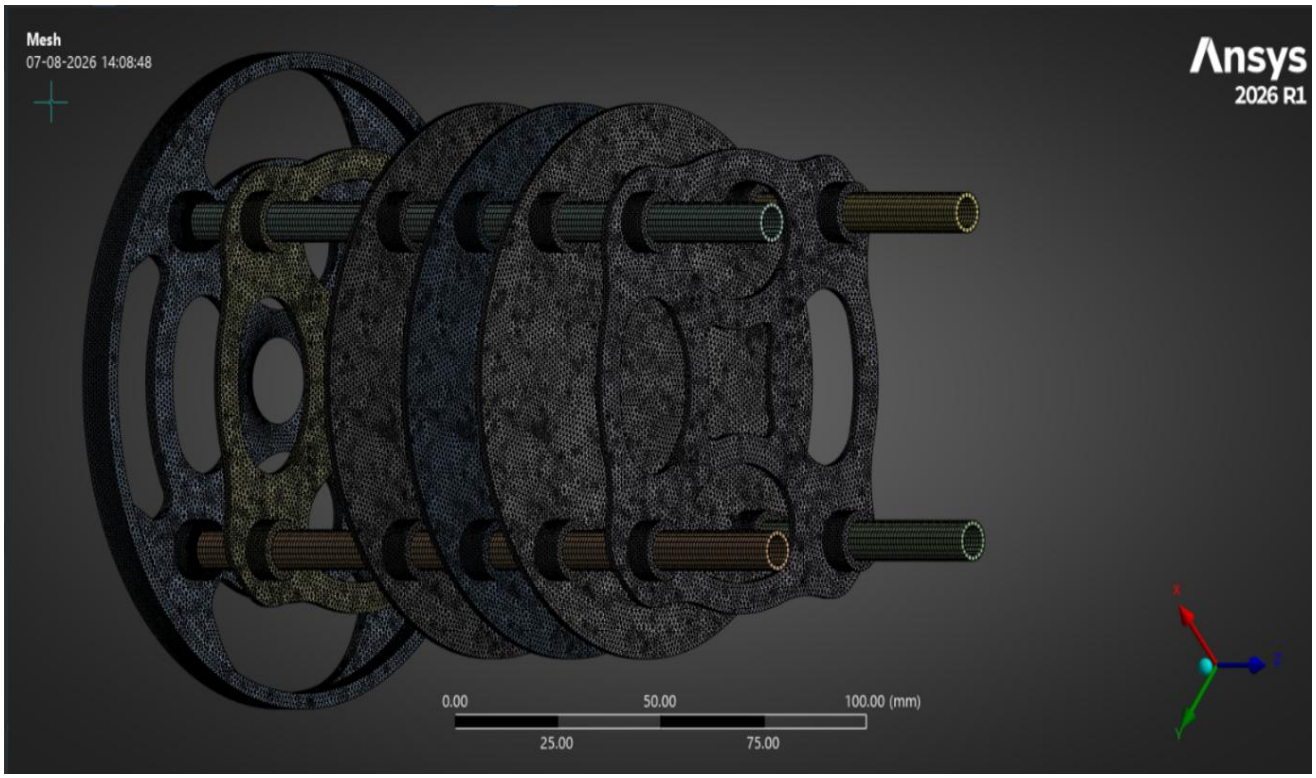


Fig 21: Meshing of CANSAT Assembly

Type: **Tetrahedral meshing.**

Refinement: Finer mesh applied to critical areas, including the holes and thrust pad.

Parameters Simulated: Von Mises Stress, Total Deformation, Buckling analysis

Results

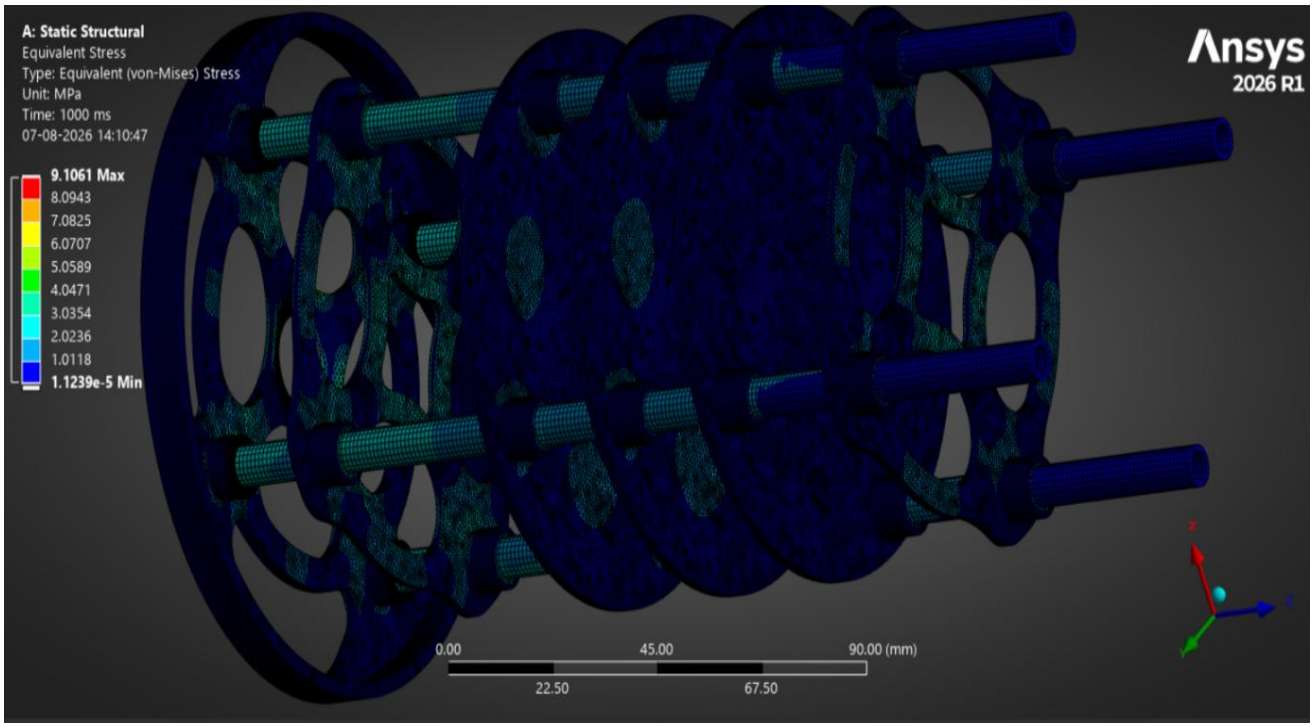


Fig 22: Stress Analysis of CANSAT Assembly

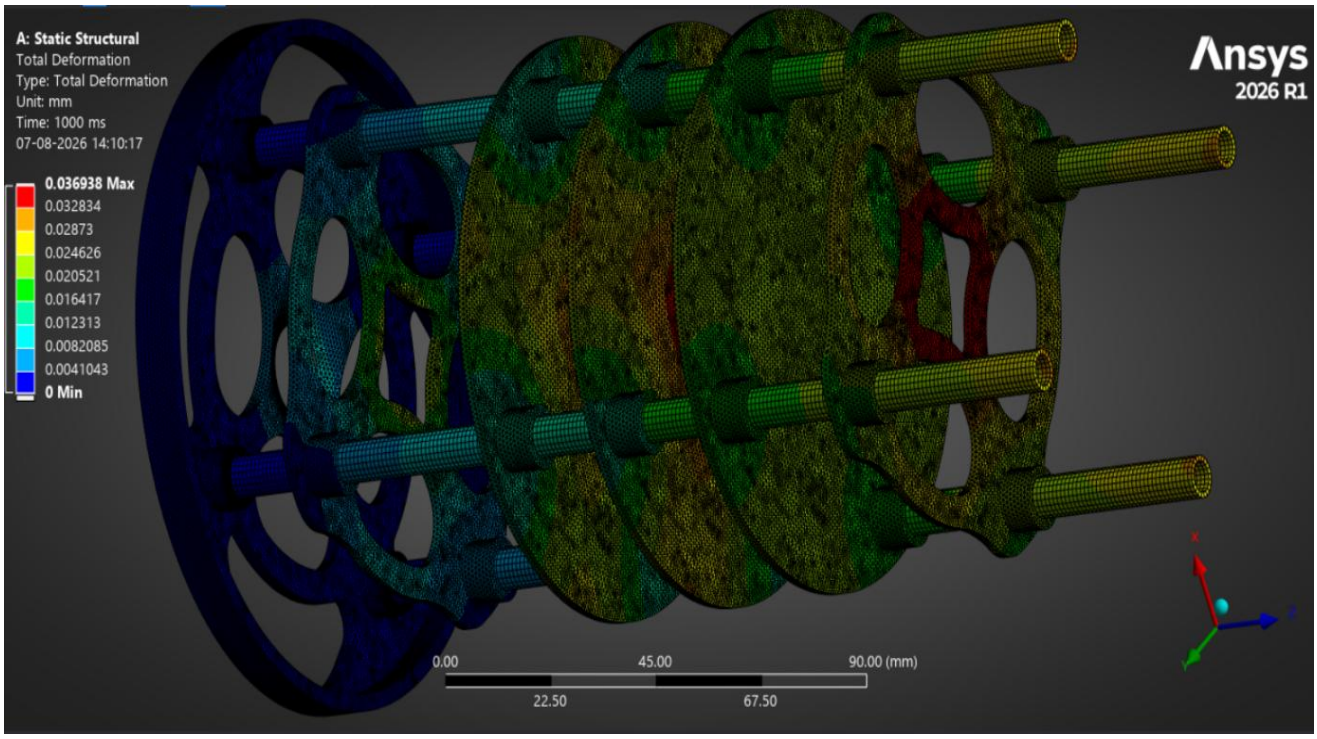


Fig 23: Total Deformation Analysis of CANSAT Assembly

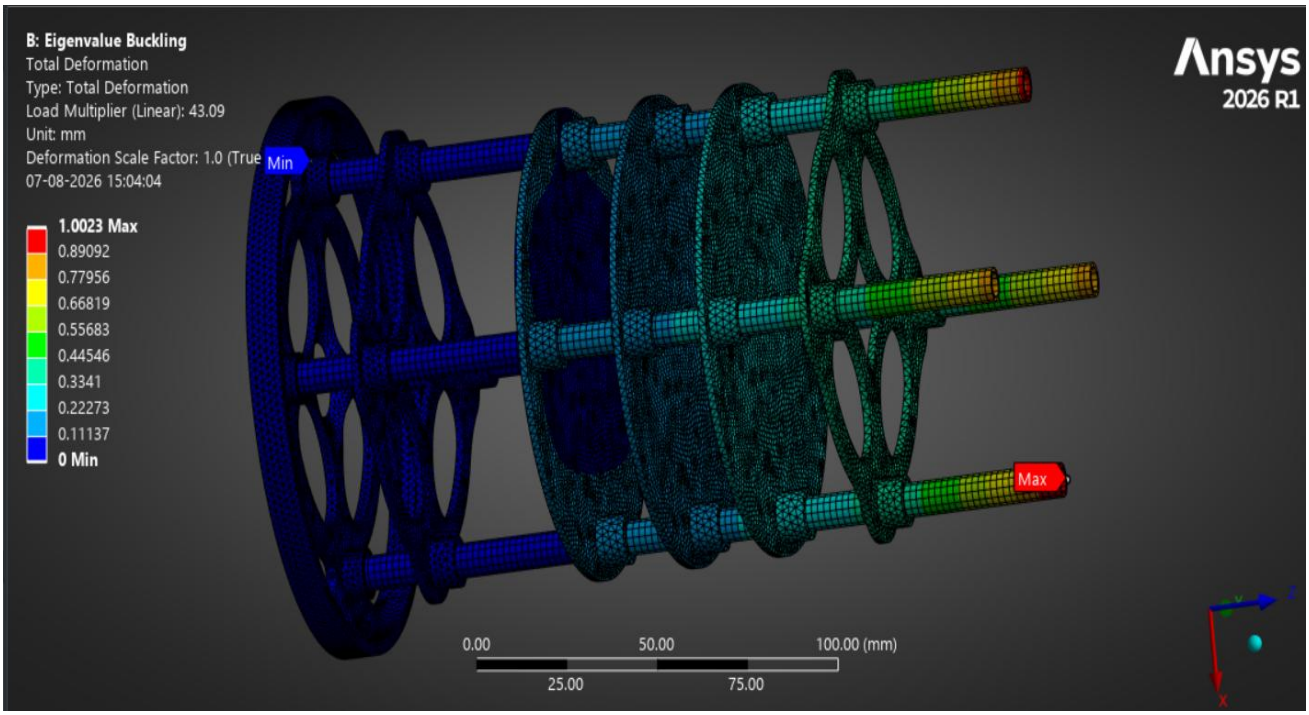


Fig 24: Buckling analysis of CANSAT Assembly

Conclusion

A full-scale structural analysis was performed under simulated launch environment conditions to determine the structural behaviour of the assembly. It is found there is good stress distribution and negligible deformation.

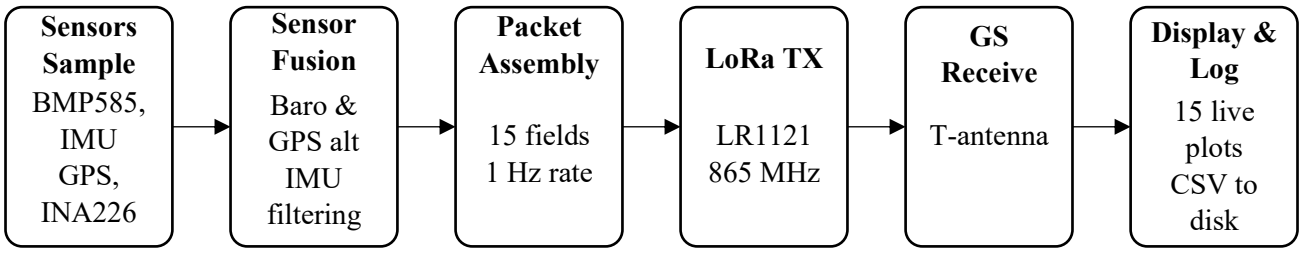
Buckling analysis was also done to ensure the assembly is structurally sound, since the critical buckling load is far more than anticipated loads the structure would face. This shows that there is sufficient margin to avoid instability during launch.

Final verdict

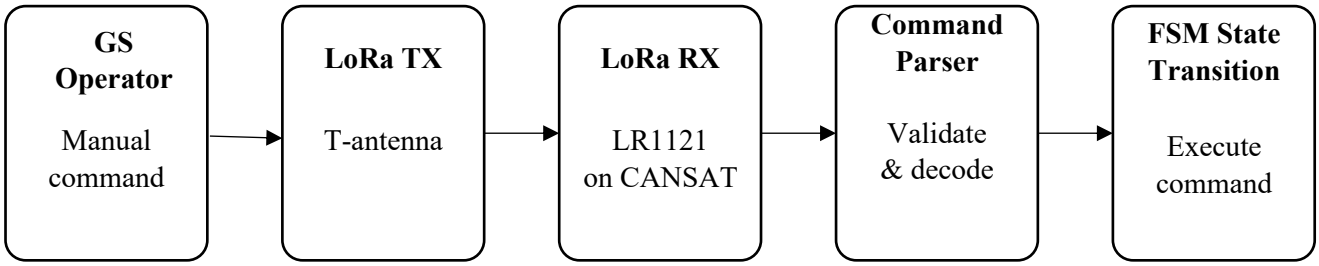
Overall, the results of structural analysis have shown that the CANSAT structure complies with all necessary requirements of strength, rigidity and stability under given loads. The optimised structure is validated for fabrication, integration and mission deployment.

4.7.2 Command & Data Handling (CDH) (Analysis & Test)

Downlink data path (CANSAT to ground station)



Uplink command path (ground station to CANSAT)



Data Handling subsystem

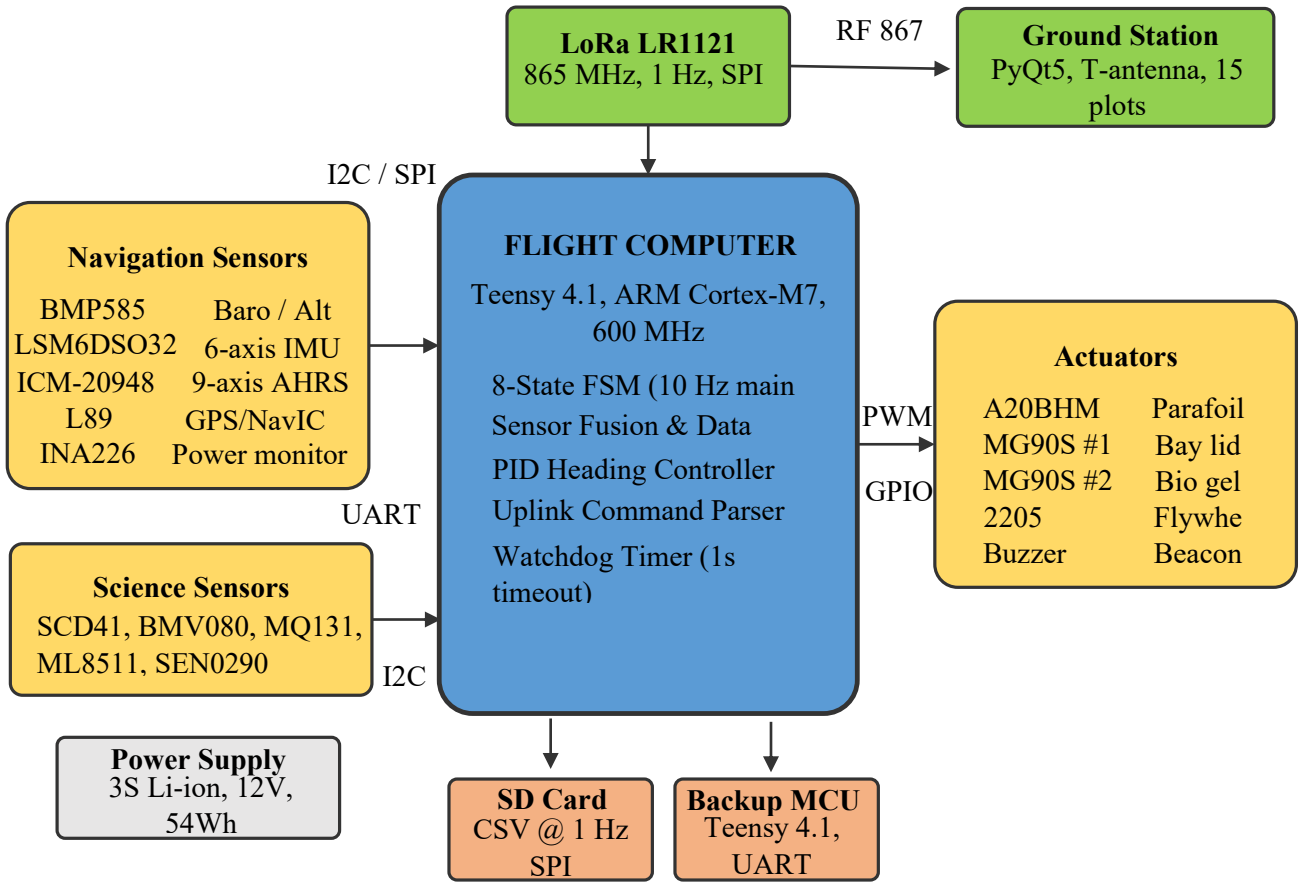


Fig 25: Command and Data handling subsystem

4.7.3 Sensor Subsystem (Analysis & Test)

To test about the sensor subsystem, we have assembled all the important components which are required for transmitting the necessary telemetry data. All of it is assembled on a zero board to test the compatibility of the

sensors and also to test proper power supply. The diameter of the board is 11 cm which will in future be placed on a Printed circuit board as a plate in our CANSAT.

The image given below is the board consisting of all the sensors on a single plate. The backup system designed will be have all its sensors on other such board. From this we have tried to test the working of sensors which is shown below in the following sections.

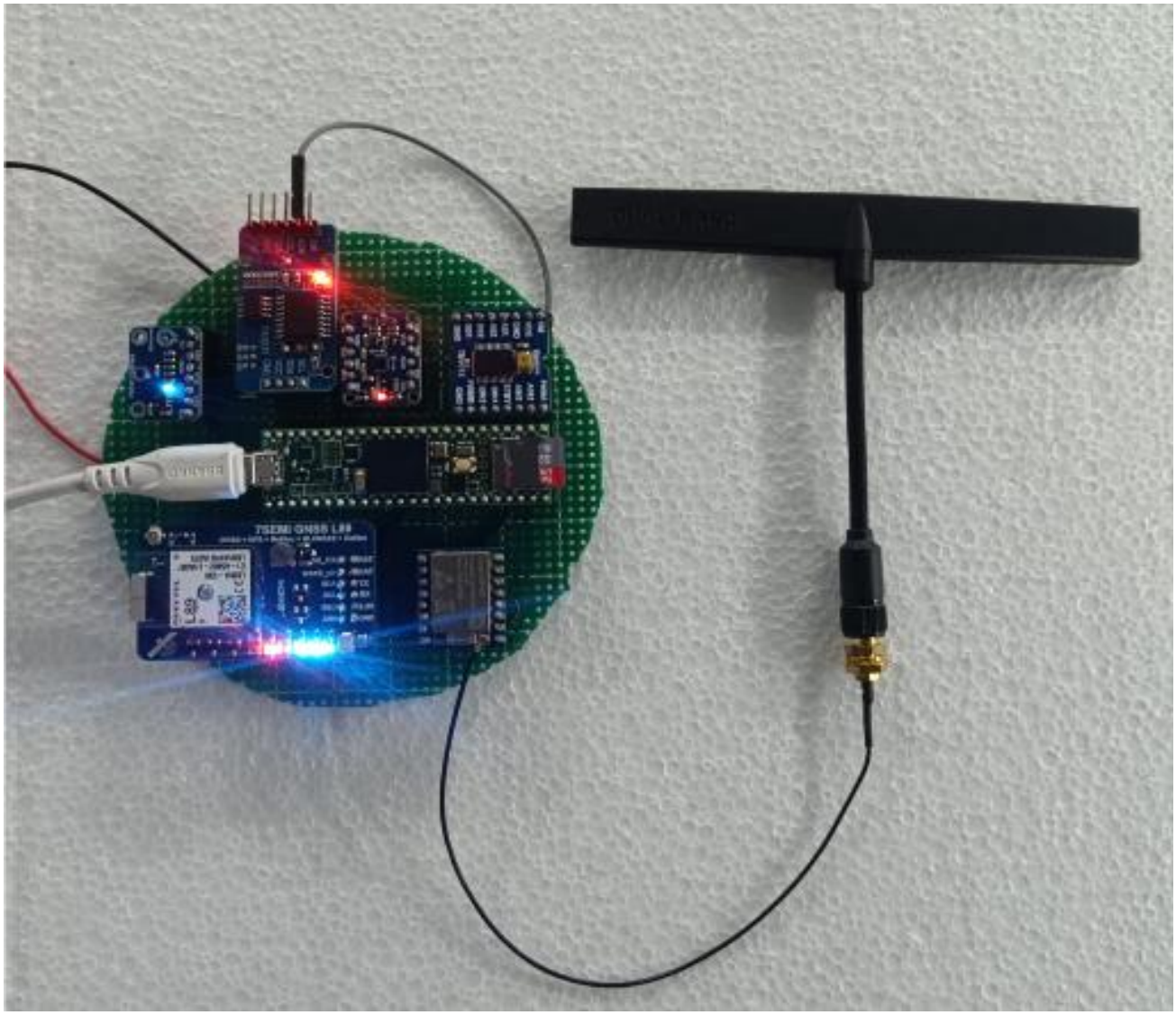


Fig 26: Sensor Board

The flight-state machine (FSM) was tested on Teensy 4.1 hardware using a synthetic telemetry injection test in which a pre-scripted flight profile such as height and vertical rate pairs was given to the state machine in place of live sensor data to work all state transitions perfectly. The given profile simulated a full mission status and provided with the results as given in the image below. The FSM correctly detected:

- Launch on provided height
- Triggered primary parachute deployment at apogee (1000.0 m peak)

- Fired the secondary parafoil at 599.3 m AGL
- Correctly registered landing

This confirms the altitude/state-decision logic (Gates 1–3 of the altitude control loop which is given below) works correctly across the full mission timeline independent of physical sensor noise, isolating the FSM logic itself as a verified component ahead of closed-loop hardware-in-the-loop testing.

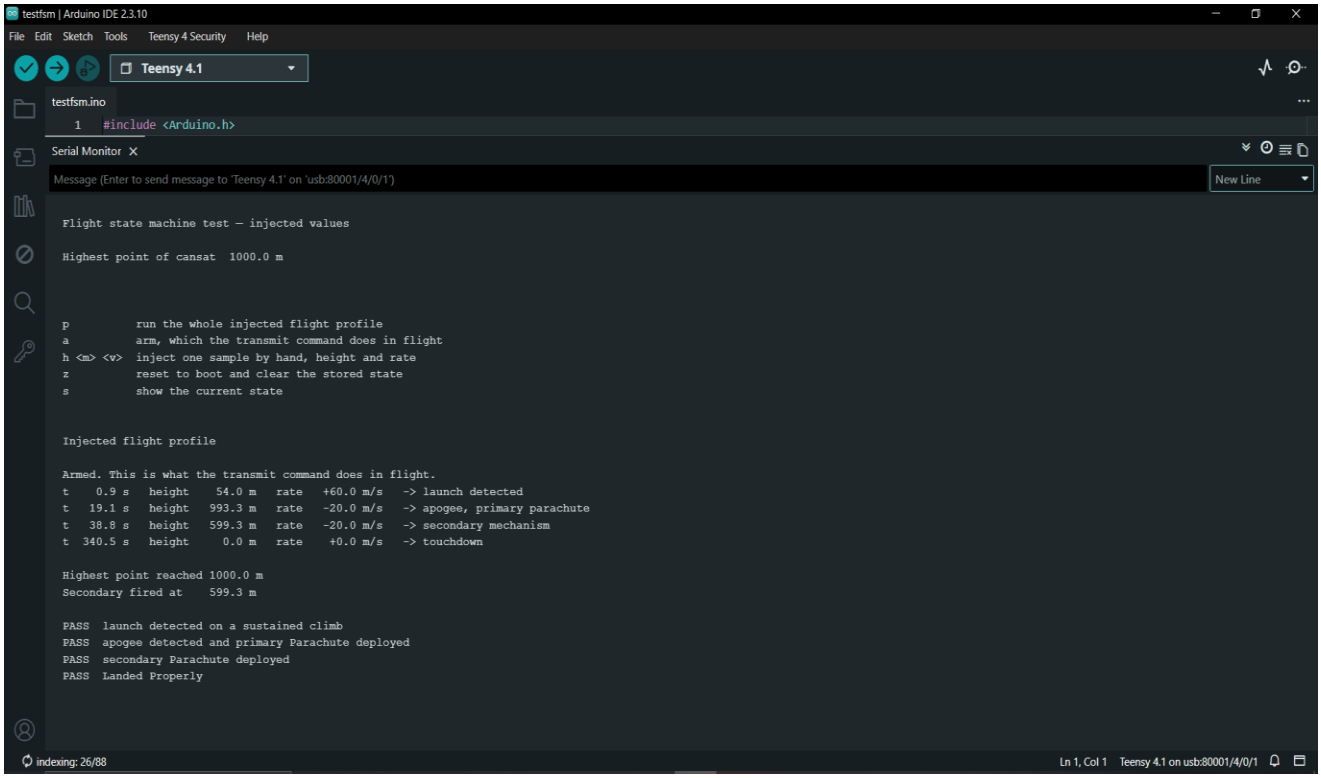


Fig 27: FSM test

The telemetry packet format was tested end-to-end by transmitting a live sample packet from the Teensy 4.1 through the ground-station serial link and confirming correct field-by-field decoding. The captured packet correctly explains into all 17 defined fields:

1. Team ID
2. Mission time (167 s)
3. Packet count (8)
4. Barometric altitude (−0.9 m, ground-level offset within calibration)
5. Pressure (97398 Pa)
6. Temperature (29.8°C)
7. Bus voltage (10.09 V)
8. GNSS time
9. Latitude/longitude (31.2510490°, 75.7052917°)
10. GPS altitude (229.9 m)

11. Satellite count (4 locked)
12. 3-axis acceleration
13. Mechanical gyro spin rate (0.4°/s)
14. Flight software state

This confirms the packet structure, field ordering, resolution/scaling, and CSV-style analysing all match the defined telemetry, and that GNSS locks 4 satellites and sensor data are being correctly calculated and transmitted over the downlink prior to flight.

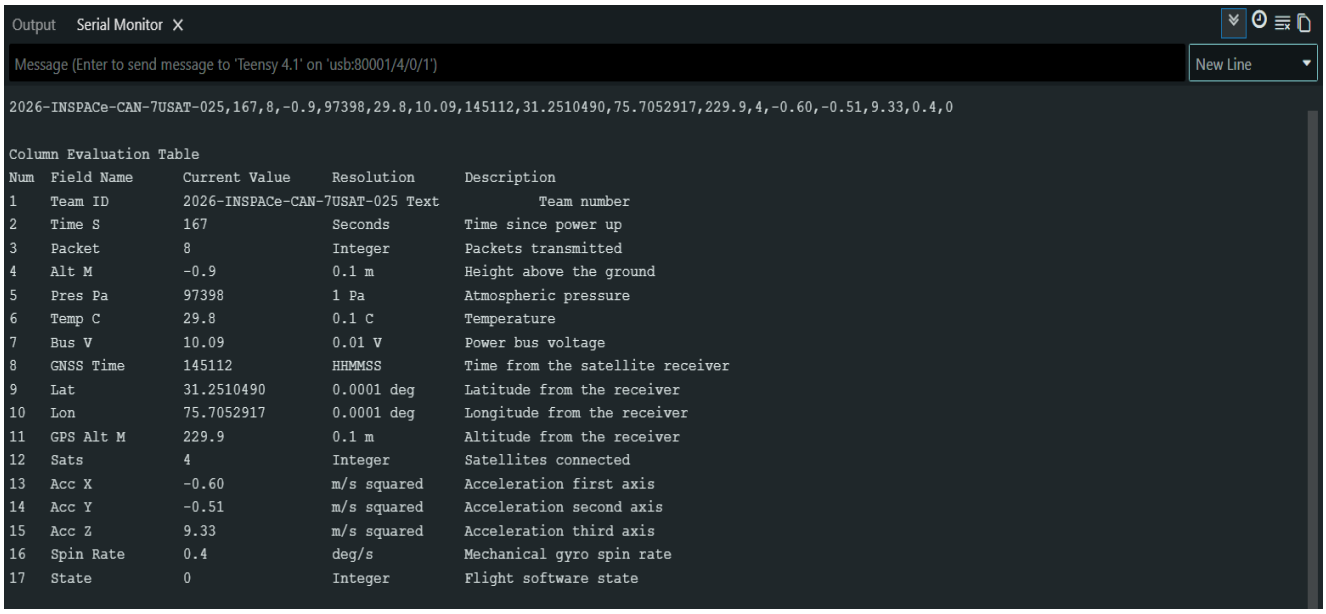


Fig 28: Data Output

Our packet length is 99 bytes. So, in this we limit data input by taking the mandatory 15 fields and the remaining fields which is optional data is actually divided in different packets so each time we get different optional sensor data and in that we use base encoding so that all data is actually compressed and reduced in size. And then it is transmitted using our LoRa module LR1121 of size 99 bytes which actually take around 971 milliseconds of a time to transmit. After that we pause for few milliseconds to listen for the command which are sent from our ground stations. In our ground station that specific command is transmitted three times. Suppose that we are transmitting our current packet then the ground station will try to send that command 3 time as soon as it receives the packet and we actually encode the specific packet field according to the standard encoding that encodes and properly compresses the information. Then that is transmitted in the base station and it decodes that packet and bring it to its original form. We can also check by that packet received is proper or not and if we have missed anything or corrupt. That millisecond gap can be also be utilized for sending and receiving acknowledgement signal.

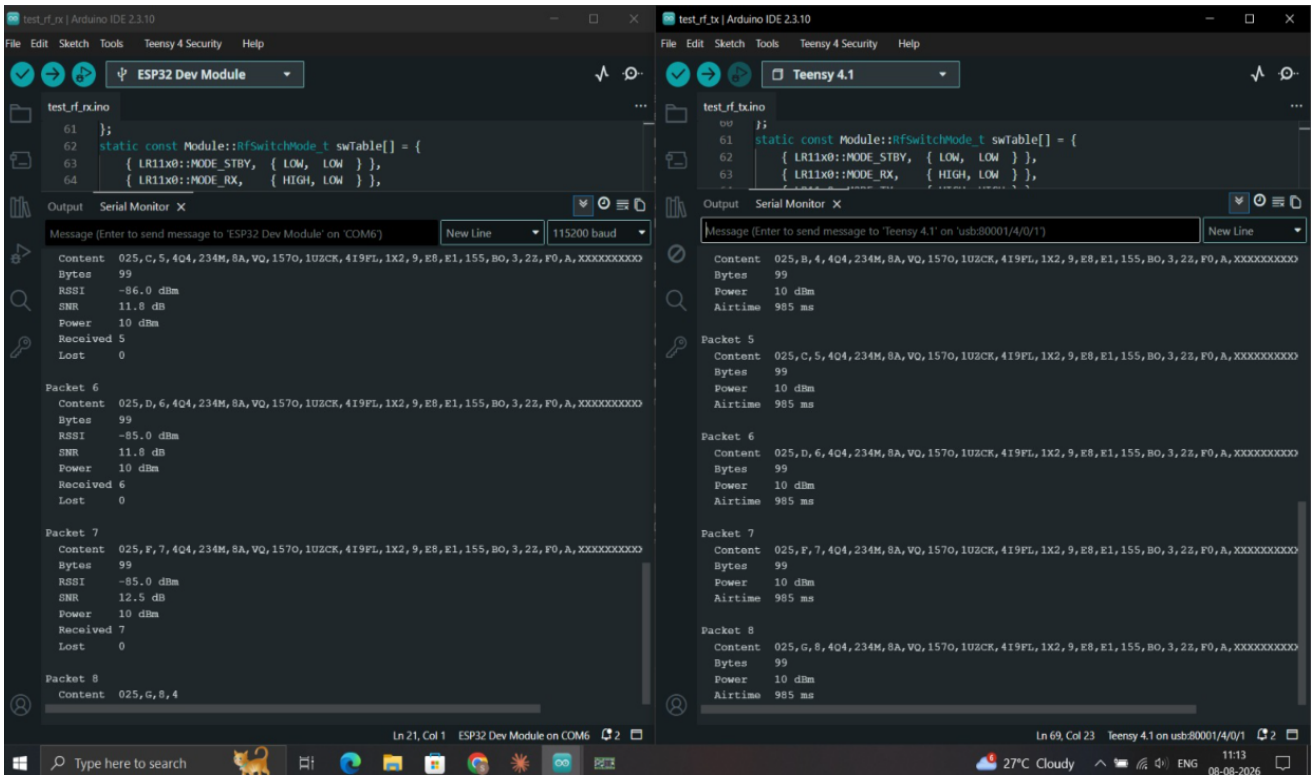


Fig 29: Packet Transmission result

After collecting all the data required, it is transmitted and even stored in MicroSD card for backup purpose so that the telemetry loss could not affect the sensor data readings for even a second. All the stored data in MicroSD is arranged in an Excel sheet in systematic order so that we can directly handover the Card to jury post flight period. The given image shows the data arrangement in a single Excel sheet:

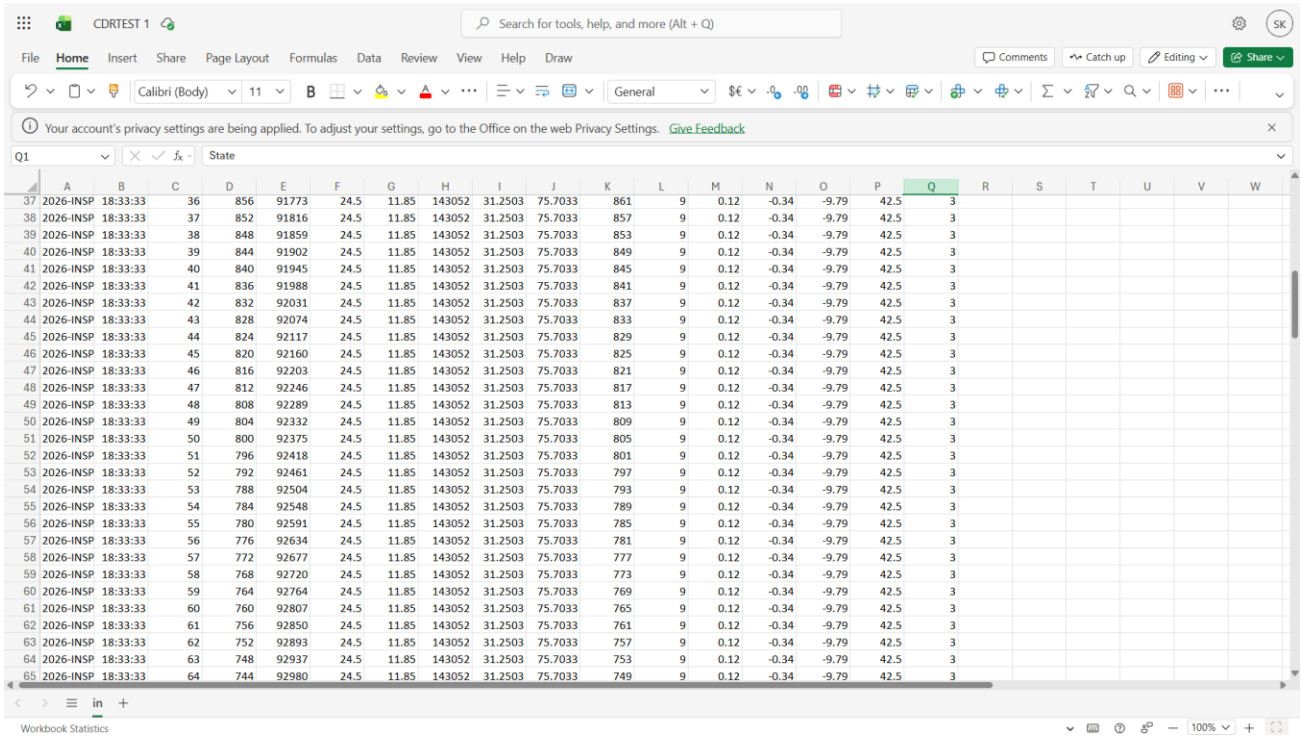


Fig 30: MicroSD Card storage Data

4.7.4 Communication Subsystem (Link Analysis & Test)

For testing communication subsystem, we are using 865 MHz Rubber duck omnidirectional antenna and a LoRa module having as LR1121. First, we made the circuit connections required for communication testing. A basic ground station platform was used to see the reading and telemetry data transfer. After finalising the circuit connections, we moved forward from breadboard and designed a PCB board for proper testing.

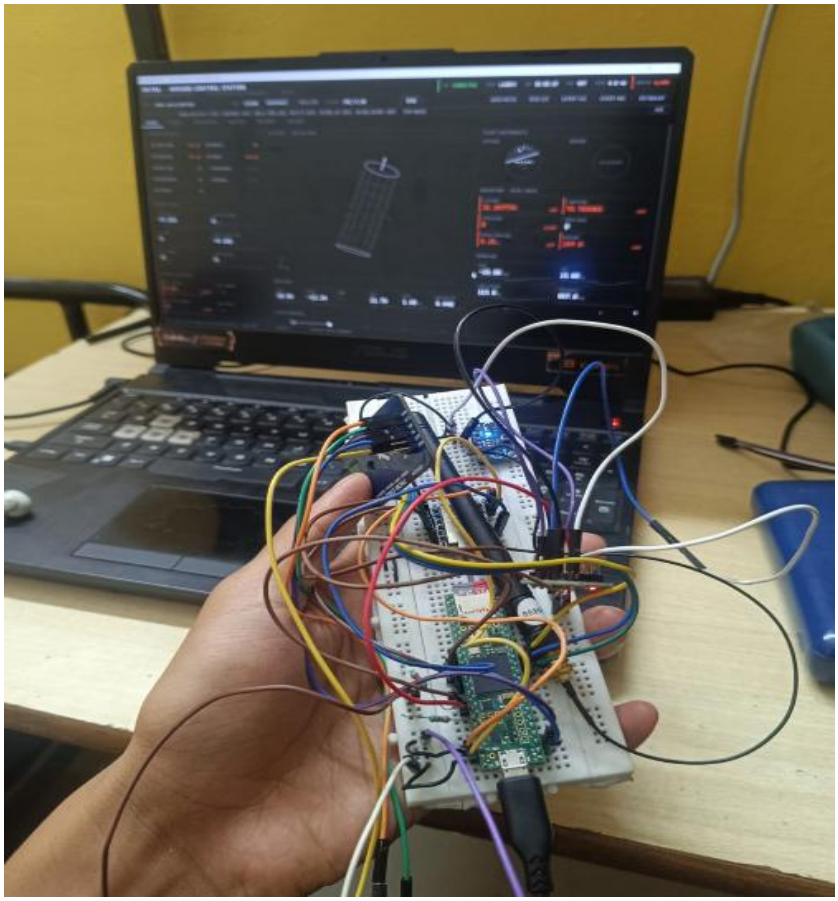
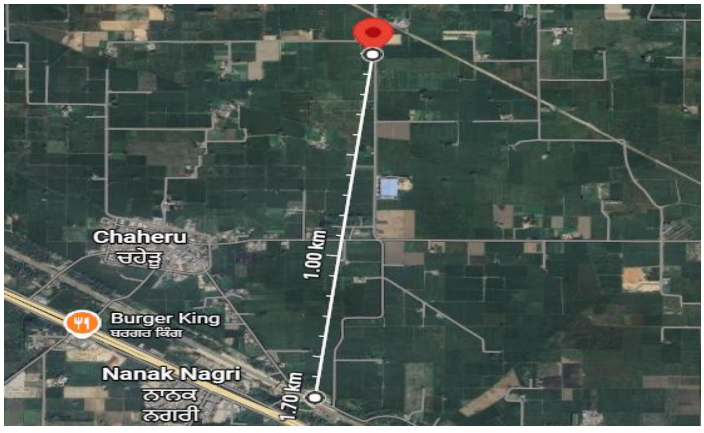


Fig 31: Communication Range Test

After testing the new selected antenna which is Dual band T-antenna from Radio master the results we got are:



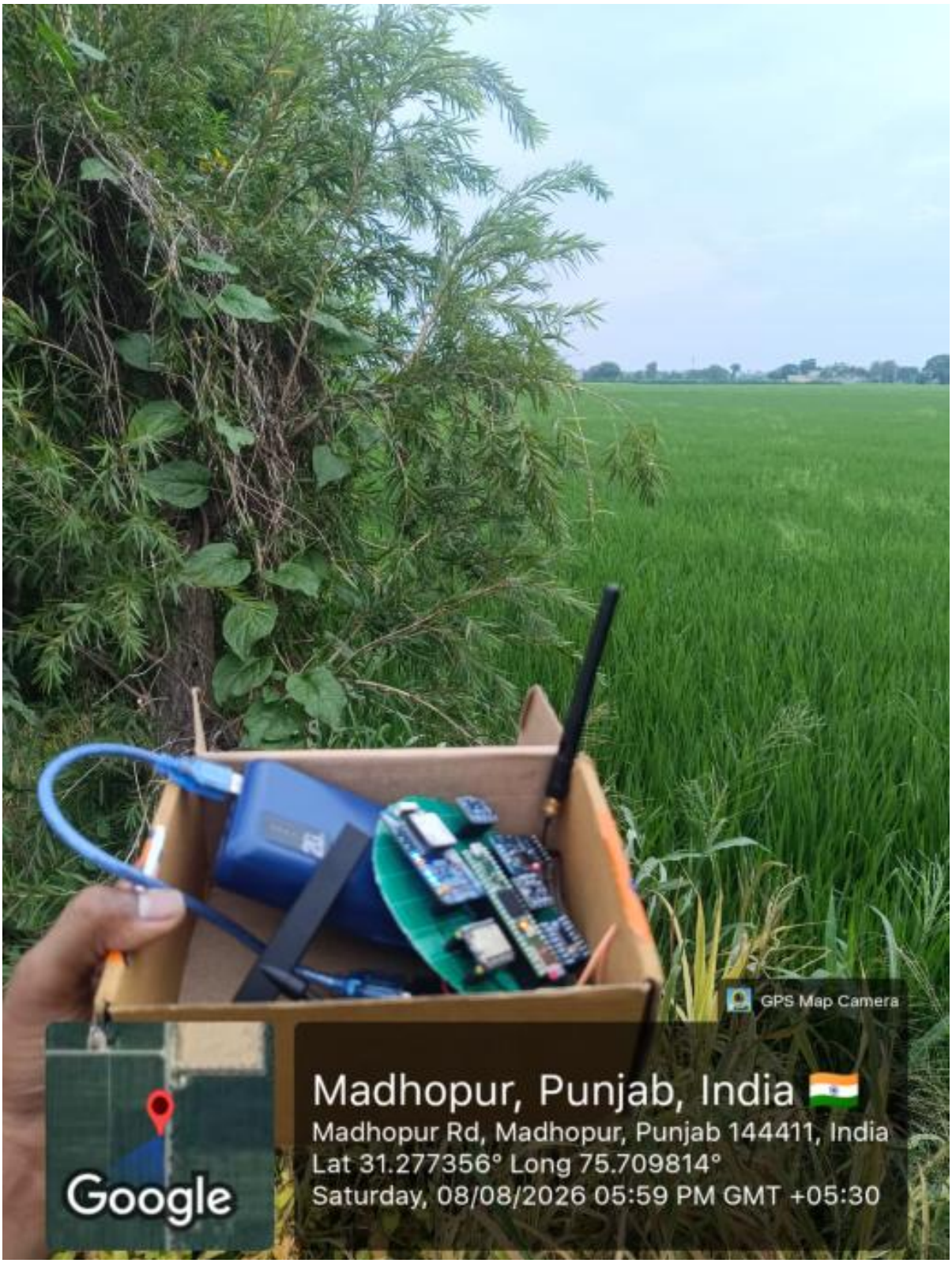


Fig 32: Telemetry Distance Test (Spot 1)



Fig 32: Telemetry Distance Test (Spot 2)

We tested our communication subsystem with both the antennas which are rubber duck 865 MHz antenna and Radio master dual band T antenna 867 MHz. According to the result we understood that it has high signal connectivity as we tested it with the distant of 1.6 KM. The received data packets were proper as shown in image given below. It was also tested on our Ground station software and the results are attached below in the upcoming sections.

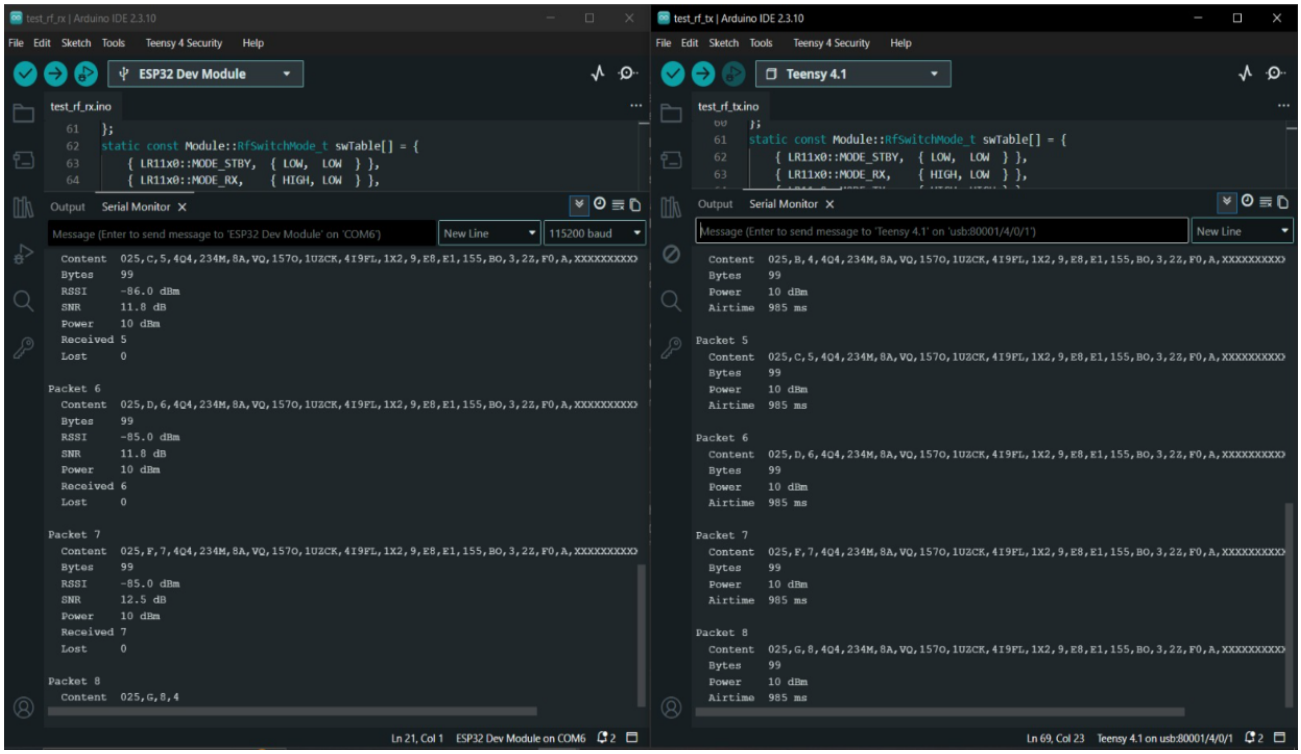


Fig 33: Telemetry Packet Received

4.7.5 Electrical Power Subsystem (Analysis & Test)

Component	Voltage	Current	Maximum Power	Duty Cycle (%)	Average Power	Source
Teensy 4.1	5	0.18	1.8	100	1.8	Existing
ESP32-S3	5	0.18	1.8	70	1.26	Existing
LR1121 LoRa (TX)	3.3	0.055	0.1815	20	0.0363	Existing
L89 GNSS	3.318	0.032	0.106176	100	0.106176	Existing
ICM-20948 IMU	3.322	0.0045	0.029898	100	0.029898	Existing
SCD41 CO2	5	0.018	0.09	100	0.09	Existing
AS3935 Lightning	3.3	0.0001	0.00033	100	0.00033	Datasheet

AS7343 Spectral	3.3	0.0x98	0.00825	100	0.00825	Datasheet
HDC3020 Temp/Humidity	3.3	0.0008	0.00264	100	0.00264	Datasheet
ML8511 UV	3.3	0.0003	0.00099	100	0.00099	Existing
MQ131 Ozone	5	0.15	0.75	50	0.375	Existing
BMV080 PM	3.3	0.018	0.0594	100	0.0594	Existing
TPL5010 Timer	2.5	0.000000035	8.75E-08	100	8.75E-08	Datasheet
W25Q64 Flash	3.3	0.025	0.0825	20	0.0165	Datasheet
INA226	3.3	0.0012	0.00396	100	0.00396	Existing
SN74LV4052A MUX	3.3	0.00008	0.000264	100	0.000264	Datasheet
DS3231 RTC	3.3	0.00011	0.000363	100	0.000363	Datasheet
Hobbywing XRotor 40A ESC	10.8	0.1	1.08	30	0.324	Datasheet
2205 2300KV BLDC Motor	10.8	4	43.2	30	12.96	Datasheet
AGFRC A20BHM Servo #1	7.4	1.5	11.1	20	2.22	Datasheet
AGFRC A20BHM Servo #2	7.4	1.5	11.1	20	2.22	Datasheet
Teensy 4.0 (DCS)	5	0.08	0.4	100	0.4	Existing
BMP585	3.3	0.001	0.0099	100	0.0099	
7.4V BEC	7.4	0.02	0.296	100	0.296	Existing
7Semi AP62301 Buck 3.3V 2A	5	0.02	0.2	100	0.2	Existing
7Semi AP64500 Buck 5V 5A	10.8	0.03	0.648	100	0.648	Estimate
				Total Maximum Power	72.9501709	
				Total Average Power	23.06797109	
Battery Capacity (mAh)	5000					

Battery Voltage (V)	10.8					
Battery Energy (Wh)	54					
Estimated Endurance (Hours)	2.341					

Table 12: Power budget

4.7.6 Descent & Recovery Subsystem (Analysis & Test)

During our parachute testing, we first started with understanding the idea behind RAM air parachute which controls the direction with the help of servos. So, we calculated about the dimensions of the parachute given above.

This parachute testing was done from a heighted building of around 120 feet. According to the test results, the RAM air parachute has a proper shape and nodes attachment. It carried 840 g of dead weight and the parachute. It had proper opening without any entanglement with in the codes.

Improvement done after testing: Length of the cord is reduced according to the calculations.



Fig 34: Ram-Air Parachute Drop Test

Electronics part for controlled RAM air parachute was tested through a simulated panel. The panel shows about the rotation of servos with respect to the direction of flow of air in virtual environment. The image given below shows about the test:

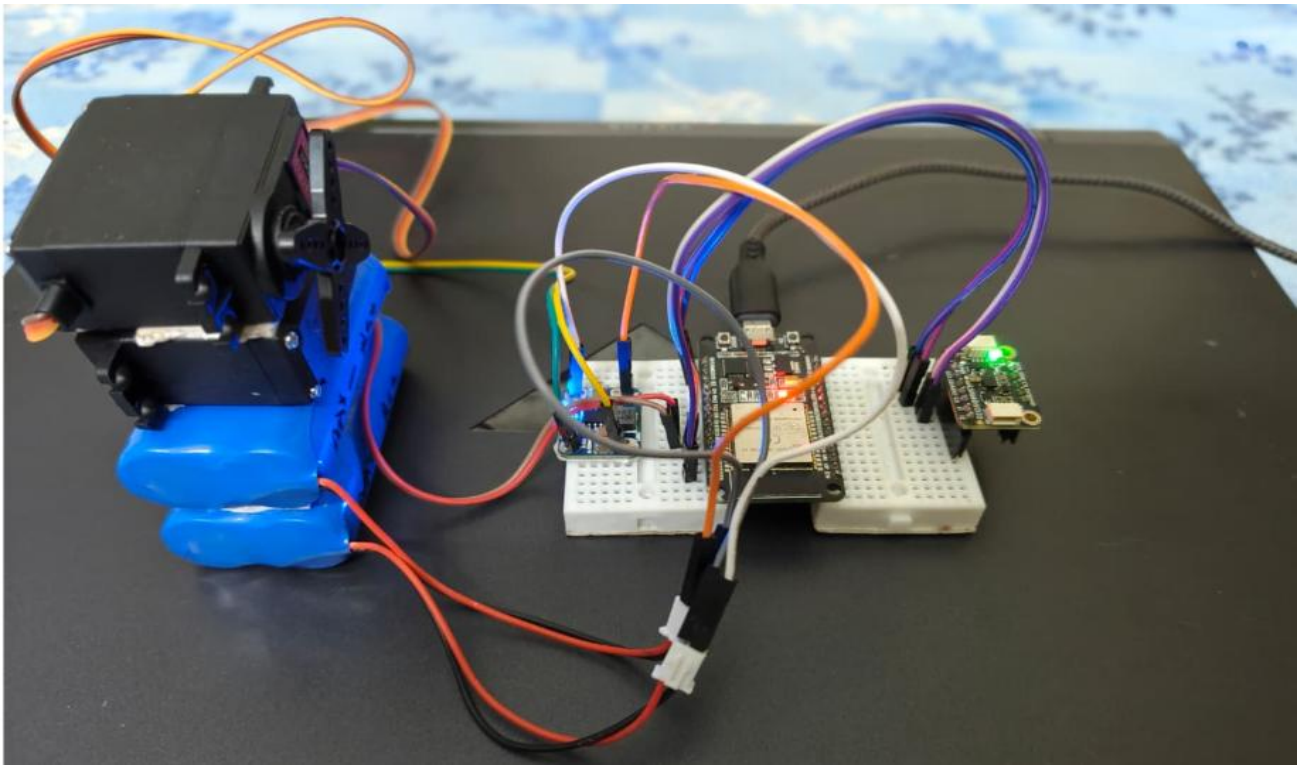


Fig 35: Secondary Mechanism



Fig 36: Secondary Mechanism Control

4.8 Flight Readiness Parameters

Flight readiness for CANSAT is assessed against a defined set of quantitative parameters spanning structural integrity, power, communication, descent performance, and software state. Each parameter carries a pass/fail threshold that must be verified during pre-launch checkout before the CANSAT is cleared for integration into the rocket payload bay. The important Go readiness parameters required are:

- All mandatory telemetry fields streaming and logging correctly at GS and onboard SD card.
- Bus voltage, structural fit, and mass should be within specified limits.
- GNSS fix acquired and target waypoint recorded.
- Primary and secondary descent systems are packed and mechanically verified.
- FSM is confirmed in TEST_MODE with successful calibration response.

4.8.1 Structural & Mechanical Readiness

PARAMETER	READINESS THRESHOLD	VERIFICATION METHOD
Total mass	≤ 1000 g (±50 g)	Pre-integration weigh-in
Envelope fit	≤ 150 mm diameter × 400 mm height	Fit-check in mock rocket fairing
Structural integrity	No visible damage, cracks, or loose standoffs	Visual inspection post-vibration test
Gyro-wheel spin-up	5000 ±500 RPM within 30 s	Hall-sensor RPM verification
Shell/lid closure	Secure latch, no exposed electronics	Physical inspection

Table 13: Structural & Mechanical Readiness

4.8.2 Electrical Power Readiness

PARAMETER	READINESS THRESHOLD	VERIFICATION METHOD
Bus voltage at power-on	≥ 8.0 V	INA226 reading on GS dashboard
Battery endurance margin	≥ 2 hours remaining at current charge	Pre-flight SOC estimate
Power indicator LED	Illuminates on power-on	Visual check
Voltage rail stability	5.0 V and 3.3 V rails within ±5%	Multimeter/INA226 cross-check

Table 14: Electric Power Readiness

4.8.3 Communication and telemetry Readiness

PARAMETER	READINESS THRESHOLD	VERIFICATION METHOD
LR1121 link status	Packets received at GS, RSSI > −120 dBm	GS dashboard link indicator
Telemetry rate	≥ 1 Hz sustained	GS packet-rate counter

All 15 mandatory TM fields	Present, correctly formatted	GS live field display
CMD_TX_ON / CMD_CALIBRATE	Uplink command acknowledged	GS command panel confirmation
GNSS fix	≥ 6 satellites locked	GS GNSS status panel
SD card logging	CSV row count incrementing at 1 Hz	GS/onboard log cross-check

Table 15: Communication and telemetry Readiness

4.8.4 Flight Software & State Readiness

PARAMETER	READINESS THRESHOLD	VERIFICATION METHOD
FSM state	TEST_MODE confirmed before CMD_TX_ON	GS state field
Sensor self-test	All sensors initialized without fault	Boot-sequence telemetry
Calibration	Baro altitude, gyro, accel zeroed at pad level	Post-CMD_CALIBRATE readback
Backup avionics	Standby, heartbeat signal nominal	Failover monitor status
Target waypoint	Launch pad GPS coordinates set	GS waypoint confirmation

Table 16: Flight Software & State Readiness

4.8.5 Descent & Recovery Readiness

PARAMETER	READINESS THRESHOLD	VERIFICATION METHOD
Primary parachute packing	Z-fold secured	Visual/physical inspection
Parafoil packing	Deployment bag pinned, no line tangle	Visual/physical inspection
Brake-line servo range	Full deflection without stall	Bench actuation check
Audio beacon	≥ 92 dB SPL on trigger	Pre-flight beacon test

Table 17: Descent & Recovery Readiness

4.9 Pre-Flight Analysis

Pre-Fight Analysis talks about the expected performance envelope of CANSAT prior to launch by using analytical models and simulation to predict system behaviour under different conditions.

4.9.1 Descent Trajectory & Wind Drift Analysis

A MATLAB-based simulation was run against expected launch-site wind conditions to understand about the parafoil's guided descent trajectory. The simulation output shows a top-down ground track and altitude-vs-time profile which is used to:

- Predict the parafoil's glide path and expected landing dispersion radius under the PID waypoint-steering logic.

- Verify that the glide ratio (2.50) and forward speed (6.37 m/s) provide sufficient cross-range authority to pay for given wind drift between 600 m activation and descent.
- Get an expected recovery search radius around the target waypoint for the ground recovery test.

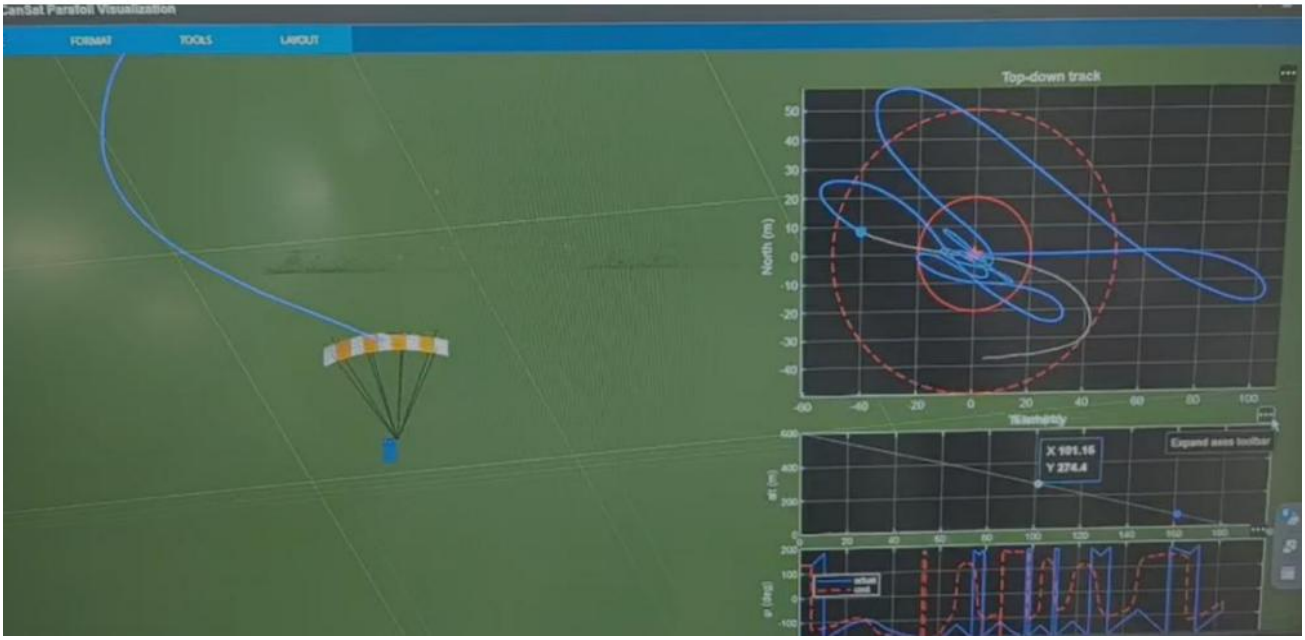


Fig 37: MATLAB parachute testing

4.9.2 Communication Link Prediction

- Calculated link margin of **71.6 dB** at 1 km (867 MHz, SF10) and is well above the 10 dB minimum needed for reliable LoRa telemetry.
- No packet loss expected under nominal line-of-sight conditions between CanSat and ground station.
- Sufficient headroom to tolerate antenna misalignment, multipath, and orientation changes near touchdown.
- NETID/PANID isolation (0x98) and SF10 orthogonality expected to prevent cross-interference with other teams' radios at the shared launch site.

4.9.3 Sensor Calibration Strategy

- Ground calibration (CMD_CALIBRATE) zeroes barometric altitude, gyroscope offset, and accelerometer bias at pad level, immediately before flight.
- Eliminates day-to-day atmospheric pressure drift from affecting the 600 m secondary-deployment trigger.
- Removes IMU bias accumulated during transport and handling, ensuring gyro-wheel stabilization and apogee detection operate against a true zero reference.
- Calibration is deliberately timed as late as possible in the pre-launch sequence to minimize the window in which drift could re-accumulate before flight.

5. SENSOR SYSTEMS SUMMARY, CHARACTERIZATION AND TEST RESULTS

CANSAT carries 11 sensors in total in which each sensors have its own flight-critical measurements such as altitude, orientation, GPS, power and some bonus sensors such as air quality, lightning, UV, ozone and CO_2, which can help team VRITRA with better scoring points. Each sensor was selected through a comparative study against its alternatives and each one carries a defined characterization method and test plan which is in detail below.

5.1 Pressure & Temperature Sensor (BMP585)

The BMP585 is a barometric pressure and temperature sensor from 7 Semi, which acts as the primary altitude measurement sensor for CANSAT. It was selected over the BMP390 and BMP581 for its better pressure accuracy and ultra-low power consumption which are important for proper performance, drift-free altitude detection throughout the descent. Onboard CANSAT will be having two units of BMP585 in which one will be for primary telemetry and other for the backup subsystem if required.



PARAMETER	VALUE
Model	BMP585
Sensor Type	Barometric Pressure & Temperature
Interface	I²C, SPI
Pressure Range	300–1250 hPa
Accuracy	±0.5 Pa
Supply Voltage	1.71–3.6 V
Power Consumption	2.5 µA
Quantity	2

Table 18: Pressure & Temperature Sensor (BMP585)

The BMP585 barometer was tested in a two-phase validation such as a ground-level reference position and then followed by a controlled height to verify AGL tracking. The results which can after testing are visible in the image given below.

This confirms the barometric altitude pipeline is correctly zeroed at the launch pad and responds accurately to small real-world height changes, validating it ahead of integration with the combination of BMP585, GPS module and Kalman filter altitude estimate used in the flight-state machine.

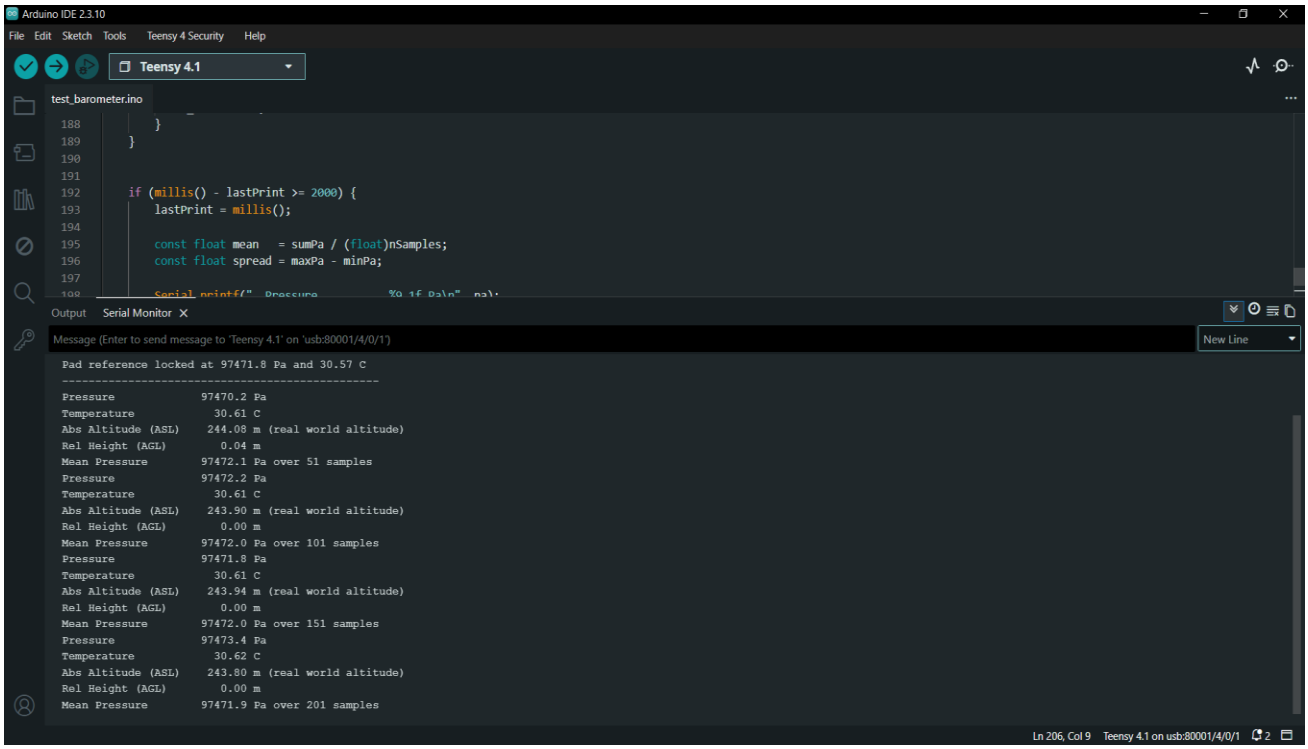


Fig 38: Ground level BMP testing

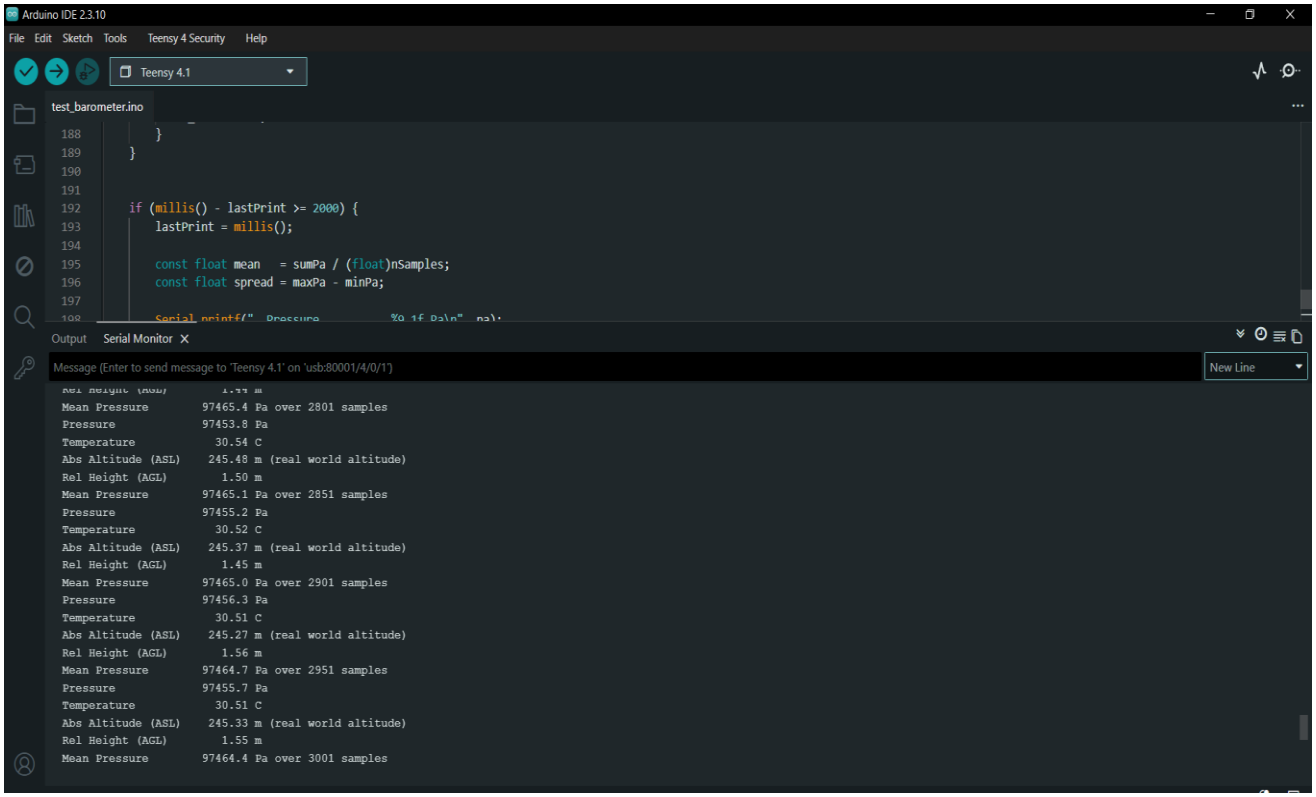
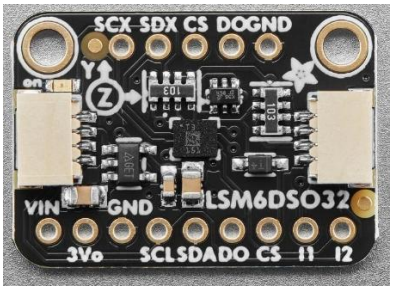


Fig 39: BMP testing from a specific height

5.2 6-Axis IMU (LSM6DSO32)

The LSM6DSO32 is the sensor manufactured by Adafruit which is used for inertial measurement which provides 3-axis acceleration and 3-axis angular rate data. It is used for controlling secondary descent mechanism of GPS guided RAM air parafoil. It was chosen over the MPU6050 and BMI270 for its wide measurement range, ensuring the sensor stays within range and does not break during the 30G ejection shock event.



PARAMETER	VALUE
Model	LSM6DSO32
Sensor Type	6-DOF IMU (Accelerometer + Gyroscope)
Interface	I²C, SPI
Accelerometer Range	±4/8/16/32 g
Gyroscope Range	±125/250/500/1000/2000/4000 dps
Supply Voltage	1.71–3.6 V
Quantity	1

Table 19: 6-Axis IMU (LSM6DSO32)

5.3 9-Axis IMU (ICM-20948)

The ICM-20948 from 7 Semi, provides orientation and motion sensing on the avionics board which is about combining accelerometer, gyroscope, and magnetometer in a single module. It was selected over the MPU9250 and BNO055 for its lower cost at 9-DOF capability. Onboard CANSAT will be having two units of BMP585 in which one will be for primary telemetry and other for the backup subsystem if required.



PARAMETER	VALUE
Model	ICM-20948 (7Semi)
Sensor Type	9-DOF IMU (Accel + Gyro + Magnetometer)
Interface	I²C, SPI
Accelerometer Range	±2/4/8/16 g
Gyroscope Range	±250/500/1000/2000 dps
Magnetometer	Yes (3-axis)
Quantity	2

Table 20: 9-Axis IMU (ICM-20948)

The IMU which is the combination of accelerometer and orientation, was tested across four static orientations such as home position (0°), and 90° tilts left, right, and front, with 5 continuous samples logged per orientation to characterize both mean reading and sample-to-sample noise.

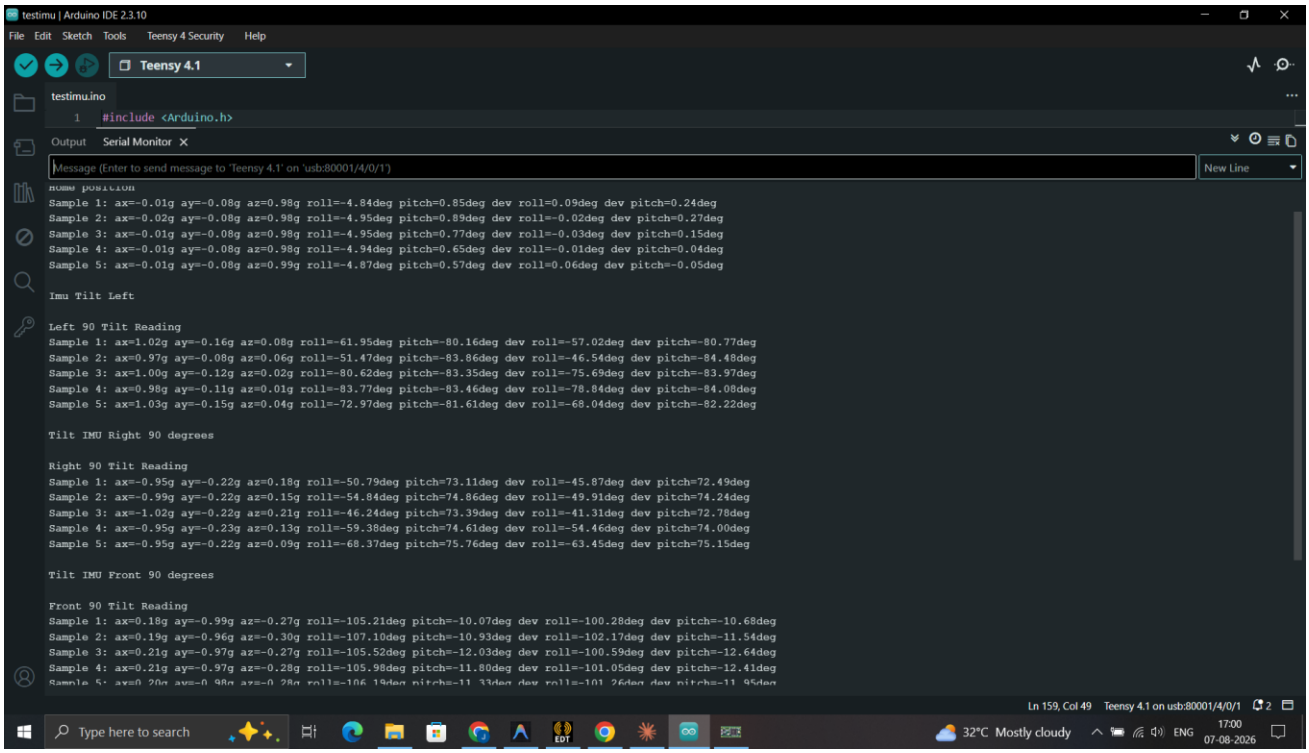


Fig 40: IMU Testing

5.4 GNSS Module (L89HA)

The L89HA GNSS from 7 Semi is a module which provides real-time position, altitude, speed, and timing data throughout the mission. It was selected over the NEO-M8N and L86-M33 for its NavIC (IRNSS) feature and

also is an earning bonus jury weightage. It is along with dual-band L1+L5 operation for faster achievement and greater positioning accuracy.



PARAMETER	VALUE
Model	L89HA (7Semi)
GNSS Support	GPS, NavIC, GLONASS, BeiDou, Galileo, QZSS
Frequency Bands	L1 + L5
Interface	I²C, UART
Position Accuracy	< 1.8 m CEP
Supply Voltage	3.3 V
Quantity	1

Table 21: GNSS Module (L89HA)

5.5 Power Monitor (INA226)

The INA226 is selected because of its features such as it continuously monitors bus voltage, current, and power draw across the CANSAT power system. This helps in tracking real-time battery health and power telemetry throughout flight. It was selected over the INA219 and ACS712 for its wider bus voltage range and continuous voltage/current/power measurement over a digital I²C interface. It also reduces noise exposure compared to analog current-sensing sensor.



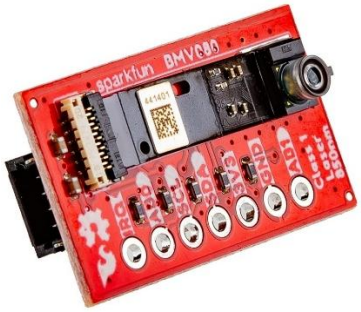
PARAMETER	VALUE
Model	INA226

Sensor Type	Power Monitor
Interface	I²C
Bus Voltage Range	0–36 V
Current Measurement	Yes (Shunt-based)
Power Calculation	Yes
Quantity	2

Table 22: Power Monitor (INA226)

5.6 Particulate Matter Sensor (BMV080)

The BMV080 sensor is an ultra-compact air quality sensor used for measuring PM1.0, PM2.5, and PM10 concentrations using laser scattering technology. This sensor is used as an extra additional telemetry as part of the bonus environmental payload. It was chosen over the PMSA003I and SPS30 for being the world's smallest particulate matter sensor with the lowest power consumption which makes it well suited to CANSAT's tight mass and power budget.

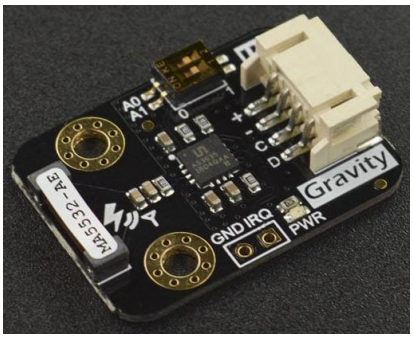


PARAMETER	VALUE
Model	BMV080 (7Semi)
Sensor Type	PM Sensor
Interface	I²C
Particle Size Detection	PM1, PM2.5, PM10
Power Consumption	55 mW
Supply Voltage	1.8–3.3 V
Quantity	1

Table 23: Particulate Matter Sensor (BMV080)

5.7 Lightning Sensor (SEN0290-AS3935)

The SEN0290 is a Gravity Lightning Sensor which detects lightning activity up to 40 km away and gives data about the distance, intensity, and strike frequency. This sensor is used as an extra additional telemetry as part of the bonus environmental payload. It uses intelligent noise filtering to minimize false detections from onboard electrical interference which makes it perfect for use CANSAT's active RF and motor-driven subsystems.

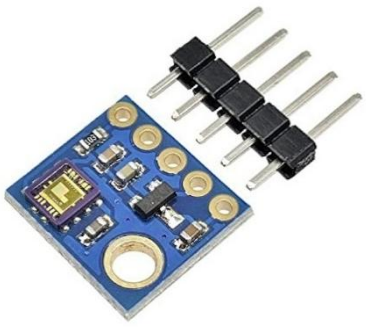


PARAMETER	VALUE
Model	SEN0290 (AS3935)
Sensor Type	Lightning Detector
Interface	I²C, SPI
Detection Range	Up to 40 km
Operating Voltage	2.4–5.5 V
Features	Lightning distance & storm warning
Quantity	1

Table 24: Lightning Sensor (SEN0290-AS3935)

5.8 UV Light Sensor (GYML8511)

The GYML8511 sensor is an analog-output ultraviolet light sensor covering the Ultraviolet B and most of the UVA spectrum. This sensor is used as an extra additional telemetry as part of the bonus environmental payload. It has a output of a voltage linearly proportional to UV intensity which provides a simple and easy way to capture UV index data during descent without added interface complexity.



PARAMETER	VALUE
Model	GYML8511
Sensor Type	UV Sensor
Interface	Analog
Spectral Range	280–390 nm

Supply Voltage	3.0–5.0 V
Output Type	Analog Voltage
Quantity	1

Table 25: UV Light Sensor (GYML8511)

5.9 Ozone Gas Sensor (MQ131)

The MQ131 sensor is a metal-oxide gas sensor whose conductivity decreases in the presence of ozone and allows ozone concentration to be indirect from the resulting signal change. It was selected for its wide detection range and fitness for atmospheric ozone sampling during descent. This sensor is used as an extra additional telemetry as part of the bonus environmental payload.



PARAMETER	VALUE
Model	MQ131
Sensor Type	Ozone Sensor
Interface	Analog, Digital
Detection Range	10–1000 ppm
Supply Voltage	5 V
Power Consumption	750 mW
Quantity	1

Table 2: Ozone Gas Sensor (MQ131)

5.10 CO_2 Sensor (SCD41)

The SCD41 sensor is a small environmental sensor that is used to measure carbon dioxide concentration with temperature and humidity. This sensor is used as an extra additional telemetry as part of the bonus environmental payload. It was chosen over the SCD30 and ENS160+SHT40 for providing true CO_2 readings with strong accuracy at low power draw, communicating over the same I^2C bus as the rest of the sensor stack.



Parameter	Value
Model	SCD41
Sensor Type	CO_2, Temperature, Humidity
Interface	I^2C
CO_2 Range	400–5000 ppm
Accuracy	±(40 ppm + 5%)
Supply Voltage	2.4–5.5 V
Quantity	1

Table 2: CO_2 Sensor (SCD41)

5.11 Real-Time Clock (DS3231M)

The DS3231M RTC maintains mission time independently of the flight computer which makes sure that the mission clock continues uninterrupted even if the Teensy 4.1 experiences a processor reset mid-flight. This directly supports the competition's requirement for steady one-second-resolution mission timing and correct flight-state recovery after any onboard reset event.



PARAMETER	VALUE
Model	DS3231M
Sensor Type	Real-Time Clock
Interface	I^2C
Function	Mission timekeeping across resets
Quantity	1

Table 28: Real- Time Clock (DS3231M)

The onboard RTC module was tested for correct timekeeping and battery-backed time holding power cycles. The results of RTC are given below in the image. This confirms the RTC is correctly wired, battery-backed, and suitable as the CANSAT determined time source for telemetry timestamping independent of GNSS lock.

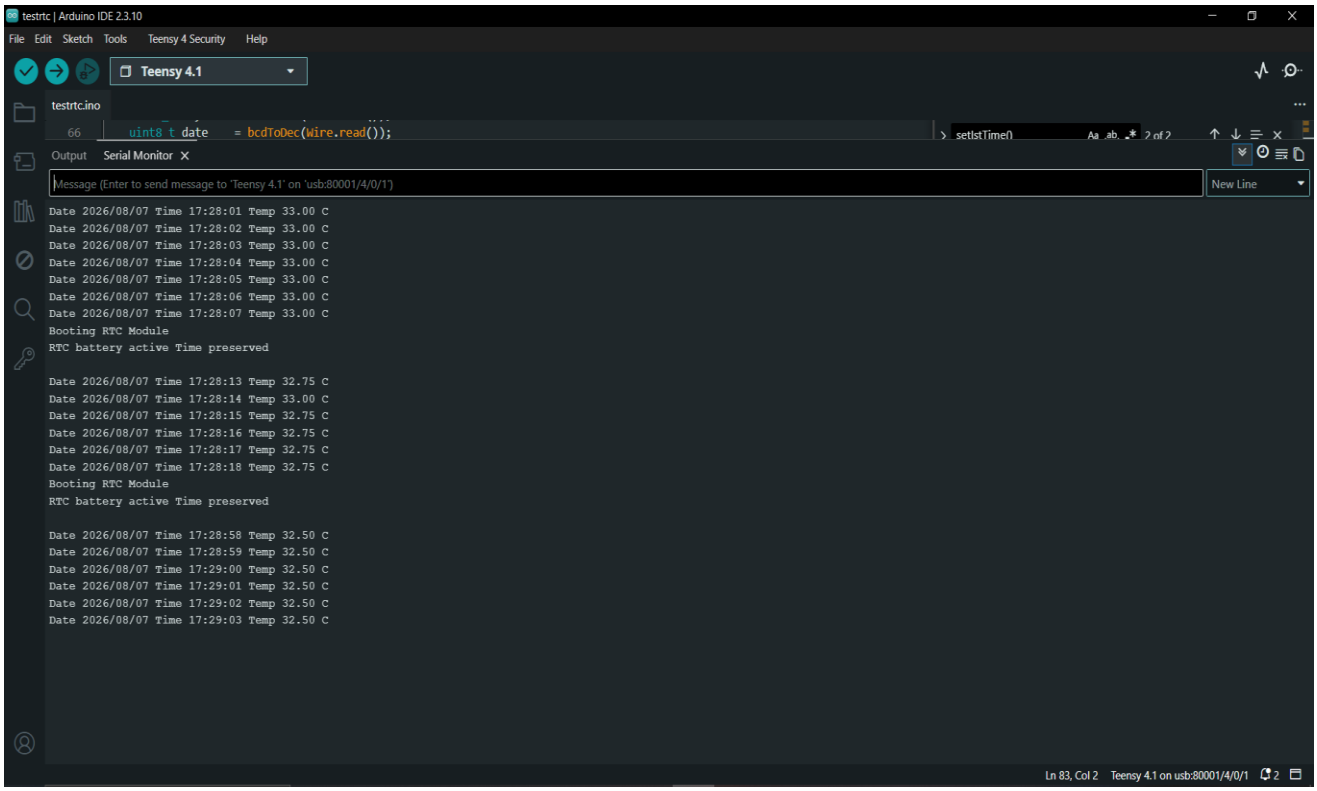


Fig 41: RTC check

5.12 SPI flash (7 Semi)

Parameter	Specification
Manufacturer	7Semi
Flash IC	Winbond W25Q-series (16 Mbit / 64 Mbit / 128 Mbit variants)
Interface	Standard SPI; Dual/Quad SPI on 64/128 Mbit variants
Max Clock Speed	Up to 104 MHz (16/64 Mbit) / 133 MHz (128 Mbit)
Operating Voltage	3.3 V (some variants down to 3.0 V)
Erase Granularity	4 KB sector / 32 KB / 64 KB block / full-chip erase
Mounting	Solder-free breakout, standard pin headers
Typical Use in CanSat	Onboard non-volatile black-box data logging, backup to LoRa downlink
Compatibility	Arduino, Teensy, STM32, ESP32, Raspberry Pi

Table 29: SPI flash (7 Semi)



The 7Semi SPI flash breakout was tested as a crash-survivable black-box data logger by continuously streaming barometric telemetry such as pressure, temperature, ASL/AGL altitude, running mean, and noise spread while a **simulated crash/power-loss event** was triggered mid-log ("Sensor Read finished (Simulated Crash)"). The results are shown in the image given below. This confirms the SPI flash successfully preserves and recovers logged flight data across an unplanned reset/power-interruption event, validating it as a viable non-volatile backup store for flight telemetry independent of the live LoRa downlink.

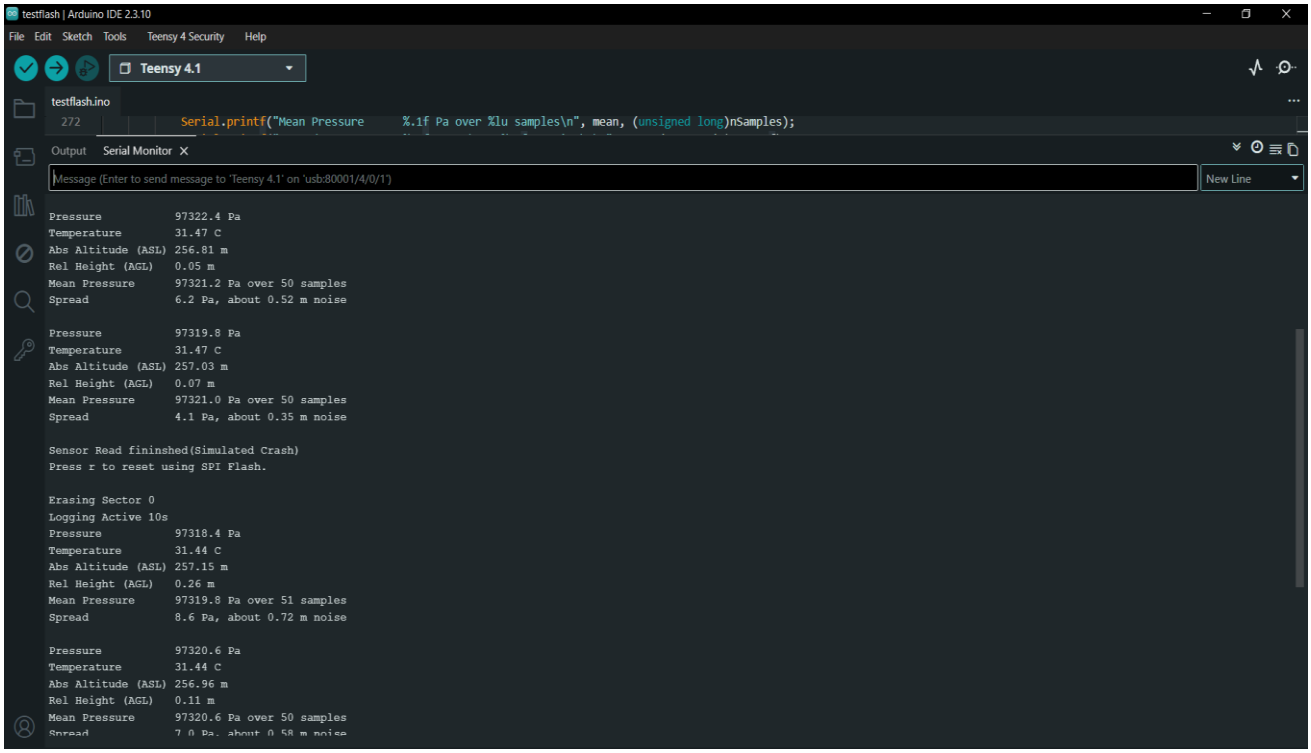


Fig 42: SPI Flash Test

5.13 Consolidated Test Verification Matrix

TEST	PASS CRITERION	STATUS
Barometer accuracy	Within ±5 m of known elevation	Planned
Barometer 600 m simulation	Deploy servo fires within 100 ms	Planned
IMU shock test	No sensor lock-up, FSM continues	Planned
GPS fix test	≥6 sats, 0.0001° position stability	Planned

Vibration test	No PCB loosening, no sensor error post-test	Planned
Power monitor validation	Accurate V/I/P readback under load	Planned
Bonus sensor functional checks	Correct telemetry field population	Planned

Table 30: Consolidated Test Verification Matrix

6. ACTUATOR SUMMARY

6.1 Control mechanism of actuator

Actuator are used in three different places in our CANSAT. The very first application for which it is used is the Innovation part of Team VRITRA for extra jury points weightage. It is about the Biochemical Characterization of Atmospheric Bacterial Isolates. The petri dish will be closed during the fight till the ejection of CANSAT. This will be done with a servo attached to the lead. The mechanism and placement are shown in the image given below.

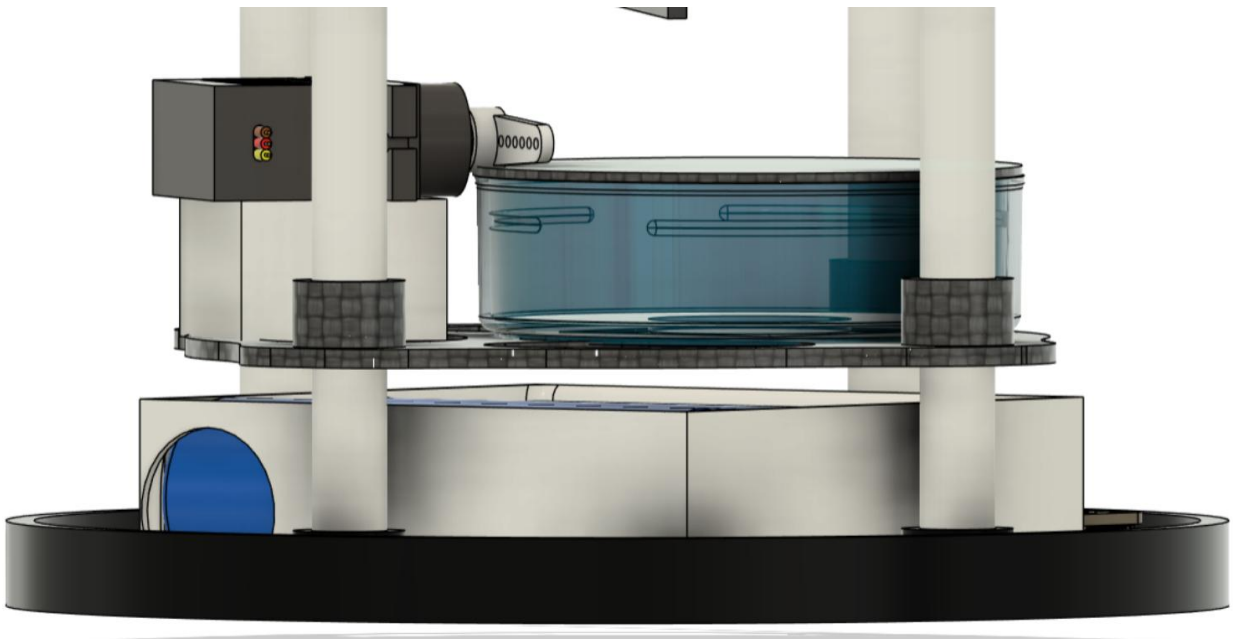


Fig 43: Actuator for Petri-dish

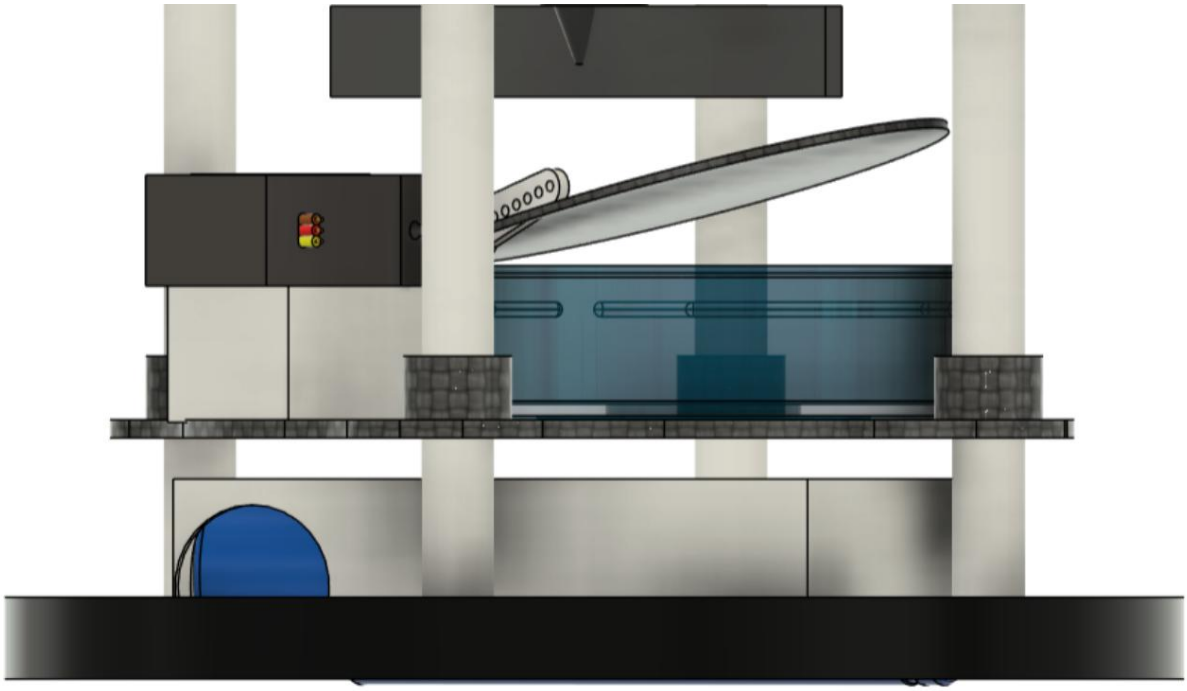


Fig 44: Servo Flip

The second actuator is used for parachute bay. The drug chute will be kept above the CANSAT during the flight readiness and the secondary RAM air parachute will be folded properly according to the steps given in further sections and kept in parachute bay. At an altitude of 600 m, the servo would flip the lead to allow RAM air to open. It also has a spring at opposite side of servo to avoid the back flip of led. The images for the same are given below.



Fig 45: Parachute Bay Actuator design

The third actuators used in our CANSAT is for controlling secondary descent GPS guided RAM-air parachute. Two servos are attached parallelly in opposite direction. The servo test results have been performed in are shown below with a ground station software.

6.2 Descent control mechanism incorporated in CAN-7USAT

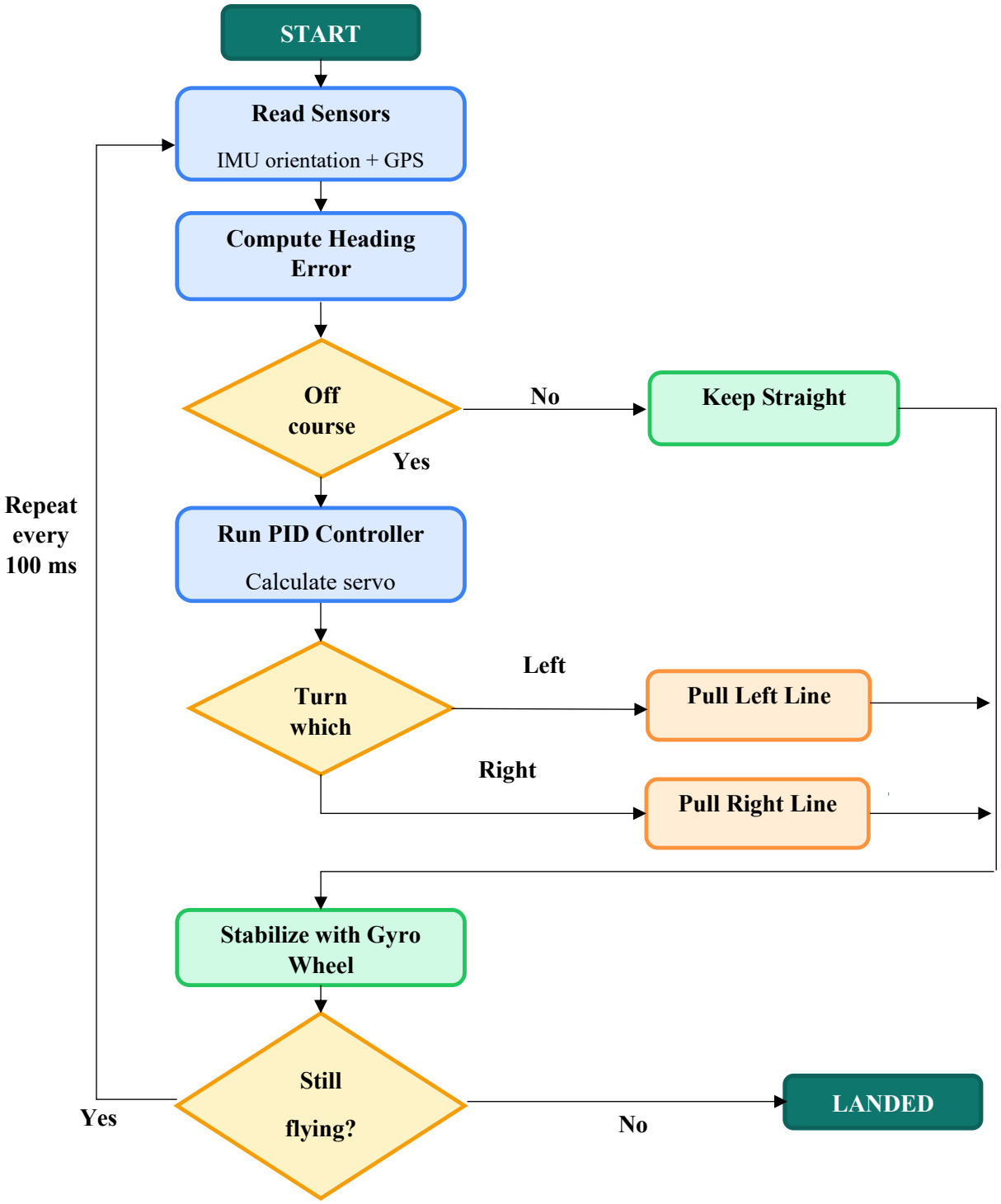


Fig 46: Descent Control Mechanism Actuator Loop

1. Sensing and error computation

The loop begins at START and, every cycle, executes:

- Read Sensors data of IMU orientation and GPS and gets current heading (IMU) and position (GPS).
- Compute Heading Error compares current heading/track against the target bearing to the landing point, producing the error signal that drives the control decision.

2. Off-course decision gate

- **If No**: It is kept Straight and no correction is applied and the loop proceeds directly to the stabilization stage.
- **If Yes**: It falls through to the correction path and Run PID Controller by Calculating servo, which calculates the required servo command from the heading error.

3. Differential brake actuation

Tells about turning to which branches on the sign of the PID output:

- **If Left**: Then pull the left line (left brake servo pulls down, right stays at/near trim) which is a asymmetric drag induces a left turn.
- **If Right**: Then pull the right line (mirror case).

4. Stabilization and termination check

- The stabilize with Gyro Wheel happen after any steering command or after "Keep Straight" command. Spinning the mechanical gyro to constant RPM before secondary parachute deployment so no reaction torque acts during guided flight, and countering any residual yaw. If it is still flying:
    - **If Yes:** Then the loop returns back to Read Sensors, repeating the whole cycle at **100 ms** (10 Hz), and matching the cadence of altitude-determination loop.
    - **If No:** It is dismissed and landed.

7. BLOCK DIAGRAM OF ALTITUDE CONTROL LOOP AND ACTUATOR CONTROL LOOP

7.1 Altitude Control Loop

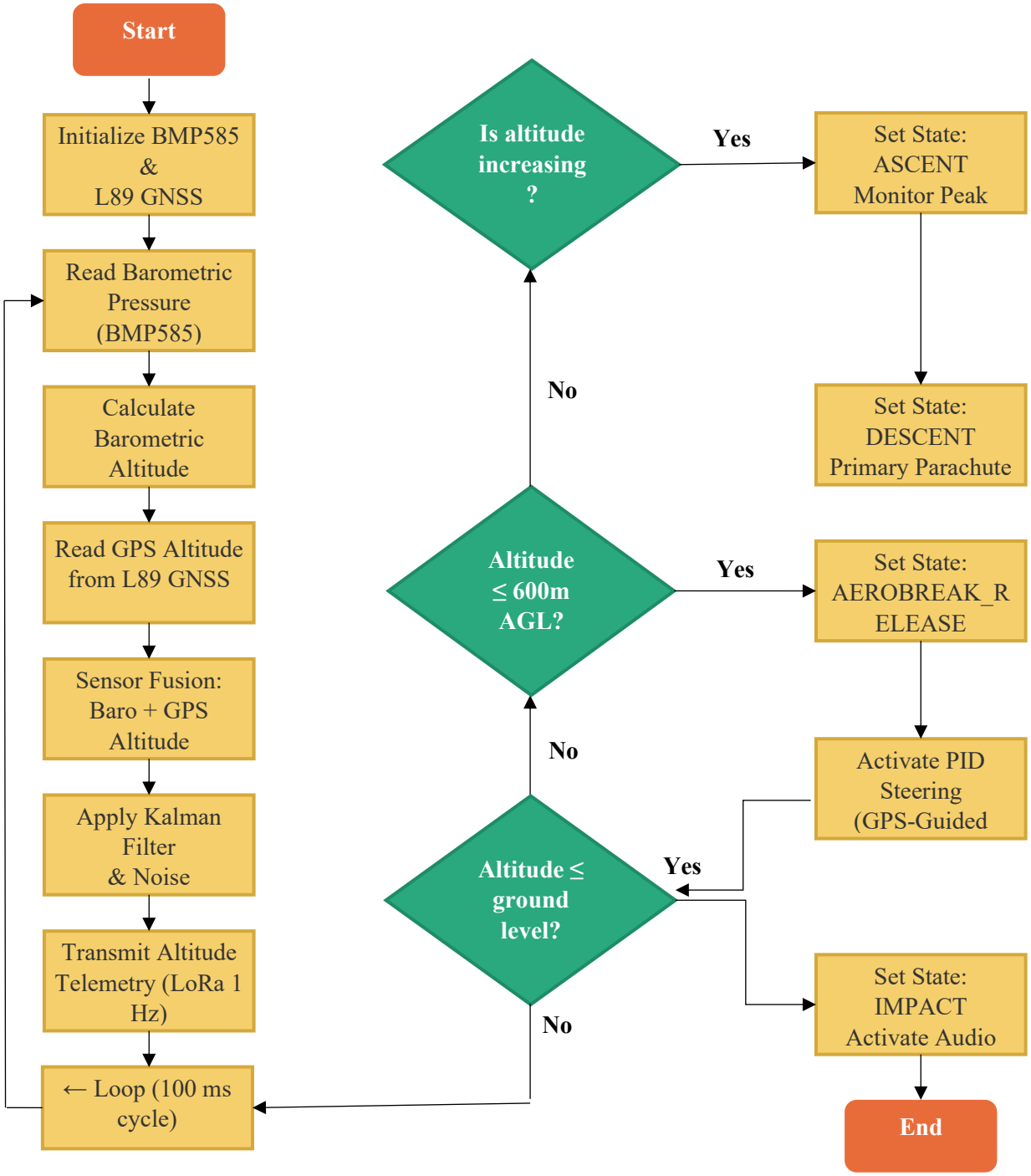


Fig 47: Altitude Control Loop

The altitude descent control loop known as ADCL have multiple steps within it. The steps are:

1. Sensor acquisition and altitude estimation:

The loop starts at **Start** and initializes the two altitude sensors which are the BMP585 barometer **and the** L89 GNSS module. Each 100 ms cycle then performs the process as:

- Read Barometric Pressure (BMP585) and calculate barometric altitude, converting raw pressure to altitude through the standard barometric formula.
- Read GPS Altitude from L89 GNSS independently of the barometric path.
- Sensor data fusion of BMP585 and GPS Altitude. This further combines the two independent altitude values, giving idleness against single sensor failure (e.g., GPS dropout or barometer drift) and improving accuracy over either source alone.
- Apply Kalman Filter & Noise [rejection] is used to smoothen the fused altitude estimate by rejecting sensor noise and reducing estimation lag which matters for the timing-critical events downstream (600 m release, ground impact).
- Transmit Altitude Telemetry (LoRa, 1 Hz) downlinks the filtered altitude at 1 Hz regardless of flight state, so ground station tracking is continuous even though the decision logic runs faster (100 ms).
- Loop (100 ms cycle) closes the sensor-acquisition loop back to the pressure read by giving a 10 Hz internal rate against a 1 Hz telemetry rate.

2. Flight-state decision chain

The filtered altitude provides three sequential decision diamonds in which each gating a distinct mission phase:

Gate 1: Is altitude increasing?

- **If Yes:** It is still ascending under boost/coast. State is set to ASCENT: Monitor Peak which then automatically goes to DESCENT: Primary Parachute once apogee is detected in which the ascent state itself watches for the peak and hands off to the descent state this is deployment trigger event from our recovery subsystem.
- **If No**: The altitude is not increasing (past apogee, under descent), so control falls through to Gate 2.

Gate 2: Altitude ≤ 600 m AGL?

- **If Yes:** State is set to AEROBRAKE_RELEASE, which triggers Activate PID Steering (GPS-Guided) that is the software-side trigger for releasing primary parachute and enabling the guided parafoil descent under closed-loop GPS/PID control.
- **If No:** It is still above 600 m, loop back through the "No" path to re-check on the next altitude update.

Gate 3: Altitude ≤ ground level?

- **If Yes**: The landing is detected. State is set to IMPACT: Activate Audio (locator beacon), and the state machine terminates at End.

- **If No**: It is not yet at ground level and the control returns to the sensor-acquisition loop for the next 100 ms cycle.

7.2 Actuators Control Loop

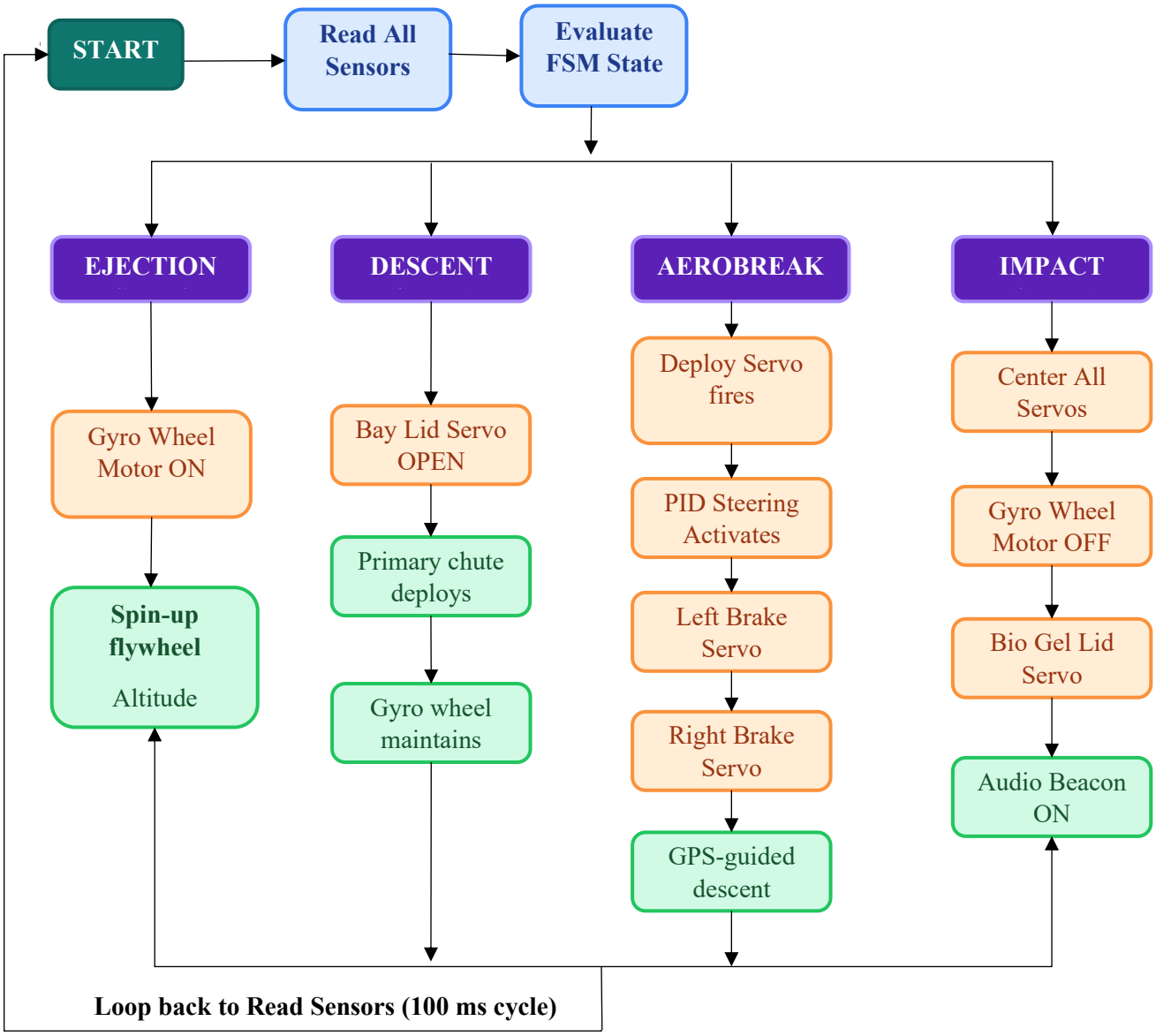


Fig 48: Actuator Control loops

The actuator control loop tells how CAN-7USAT's flight computer changes the current FSM state into physical actuator commands. Running on the same 100 ms (10 Hz) cycle as the main sensor loop, the FSM evaluates its current state after every sensor read and branches into one of four actuator response paths which are Ejection, Descent, Aerobrake, and Impact.

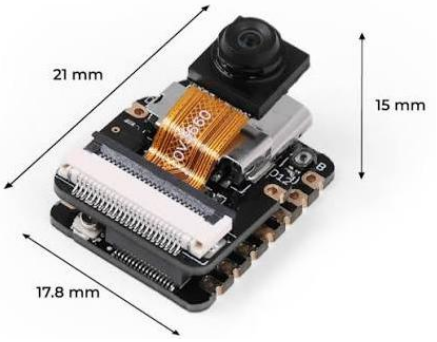
8. DETAILED DESCRIPTION OF REALIZED SUBSYSTEM

8.1 Payload Subsystem

Team VRITRA has included two payload elements other than the required telemetry data in our CAN-7USAT. One of the payloads is an imaging payload for capturing video of ground from the height of 1 KM and for descent/ejection documentation. The second additional payload is an innovation payload for atmospheric biosciences sampling. Both of this are selected to earn bonus jury weightage while remaining fully different from flight-critical control so that a payload fault cannot affect the primary mission.

8.1.1 Imaging Payload (XIAO ESP32S3 Camera Module)

The imaging payload captures video footage of two types such as one for the ejection till the CANSAT recovery stage and the other is for video capturing. This captured video will be saved in MicroSD card and will be later used for AI-based Urban planning system. It operates on an entirely independent processing path as XIAO Camera has its own processor as ESP32S3 which was selected over MIPI CSI-2 for real-time image acquisition.



PARAMETER	DETAIL
Module	XIAO ESP32S3 Sense
Function	Onboard video capture
Interface	MIPI CSI-2 (to ESP32-P4 companion computer)
Offload	Wi-Fi 6, post-flight
Mission criticality	Non-critical (isolated from flight control/telemetry)
Mass	5.6 g
Resolution	1600x1200 pixels

Table 31: Imaging Payload (XIAO ESP32S3 Camera Module)

Design rationale: By keeping the camera subsystem fully different from the FSM, telemetry chain, and descent control loop, a camera or main processor failure which is teensy 4.1, has zero impact on flight safety, telemetry data transfer, or descent operation.

8.1.2 Innovation Payload (Biochemical Characterization of Atmospheric Bacterial Isolates)

Team VRITRA has a unique innovation payload which is the biochemical characterization of atmospheric bacterial isolates that will be present during the CANSAT descent through the atmosphere. Petri dishes with culture media are exposed during flight and are tested after the recovery. This is to identify and characterize bacterial isolates collected at specific altitude.

PARAMETER	DETAIL
Payload type	Passive biological sampling (Petri plate with media)
Mounting location	Dedicated Petri dish bay within the electronics stack
Analysis method	Gram staining, biochemical test panel, carbohydrate fermentation
Isolates characterized	5 (ATB-01 through ATB-05)
Mission criticality	Non-critical (innovation/bonus scoring element)
Mass	10 g

Table 32: Innovation Payload (Biochemical Characterization of Atmospheric Bacterial Isolates)

Design rationale: The Petri dish is placed below the electronics bay housing (as shown in physical structure) but structurally kept different from sensitive electronics to avoid contamination risk during handling. After recovery, isolates undergo standard microbiological characterization such as morphology, Gram reaction, catalase/oxidase, carbohydrate fermentation as per the methodology and baseline results.

8.2 Housekeeping Subsystem

The housekeeping subsystem have the six supporting systems that keep CAN-7USAT structurally sound, powered, communicating, sensing, actuating, and recovering. Full design and specifications for each are presented in their respective sections given below.

8.2.1 Mechanical Subsystem

- **Manufacturing approach** — Shell and parachute bay are additively manufactured (3D printed) for complex lattice geometry, while plates and rods are machined/cut for dimensional precision at load-bearing interfaces.
- **Assembly & maintainability** — the rod-and-plate stack allows tool-less or minimal-tool disassembly, enabling individual PCB layers to be pulled for testing, rework, or replacement without disturbing the rest of the stack.
- **Mass efficiency** — the lattice/perforated shell pattern removes non-structural material to minimize weight while the load-bearing members (rods, plates) retain full cross-section, keeping the structure light without under-designing the critical path.
- **Thermal/venting consideration** — the perforated shell and lattice pattern also aid passive airflow around internal electronics and allow clean strip-away of the parachute deployment bag without snag points.

- **Alignment & repeatability** — the four-rod symmetric layout constrains all plates to a fixed radial position, ensuring consistent PCB stack alignment across builds and simplifying connector/wiring routing between layers.
- **Interface with recovery system** — the parachute bay geometry and internal lid/spring mechanism is mechanically hosted at the top of this stack, so the structure directly interfaces with the release device (R1) and P2 deployment bag housing.

8.2.2 Actuators

CAN-7USAT has five actuator functions across four actuator types: dual brake-line servos for parafoil steering, two independent MG90S servos for bay lid and bio-gel lid release, a brushless DC motor driving the stabilization flywheel, and a buzzer for post-landing recovery.

1. AGFRC A20BHM (×2) Parafoil Brake-Line Steering

The AGFRC A20BHM is a high-voltage-compatible brushless micro servo which will be used in pairs for differential brake-line deflection during GPS-guided parafoil steering. Its brushless motor design (the first of its kind in a 12 mm-thick RC servo form factor) gives it significantly higher torque and speed than a traditional coreless servo of the same size, which is necessary to overcome brake-line tension under load during active steering corrections.

PARAMETER	VALUE
Model	AGFRC A20BHM
Motor type	Brushless
Torque	11.4 kg-cm at 8.4 V
Speed	0.068 sec/60° at 8.4 V
Operating voltage	4.8–8.4 V (HV compatible)
Gear material	Steel gear, CNC aluminium case
Weight	21 g
Quantity	2 (left brake, right brake)

Table 33: AGFRC A20BHM (x2) Parafoil Brake-Line Steering



2. MG90S #1 Parachute Bay Lid Servo

The first MG90S micro servo actuates the parachute bay lid latch. Its compact size and adequate torque for a lightweight latch mechanism make it well suited to this single-shot, low-load release function.

PARAMETER	VALUE
Model	MG90S
Motor type	Coreless DC, metal gear
Torque	1.8 kg-cm @ 4.8 V / 2.2 kg-cm @ 6 V
Speed	0.1 sec/60° @ 4.8 V
Operating voltage	4.8–6 V
Weight	13.4 g
Quantity	1

Table 34: MG90S #1 Parachute Bay Lid Servo



3. MG90S #2 Bio-Gel Lid Servo

The second MG90S servo actuates the bio-gel plate lid, exposing the Petri dish sampling surface for the biochemical characterization payload. This actuation is purposely sequenced into the IMPACT state, ensuring the sampling plate is only exposed to the atmosphere for its intended descent-through-landing collection window rather than throughout the entire flight.

4. 2205 2300KV BLDC Motor — Stabilization Flywheel

The 2205 2300KV brushless DC motor works for CAN-7USAT's reaction flywheel, spinning up during the EJECTION state to provide gyroscopic Altitude stabilization ahead of parachute deployment shock. Its brushless design offers substantially higher achievable RPM and torque density than a brushed DC motor of similar size, giving the flywheel a faster spin-up time and greater stored angular momentum for stabilization authority.

PARAMETER	VALUE

Model	2205, 2300KV
Motor type	Brushless (outrunner)
KV rating	2300 KV (RPM per volt)
Compatible cells	3–4S LiPo
Stator dimensions	22 mm diameter × 5 mm length
Max continuous current	32 A
Max continuous power	~500 W
Weight	~28–30 g
Quantity	1

Table 35: 2205 2300KV BLDC Motor- Stabilization Flywheel



5. Buzzer Recovery Beacon

The onboard buzzer serves as CAN-7USAT's audio recovery beacon, activating continuously upon reaching the IMPACT state to guide the ground recovery team to the landed CANSAT. It is a general-purpose magnetic speaker rather than a piezo element, giving it a broad audible frequency range suited to reliable detection in open-field recovery conditions.

8.2.3 Communication and Data Handling Subsystem (CDH)

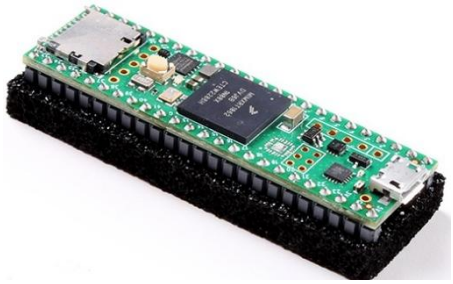
The CDH subsystem is the main subsystem of CAN-7USAT which is used to collects data from every onboard sensor and manages telemetry transmission and onboard logging. It is built with the help of a primary flight computer, an independent camera and recording path, and storage layer.

1. Primary Flight Computer (Teensy 4.1)

PARAMETER	DETAILS
Role	Central processing unit

Core	ARM Cortex-M7, 600 MHz
Interfaces	55 Digital I/O, 35 PWM, 8 UART, 3 SPI, 3 I²C
Memory	8 MB Flash, 1 MB RAM, 4284 KB EEPROM
Operating voltage	3.3–5 V
Boot time	0.005 s
Loop rate	100 ms (10 Hz) main loop, 1 Hz telemetry sub-rate
Deployment	One unit primary, one unit backup (secondary avionics board)

Table 36: Communication and Data Handling Subsystem (CDH)

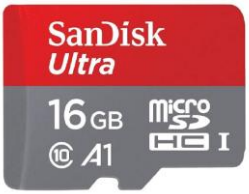


Teensy 4.1 was chosen because of the fastest boot time and clock speed. It is the largest available memory for telemetry, sufficient I/O count for the full sensor/actuator connections. The footprint is small enough not to strain the mass budget.

2. Onboard Data Storage

PARAMETER	DETAILS
Role	Continuous onboard backup
Hardware	SanDisk Ultra MicroSDHC, 16 GB
Interface	SPI1
Logging rate	1 Hz, CSV format
Purpose	Recovers full mission dataset even if telemetry link is lost mid-flight

Table 37: Onboard Data Storage



3. Telemetry Radio (LR1121 LoRa)

Parameter	Details
Role	Bidirectional telemetry downlink and telecommand uplink

Device type	Multi-band LoRa transceiver (Sub-GHz + 2.4 GHz)
Configured mode	LoRa, 865 MHz (India ISM sub-band)
Network ID	NETID/PANID set to team number — 0x98
Transmit power	+10 dBm pad / +14 dBm flight
Receiver sensitivity	~ −130 dBm (SF9, 125 kHz)
Fallback mode	Reconfigurable to 2.4 GHz LoRa if sub-GHz band congested
Deployment	One primary unit, one backup unit (independent failover link)

Table 38: Telemetry Radio (LR1121 LoRa)



4. T- Antenna 865 MHz

The RadioMaster Bandit T Antenna is a omnidirectional whip antenna selected as an alternative to the team's primary onboard antenna, covering the 860–930 MHz band that fully includes CAN-7USAT's 867 MHz LoRa telemetry frequency. It uses the same heavy-duty RG402 coaxial line and 360° swivel RP-SMA connector as the rest of the Bandit antenna series, in a lighter, smaller housing suited to onboard mounting.

PARAMETER	VALUE
Model	RadioMaster Bandit 915/868MHz T Antenna
Bandwidth	860–930 MHz
Polarisation	Linear
Radiation direction	Omnidirectional
Gain	2.7 dBi
VSWR	≤ 1.5:1
Connector	RP-SMA
Input impedance	50 Ω
Cable	RG402 coax, 80 mm
Dimensions	107 × 98 mm
Weight	14.00 g

Table 39: T – Antenna 865 MHz

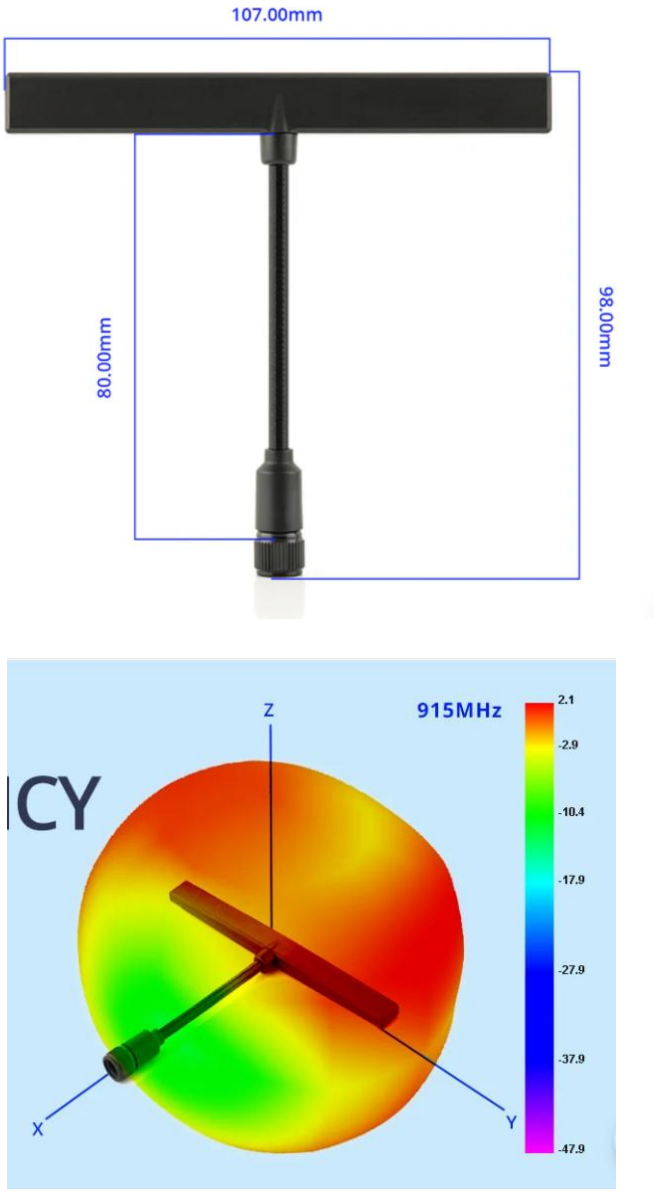


Fig 49: Radio Master T-Anteena

8.2.4 Electrical Power Subsystem

The EPS helps us with stable, protected power to every onboard subsystem by converting raw battery voltage into regulated rails while providing real-time health monitoring and multi-layer protection against overcharge, over-discharge, and short-circuit conditions.

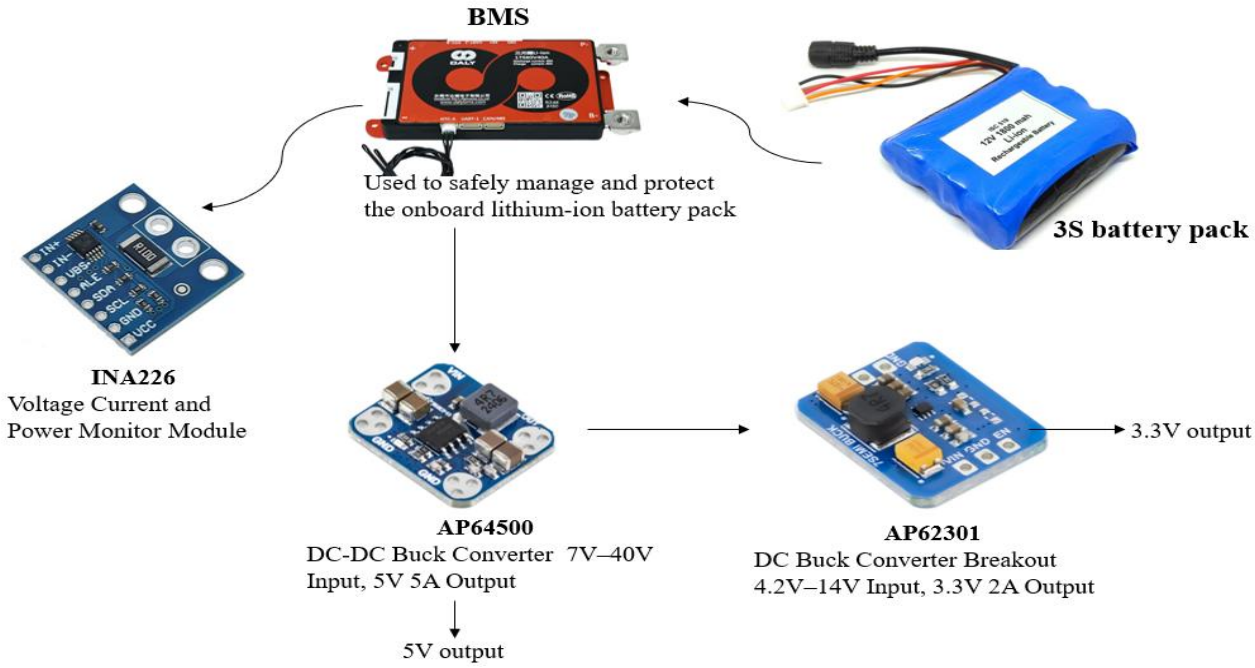


Fig 50: Power Subsytem

1. Molicel INR-21700-P50B Battery Pack

PARAMETER	DETAILS
Role	Primary energy source for the entire CAN-7USAT power system
Configuration	3 × Molicel INR-21700-P50B cells (3S)
Nominal voltage	10.8 V
Fully charged voltage	12.6 V
Capacity	5000 mAh
Total energy	54 Wh (3 × 18 Wh)
Usable energy (80% DoD)	~43.2 Wh
Deployment	Single pack, feeding VBAT rail

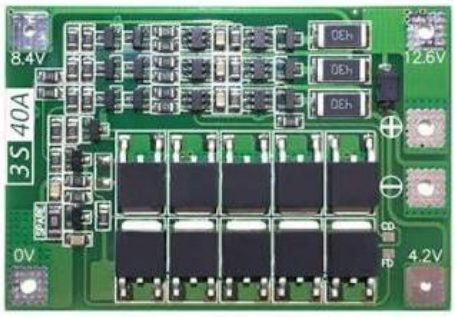
Table 40: Molicel INR-21700-P50B Battery Pack



2. Battery Protection (3S 40A Li-ion BMS)

PARAMETER	DETAILS
Role	Safeguards the battery pack during charge and discharge operation
Protection functions	Overcharge, over-discharge, overcurrent, short-circuit
Current rating	40 A
Additional feature	Built-in cell balancing for extended battery life
Interface	In line with battery pack, upstream of power distribution

Table 41: Battery Protection (3S 40A Li-ion BMS)



3. 5V Power Regulation (AP64500 Buck Converter-7Semi)

PARAMETER	DETAILS
Role	Converts battery voltage to a stable 5V rail
Input range	7–40 V
Output	5.0 V, up to 5 A
Efficiency	Up to 95%
Consumers powered	Teensy 4.1, servos, XIAO ESP32S3 camera module, gyro motor driver
Interface	Direct 5V power rail (5V_MAIN)

Table 42: 5V Power Regulation (AP64500 Buck Converter – 7Semi)



4. 3.3V Power Regulation (AP62301 Buck Converter-7Semi)

PARAMETER	DETAILS
Role	Converts battery voltage to a stable 3.3V rail
Output	3.3 V, up to 2 A
Consumers powered	LR1121, BMP585, ICM-20948, INA226, DS3231M RTC, L89 GPS
Interface	Direct 3.3V power rail (3V3_MAIN)
Advantage over linear regulation	Higher efficiency than AMS1117-type LDOs

Table 43: 3.3V Power Regulation (AP62301 buck Converter -7Semi)



5. Power-On Control (External Switch & LED Indicator)

PARAMETER	DETAILS
Role	Manual power control and visual power-state confirmation
Switch type	External SPST switch (SW1)
Indicator	LED (illuminates when supply is active, off when no current flows)
Requirement satisfied	External power switch with visual on/off indication

Table 44: Power-On Control (External Switch & LED Indicator)

6. Power Budget

Component	Voltage	Current	Maximum Power	Duty Cycle (%)	Average Power	Source
Teensy 4.1	5	0.18	1.8	100	1.8	Existing

ESP32-S3	5	0.18	1.8	70	1.26	Existing
LR1121 LoRa (TX)	3.3	0.055	0.1815	20	0.0363	Existing
L89 GNSS	3.318	0.032	0.106176	100	0.106176	Existing
ICM-20948 IMU	3.322	0.0045	0.029898	100	0.029898	Existing
SCD41 CO2	5	0.018	0.09	100	0.09	Existing
AS3935 Lightning	3.3	0.0001	0.00033	100	0.00033	Datasheet
AS7343 Spectral	3.3	0.0x98	0.00825	100	0.00825	Datasheet
HDC3020 Temp/Humidity	3.3	0.0008	0.00264	100	0.00264	Datasheet
ML8511 UV	3.3	0.0003	0.00099	100	0.00099	Existing
MQ131 Ozone	5	0.15	0.75	50	0.375	Existing
BMV080 PM	3.3	0.018	0.0594	100	0.0594	Existing
TPL5010 Timer	2.5	0.000000035	8.75E-08	100	8.75E-08	Datasheet
W25Q64 Flash	3.3	0.025	0.0825	20	0.0165	Datasheet
INA226	3.3	0.0012	0.00396	100	0.00396	Existing
SN74LV4052A MUX	3.3	0.00008	0.000264	100	0.000264	Datasheet
DS3231 RTC	3.3	0.00011	0.000363	100	0.000363	Datasheet
Hobbywing XRotor 40A ESC	10.8	0.1	1.08	30	0.324	Datasheet
2205 2300KV BLDC Motor	10.8	4	43.2	30	12.96	Datasheet
AGFRC A20BHM Servo #1	7.4	1.5	11.1	20	2.22	Datasheet
AGFRC A20BHM Servo #2	7.4	1.5	11.1	20	2.22	Datasheet
Teensy 4.0 (DCS)	5	0.08	0.4	100	0.4	Existing
BMP585	3.3	0.001	0.0099	100	0.0099	
7.4V BEC	7.4	0.02	0.296	100	0.296	Existing
7Semi AP62301 Buck 3.3V 2A	5	0.02	0.2	100	0.2	Existing
7Semi AP64500 Buck 5V 5A	10.8	0.03	0.648	100	0.648	Estimate
				Total Maximum Power	72.9501709	

				Total Average Power	23.06797109	
Battery Capacity (mAh)	5000					
Battery Voltage (V)	10.8					
Battery Energy (Wh)	54					
Estimated Endurance (Hours)	2.341					

Table 45: Power Budget

8.2.5 Sensor Subsystem

The sensor subsystem provides CAN-7USAT with all mandatory flight-critical measurements such as altitude, orientation, position, and power state. Additionally, Team VRITRA have added five-sensor bonus environmental payload for atmospheric characterization during descent.

1. Sensor Bus Allocation

BUS	SENSORS ON BUS
I2C0	BMP585, DS3231M RTC, SCD41, SEN0290, BMV080
Backup I2C0	ICM-20948
UART1	L89HA GNSS
Analog	GYML8511, MQ131
SPI (shared with LR1121/SD)	LSM6DSO32

Table 46: Sensor Bus Allocation

2. Telemetry Field Mapping

SR NO.	TM PARAMETER	SENSOR	FUNCTION	RESOLUTION/ FORMAT
1.	<TEAM ID>	-	Team Name	2026-IN-SPACeCAN-7USAT-0x98
2.	<TIME STAMPING>	DS3231M RTC	Time since the initial power	Seconds
3.	<PACKET COUNT>	EEPROM counter	Count of transmitted packets	

4.	<ALTITUDE>	BMP585	Altitude in units of meters and must be relative to ground	0.1 meters
5.	<PRESSURE>	BMP585	Measurement of atmospheric pressure	1 pascal
6.	<TEMP>	BMP585	Temperature in Celsius	0.1 oC
7.	<VOLTAGE>	INA226	Voltage of the CAN-7USAT power bus	0.01 Volts
8.	<GNSS TIME>	L89_GPS	Time generated by the GNSS receiver	Seconds
9.	<GNSS LATITUDE>	L89_GPS	Latitude generated by the GNSS receiver	0.0001 degrees
10.	<GNSS LONGITUDE>	L89_GPS	Longitude generated by the GNSS receiver	0.0001 degrees
11.	<GNSS ALTITUDE>	L89_GPS	Altitude generated by the GNSS receiver	0.1 meters
12.	<GNSS SATS>	L89_GPS	GNSS satellites connected	integer number
13.	<ACCELEROMETER DATA>	LSM6DSO32	Data received from the gyroscopic sensor	m/s2
14.	<GYRO SPIN RATE>		Spin rate of Mechanical Gyro wrt. CAN-7USAT	deg/s
15.	<FLIGHT SOFTWARE STATE>	Teensy FSM	Operating state of the software	(boot, idle, launch detect, deploy, etc.)
16.	< AIR QUALITY SENSING>	BMV080	Measures fine particulate matter concentration	PM1.0, PM2.5, and PM 10
17.	<GRAVITY LIGHTNING SENSOR>	SEN0290	Measures lightning distance, intensity, and frequency	21-bit resolution
18.	<UV LIGHT SENSOR MODULE>	GYML8511	Detects ultraviolet light in the 280 nm – 390 nm wavelength	peak sensitivity at 365 nm
19.	<OZONE GAS SENSOR>	MQ131	Detects Ozone particles	10 – 1000 ppm
20.	<CARBON DIOXIDE CONCENTRATION>	SCD41	Measure carbon dioxide (CO_2), temperature, and humidity	400 – 5000 ppm

Table 47: Telemetry Field Mapping

8.2.6 Parachute deployment and recovery



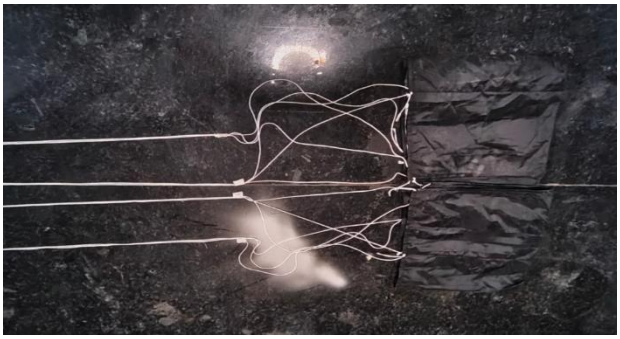
Stage 1



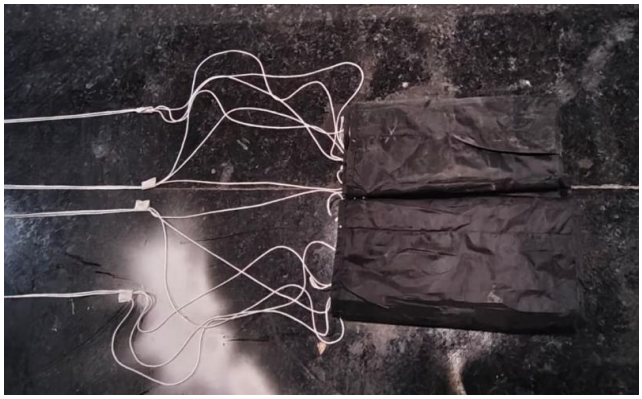
Stage 2



Stage 3



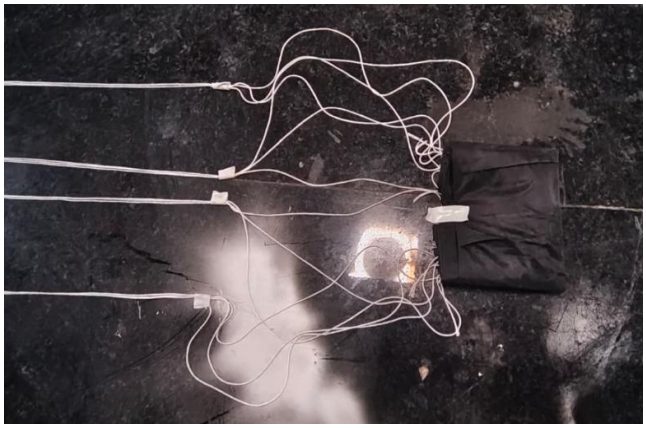
Stage 4

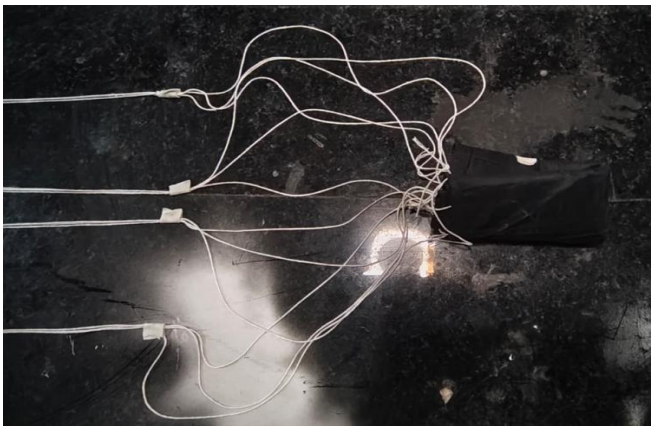


Stage 5



Stage 6





Stage 7 Stage 8



Stage 9

Above is the step-by-step process of folding a RAM air parachute. This folding method is done for proper adjustment of parachute in parachute bay and to avoid entanglement of cords during deployment.

This parachute testing was done from a heighted building of around 120 feet. According to the test results, the RAM air parachute has a proper shape and nodes attachment. It carried 840 g of dead weight and the parachute. It had proper opening without any entanglement with in the codes.

Improvement done after testing: Length of the cord is reduced according to the calculations.



Fig 51: Parachute dead weight



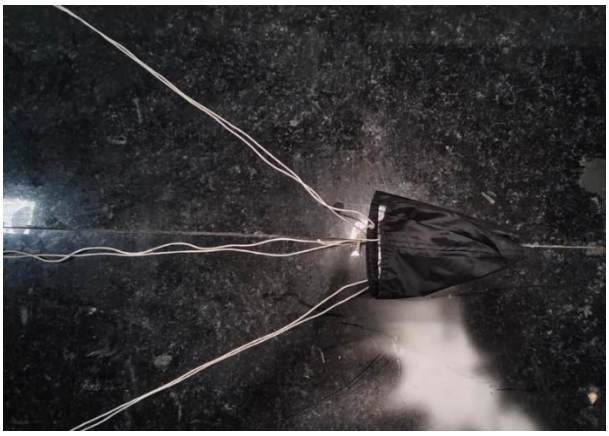
Stage 1



Stage 2



Stage 3



Stage 4



Stage 5



Stage 6



Above given is the step-by-step guid of folding out primary parachute.

9. CAN-7USAT ALGORITHM, FINAL TEST RESULTS, ALONG WITH ANY DEVIATIONS

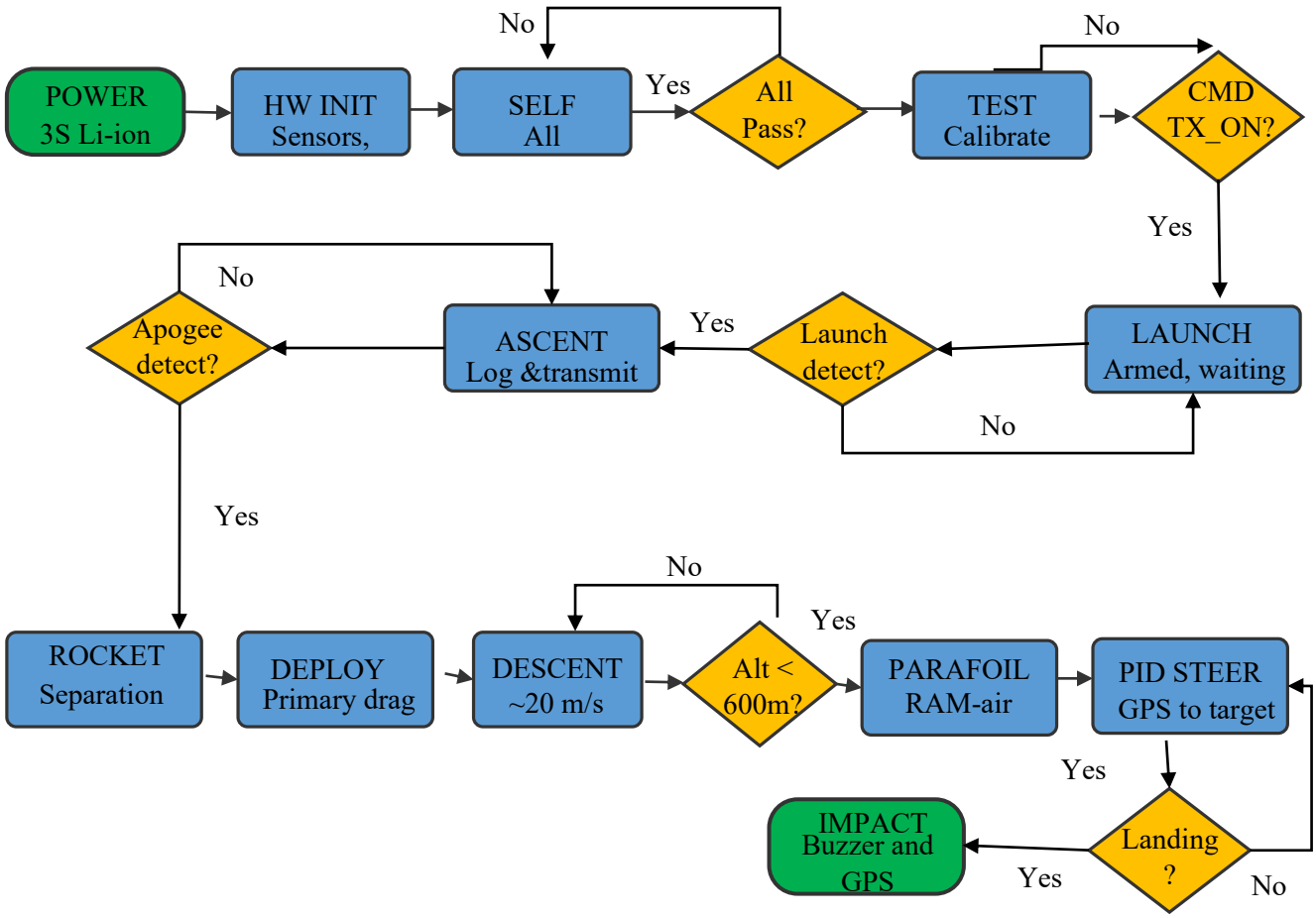


Fig 52: CANSAT Algorithm

10. TTLC LINK MARGIN REALIZED

Given Parameters according to the datasheet and instruction manual:

TX Power = 22 dBm

Onboard Antenna Gain = 2.7 dBi

Ground Antenna Gain = 2.7 dBi

Frequency (f) = 867 MHz

Distance (d) = 1 km

Spreading Factor = SF10

Bandwidth = 125 kHz

Receiver Sensitivity at SF10

Sensitivity at SF7, BW 125 kHz = −127 dBm

Sensitivity at SF12, BW 125 kHz = −141 dBm

$$\text{Sensitivity per SF step} = \frac{-141-(-127)}{(12-7)}$$

$$\text{Sensitivity per SF step} = -14 / 5 = -2.8 \text{ dB per step}$$

$$\text{S_RX (SF10)} = \text{S_RX (SF7)} + (\text{SF10} - \text{SF7}) \times (-2.8)$$

$$\text{S_RX (SF10)} = -127 + (10 - 7) \times (-2.8)$$

$$\text{S_RX (SF10)} = -127 - 8.4$$

S_RX = −135.4 dBm

Free Space Path Loss (FSPL)

$$\text{FSPL} = 20\ log_{10}(\text{d}) + 20\ log_{10}(\text{f}) + 32.44$$

$$\text{FSPL} = 20\ log_{10}(1) + 20\ log_{10}(867) + 32.44$$

$$\text{FSPL} = 0 + 58.76 + 32.44$$

FSPL = 91.2 dB

Received Power

$$P_RX = P_TX + G_TX + G_RX - FSPL$$

$$P_RX = 22 + 2.7 + 2.7 - 91.2$$

$$\mathbf{P_RX = -63.8\ dBm}$$

Link Margin

$$\text{Link Margin} = P_RX - S_RX$$

$$\text{Link Margin} = -63.8 - (-135.4)$$

$$\mathbf{\text{Link Margin} = 71.6\ dB}$$

Results: The calculated received signal strength of around **−63.8 dBm** is higher than the LR1121 receiver sensitivity of **−135.4 dBm** at SF10, 125 kHz bandwidth. This results in an estimated link margin of **71.6 dB** is confirming that the telemetry system holds large signal headroom for proper communication between CAN-7USAT and the ground station over the full 1 km design range.

10.1 Radio Interference Mitigation Plan:

- NETID/PANID unique to team number (provided as team number-25) prevents packet collision with other teams' LR1121 units. (PANID- 0x98)
- Pre-launch RF checks will be performed using the ground station receiver to verify signal quality and identify nearby interference sources before launch.
- Critical telemetry packets will include packet counters and status information to help detect packet loss during communication.
- Ground station T-antenna provides better gain, reducing off-axis interference
- LR1121 LoRa SF10 spreading factor is orthogonal to SF7/SF8.

11. GROUND STATION FINAL TEST RESULTS, ALONG WITH ANY DEVIATIONS

The VRITRA Ground Control Station was tested end-to-end against a simulated flight by performing all six operational tabs while decoding live telemetry at 3.8 Hz over the 867 MHz downlink. The Flight tab confirms correct 3D CANSAT visualization, real-time Altitude /heading display, GPS fix, and full flight-sequence state tracking. Across all tabs the GCS correctly reflected realistic degraded-link without crashing or losing packet sync, confirming the ground software is stable and ready to support live flight operations.

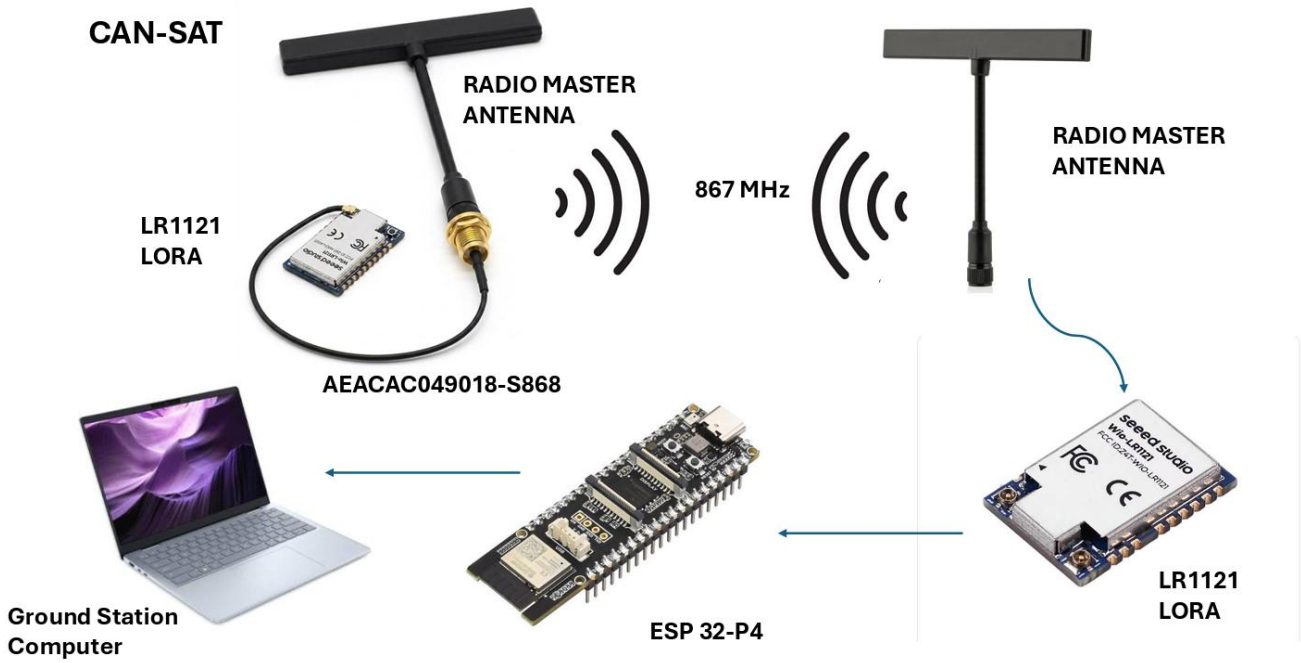
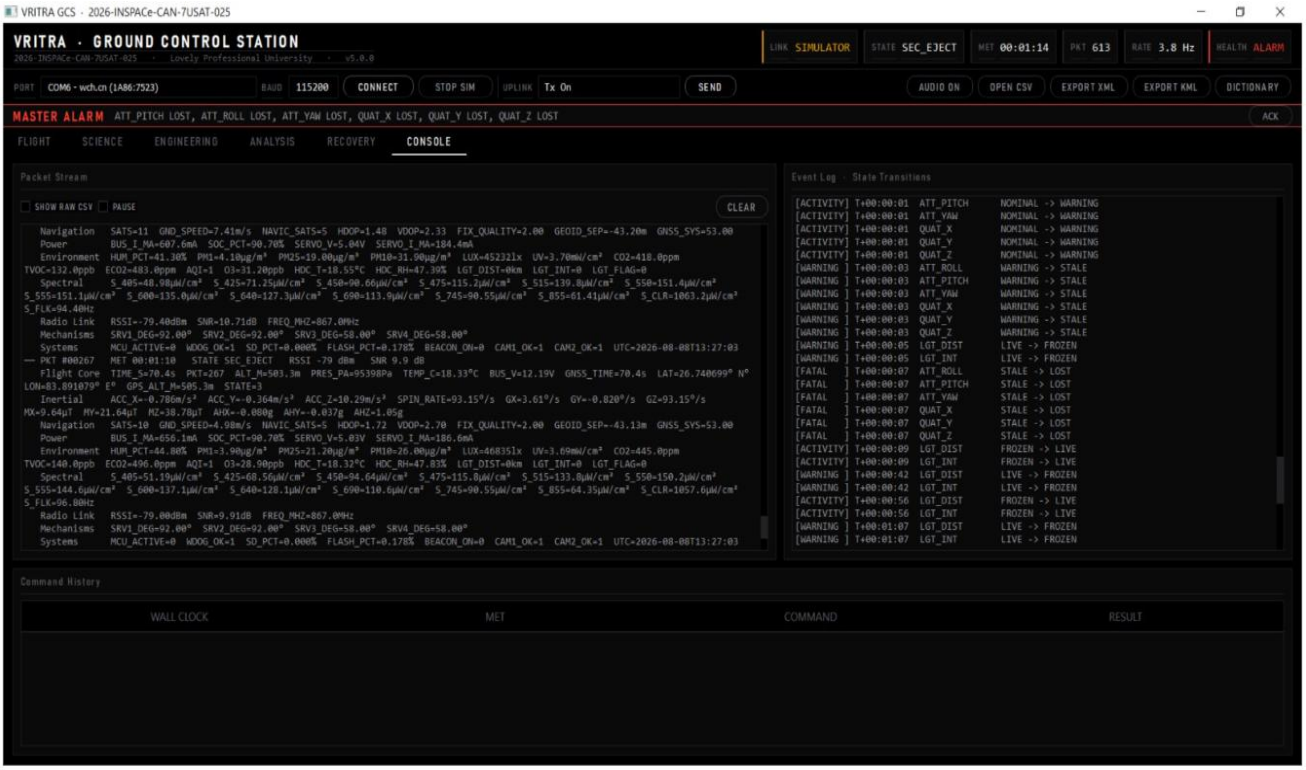
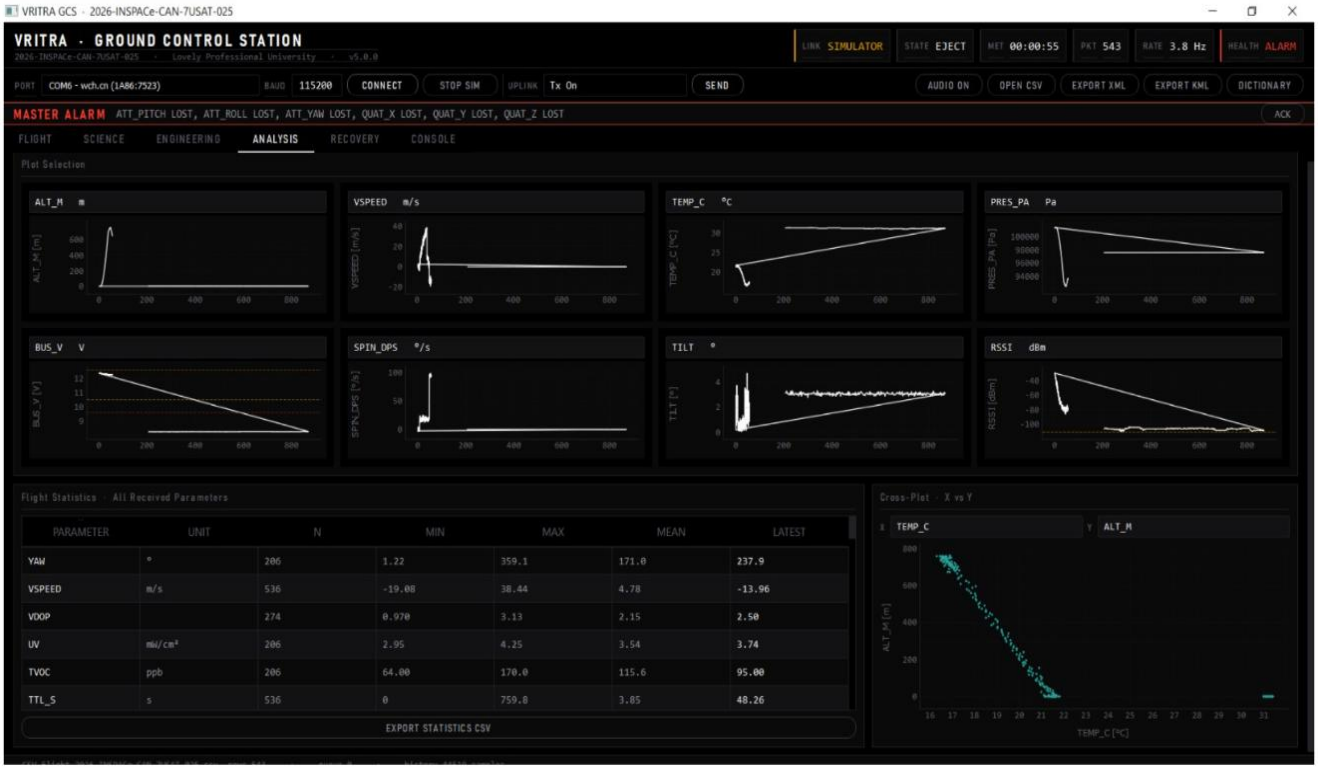
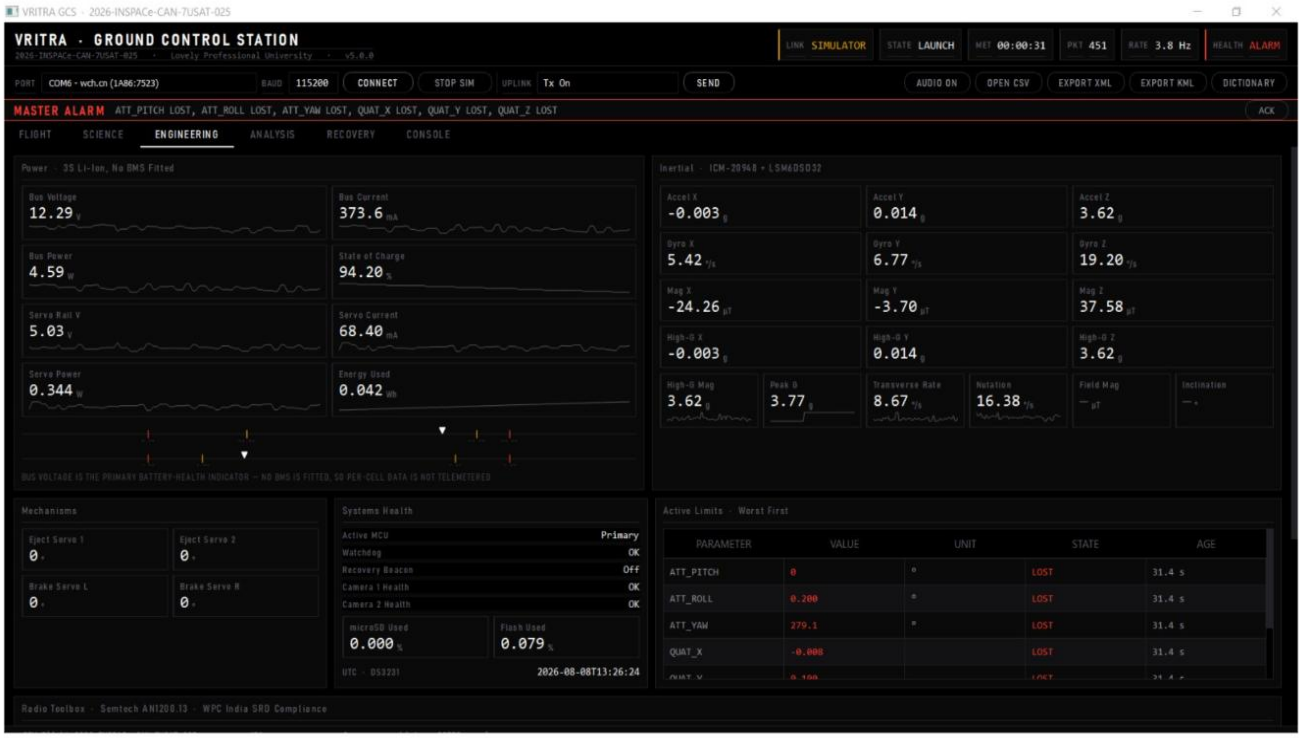


Fig 53: Communication Subsystem


VRITRA GCS · 2026-INSPACe-CAN-7USAT-025
VRITRA · GROUND CONTROL STATION
2026-INSPACe-CAN-7USAT-025 · Lovely Professional University · v5.0.0
LINK SIMULATOR
STATE SEC_EJECT
MET 00:01:14
PKT 613
RATE 3.8 Hz
HEALTH ALARM
PORT COM6 - wch.cn (1A86:7523)
BAUD 115200
CONNECT
STOP SIM
UPLINK Tx On
SEND
AUDIO ON
OPEN CSV
EXPORT XML
EXPORT KML
DICTIONARY
MASTER ALARM ATT_PITCH LOST, ATT_ROLL LOST, ATT_YAW LOST, QUAT_X LOST, QUAT_Y LOST, QUAT_Z LOST
ACK
FLIGHT
SCIENCE
ENGINEERING
ANALYSIS
RECOVERY
CONSOLE
Packet Stream
SHOW RAW CSV
PAUSE
CLEAR
Navigation SATS=11 GND_SPEED=7.41m/s NAVIC_SATS=5 HDOP=1.48 VDOP=2.33 FIX_QUALITY=2.00 GEOID_SEP=-43.20m GNSS_SYS=53.00
Power BUS_I_MA=607.6mA SOC_PCT=90.70% SERVO_V=5.04V SERVO_I_MA=184.4mA
Environment HUM_PCT=41.30% PM1=4.10µg/m³ PM25=19.00µg/m³ PM10=31.90µg/m³ LUX=452321lx UV=3.70mW/cm² CO2=418.0ppm TVOC=132.0ppb ECO2=483.0ppm AQI=1 O3=31.20ppb HDC_T=18.55°C HDC_RH=47.39% LGT_DIST=0km LGT_INT=0 LGT_FLAG=0
Spectral S_405=48.98µW/cm² S_425=71.25µW/cm² S_450=90.66µW/cm² S_475=115.2µW/cm² S_515=139.8µW/cm² S_550=151.4µW/cm² S_555=151.1µW/cm² S_600=135.0µW/cm² S_640=127.3µW/cm² S_690=113.9µW/cm² S_745=90.55µW/cm² S_855=61.41µW/cm² S_CLR=1063.2µW/cm² S_FLK=94.40Hz
Radio Link RSSI=-79.40dBm SNR=10.71dB FREQ_MHZ=867.0MHz
Mechanisms SRV1_DEG=92.00° SRV2_DEG=92.00° SRV3_DEG=58.00° SRV4_DEG=58.00°
Systems MCU_ACTIVE=0 WDOG_OK=1 SD_PCT=0.000% FLASH_PCT=0.178% BEACON_ON=0 CAM1_OK=1 CAM2_OK=1 UTC=2026-08-08T13:27:03
— PKT #00267 MET 00:01:10 STATE SEC_EJECT RSSI -79 dBm SNR 9.9 dB
Flight Core TIME_S=70.4s PKT=267 ALT_M=503.3m PRES_PA=95398Pa TEMP_C=18.33°C BUS_V=12.19V GNSS_TIME=70.4s LAT=26.740699° N° LON=83.891079° E° GPS_ALT_M=505.3m STATE=3
Inertial ACC_X=-0.786m/s² ACC_Y=-0.364m/s² ACC_Z=10.29m/s² SPIN_RATE=93.15°/s GX=3.61°/s GY=-0.820°/s GZ=93.15°/s MX=9.64µT MY=21.64µT MZ=38.78µT AHX=-0.080g AHY=-0.037g AHZ=1.05g
Navigation SATS=10 GND_SPEED=4.98m/s NAVIC_SATS=5 HDOP=1.72 VDOP=2.70 FIX_QUALITY=2.00 GEOID_SEP=-43.13m GNSS_SYS=53.00
Power BUS_I_MA=656.1mA SOC_PCT=90.70% SERVO_V=5.03V SERVO_I_MA=186.6mA
Environment HUM_PCT=44.80% PM1=3.90µg/m³ PM25=21.20µg/m³ PM10=26.00µg/m³ LUX=468351lx UV=3.69mW/cm² CO2=445.0ppm TVOC=140.0ppb ECO2=496.0ppm AQI=1 O3=28.90ppb HDC_T=18.32°C HDC_RH=47.83% LGT_DIST=0km LGT_INT=0 LGT_FLAG=0
Spectral S_405=51.19µW/cm² S_425=68.56µW/cm² S_450=94.64µW/cm² S_475=115.8µW/cm² S_515=133.8µW/cm² S_550=150.2µW/cm² S_555=144.6µW/cm² S_600=137.1µW/cm² S_640=128.1µW/cm² S_690=110.6µW/cm² S_745=90.55µW/cm² S_855=64.35µW/cm² S_CLR=1057.6µW/cm² S_FLK=96.80Hz
Radio Link RSSI=-79.00dBm SNR=9.91dB FREQ_MHZ=867.0MHz
Mechanisms SRV1_DEG=92.00° SRV2_DEG=92.00° SRV3_DEG=58.00° SRV4_DEG=58.00°
Systems MCU_ACTIVE=0 WDOG_OK=1 SD_PCT=0.000% FLASH_PCT=0.178% BEACON_ON=0 CAM1_OK=1 CAM2_OK=1 UTC=2026-08-08T13:27:03
Event Log · State Transitions
[ACTIVITY] T+00:00:01 ATT_PITCH NOMINAL -> WARNING
[ACTIVITY] T+00:00:01 ATT_YAW NOMINAL -> WARNING
[ACTIVITY] T+00:00:01 QUAT_X NOMINAL -> WARNING
[ACTIVITY] T+00:00:01 QUAT_Y NOMINAL -> WARNING
[ACTIVITY] T+00:00:01 QUAT_Z NOMINAL -> WARNING
[WARNING] T+00:00:03 ATT_ROLL WARNING -> STALE
[WARNING] T+00:00:03 ATT_PITCH WARNING -> STALE
[WARNING] T+00:00:03 ATT_YAW WARNING -> STALE
[WARNING] T+00:00:03 QUAT_X WARNING -> STALE
[WARNING] T+00:00:03 QUAT_Y WARNING -> STALE
[WARNING] T+00:00:03 QUAT_Z WARNING -> STALE
[WARNING] T+00:00:05 LGT_DIST LIVE -> FROZEN
[WARNING] T+00:00:05 LGT_INT LIVE -> FROZEN
[FATAL] T+00:00:07 ATT_ROLL STALE -> LOST
[FATAL] T+00:00:07 ATT_PITCH STALE -> LOST
[FATAL] T+00:00:07 ATT_YAW STALE -> LOST
[FATAL] T+00:00:07 QUAT_X STALE -> LOST
[FATAL] T+00:00:07 QUAT_Y STALE -> LOST
[FATAL] T+00:00:07 QUAT_Z STALE -> LOST
[ACTIVITY] T+00:00:09 LGT_DIST FROZEN -> LIVE
[ACTIVITY] T+00:00:09 LGT_INT FROZEN -> LIVE
[WARNING] T+00:00:42 LGT_DIST LIVE -> FROZEN
[WARNING] T+00:00:42 LGT_INT LIVE -> FROZEN
[ACTIVITY] T+00:00:56 LGT_DIST FROZEN -> LIVE
[ACTIVITY] T+00:00:56 LGT_INT FROZEN -> LIVE
[WARNING] T+00:01:07 LGT_DIST LIVE -> FROZEN
[WARNING] T+00:01:07 LGT_INT LIVE -> FROZEN
Command History
WALL CLOCK
MET
COMMAND
RESULT


VRITRA GCS · 2026-INSPACe-CAN-7USAT-025
VRITRA · GROUND CONTROL STATION
2026-INSPACe-CAN-7USAT-025 · Lovely Professional University · v5.0.0
LINK SIMULATOR
STATE EJECT
MET 00:00:55
PKT 543
RATE 3.8 Hz
HEALTH ALARM
PORT COM6 - wch.cn (1A86:7523)
BAUD 115200
CONNECT
STOP SIM
UPLINK Tx On
SEND
AUDIO ON
OPEN CSV
EXPORT XML
EXPORT KML
DICTIONARY
MASTER ALARM ATT_PITCH LOST, ATT_ROLL LOST, ATT_YAW LOST, QUAT_X LOST, QUAT_Y LOST, QUAT_Z LOST
ACK
FLIGHT
SCIENCE
ENGINEERING
ANALYSIS
RECOVERY
CONSOLE
Plot Selection
ALT_M m
VSPEED m/s
TEMP_C °C
PRES_PA Pa
BUS_V V
SPIN_DPS °/s
TILT °
RSSI dBm
Flight Statistics · All Received Parameters
PARAMETER UNIT N MIN MAX MEAN LATEST
YAW ° 206 1.22 359.1 171.0 237.9
VSPEED m/s 536 -19.08 38.44 4.78 -13.96
VDOP 274 0.970 3.13 2.15 2.50
UV mW/cm² 206 2.95 4.25 3.54 3.74
TVOC ppb 206 64.00 170.0 115.6 95.00
TTL_S s 536 0 759.8 3.85 48.26
EXPORT STATISTICS CSV
Cross-Plot · X vs Y
X TEMP_C
Y ALT_M
ALT_M [m]
TEMP_C [°C]



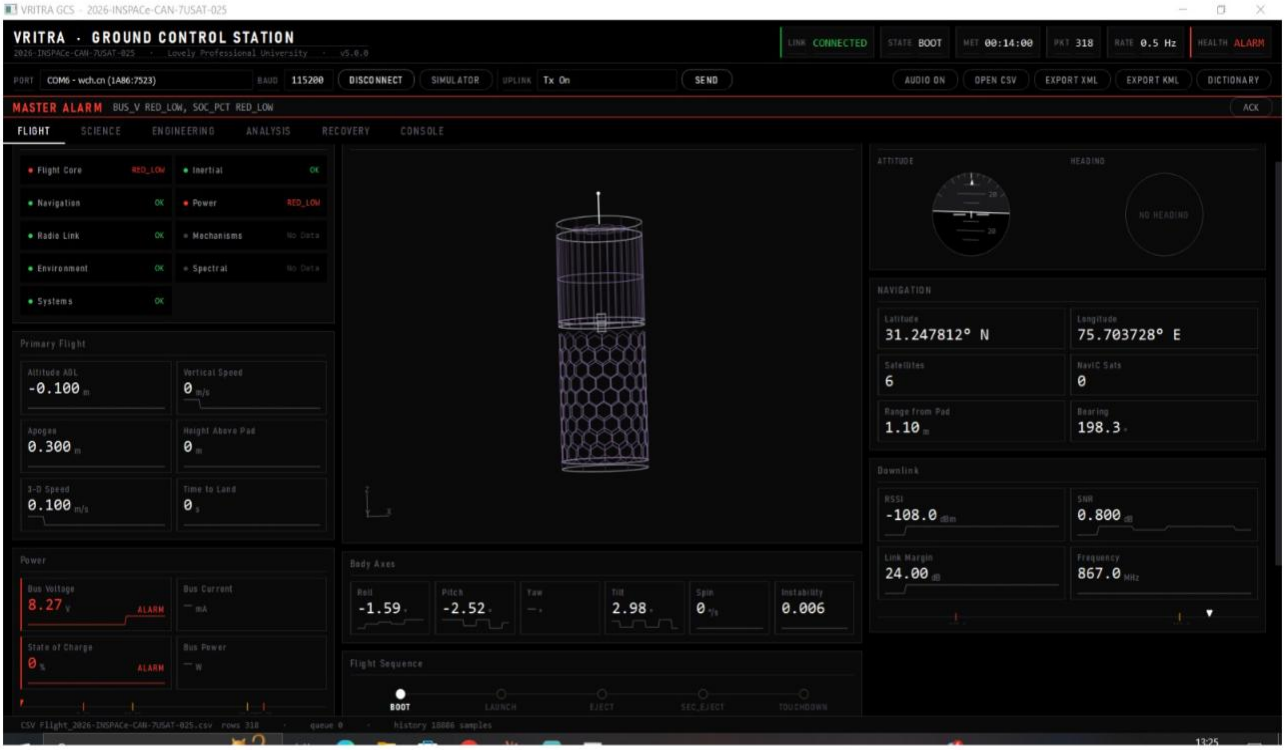


Fig 54: Ground Station Test

12. MISSION OPERATIONS & ANALYSIS

This section defines how Team VRITRA is going to operates CAN-7USAT across the mission timeline from ground preparation through flight execution to recovery.

Phase	Duration (approx.)	Primary Operator Role
Ground station setup	30–45 min	GCS lead + antenna crew
Pre-flight checkout & calibration	20–30 min	Avionics lead
Pad wait	Up to 2 hours	Full team, telemetry monitoring
Powered ascent	~15–30 s	Passive monitoring
Primary descent	~50–70 s (est.)	Telemetry monitoring
Secondary guided descent	~120–200 s (est.)	Telemetry monitoring + recovery team staging
Recovery	Beacon-guided	Recovery team

Table 48: Mission Operations & Analysis

12.1 Launch Day Sequence

1. Arrive at launch site ≥ 3 hours before launch slot
2. Set up ground station: laptop, LR1121 receiver, and antenna
3. Power on GS laptop — open Python GS software, verify LR1121 receiver detected on COM port
4. Install CANSAT batteries — verify voltage reading on INA226 (≥ 8.0 V), close battery hatch
5. Power on CANSAT via external switch — green LED illuminates, GS software shows BOOT packets
6. Send CMD_CALIBRATE from GS — CANSAT zeros baro altitude, gyro offsets, accel offsets at pad level
7. Verify TEST_MODE telemetry: all sensors reading correctly, GPS fix acquired (≥ 6 sats)
8. Record launch pad GPS coordinates as target waypoint for parafoil PID (CMD_SET_WAYPOINT)
9. Fold primary chute in Z-fold, secure against outer body with Velcro tab per packing procedure
10. Pack parafoil in deployment bag, insert pin, place in Parachute Bay
11. Hand CANSAT to launch crew for placement in rocket payload bay
12. Send CMD_TX_ON from GS — CANSAT transitions to LAUNCH_PAD, confirms telemetry streaming
13. Verify GS CSV is logging: Flight_[TEAM_ID].csv row count incrementing at 1 Hz
14. Launch — monitor all TM fields in real-time on GS dashboard
15. At AEROBREAK_RELEASE (state 6): observe GPS track converging toward launch pad waypoint
16. At IMPACT (state 7): note final GPS position, move to recovery with audio beacon guidance
17. Retrieve CANSAT — download full CSV and video from ESP32-P4.

13. PRE-FLIGHT REQUIREMENT-CHECK ANALYSIS

Before CAN-7USAT is cleared for integration into the launch vehicle, every mandatory competition requirement is checked against the as-built system through a defined verification method such as analysis, test, inspection, or demonstration.

Sr.	Category	Requirement	Status
1	Structural	No sharp edges on container body	Checked
2	Structural	Fluorescent body color + Indian flag	Checked
3	Structural	No use of rocket airframe as CanSat structure	Checked
4	Structural	Team details in English, Hindi, and regional language	Checked
5	Structural	Electronics enclosed/shielded (sensors excepted)	Checked
6	Structural	Circuit boards hard-mounted (standoffs/screws)	Checked
7	Power	External switch with LED and sound indicator	Tested
8	Power	Battery type compliance (Li-ion)	Checked
9	Power	Battery accessible/removable in <1 minute	Planned
10	Power	No spring-contact battery connections	Checked
11	Descent & Recovery	Primary descent parachute	Tested
12	Descent & Recovery	Primary chute deploys immediately post-ejection	Checked
13	Descent & Recovery	Primary chute attached to body, no separate ejection mechanism	Planned
14	Descent & Recovery	No pyrotechnic/explosive/hazardous chemical deployment	Checked
15	Descent & Recovery	Gyroscopic wheel stabilization during descent	Planned
16	Communication	Radio XBEE or LoRa	Tested
17	Communication	NETID/PANID set to team number	Tested
18	Communication	No broadcast mode used	Checked
19	Communication	No interference with other teams' radios	Checked
20	Communication	RF-commanded telemetry start/stop capability	Checked
21	Communication	Radio interference mitigation plan presented	Checked
22	Ground Station	Own ground station developed by team	Tested
23	Ground Station	Real-time telemetry display, SI units	Tested
24	Ground Station	GS commands CanSat to start telemetry	Tested

25	Ground Station	Elevated ground antenna	Tested
26	Ground Station	CanSat does not transmit until commanded	Tested
27	Ground Station	Ground calibration command (gyro/baro/accel zero)	Checked
28	Ground Station	CSV generation, mission time resolution	Tested
29	Ground Station	Mission clock/state persists across processor reset	Tested
30	Ground Station	Portable ground station (<5 min setup)	Checked
31	Flight Software	FSM state maintained and telemetered	Checked
32	Flight Software	Correct state determination after processor reset	Tested
33	Flight Software	All states described in review presentation	Checked
34	Telemetry	Full mission telemetry saved as CSV on GS	Tested
35	Telemetry	Onboard data storage utilized	Tested

Table 49: Pre-Flight Requirement-Check Analysis

14. LOGISTICS AND TRANSPORTATION

14.1 CAN-7USAT transportation and handling

CAN-7USAT is designed to be transported and handled with minimum risk to its structural, electronic, and descent-system integrity between Lovely Professional University and the competition launch site in khushinagar, UP. The following handling protocol is followed by Team VRITRA:

14.1.1 Pre-Transport Preparation

- Battery will be disconnected/powered off via the external switch prior to transport, as per the competition's power-safety requirement.
- Primary parachute and RAM-air parafoil will be packed into their respective bays and will be opened only at the launch site immediately before integration to avoid canopy compression damage or line tangling over an extended packed duration.
- The CANSAT will be placed in a custom-fitted protective transport case with foam cutouts matching its dimensions, cushioning the structure against shock and vibration during vehicle transport.
- All exposed sensor apertures (GPS antenna, air-quality sensor inlets, UV sensor window) will be covered with protective caps/tape during transit and removed only during final pre-launch checkout.
- Spare batteries, tools, and the Petri dish culture media for the biochemical payload will be transported separately in a dedicated accessory case to avoid contamination or accidental damage to the primary structure.

14.1.2 Transport Method

CAN-7USAT and the ground station equipment are transported by road from LPU to the launch site by team members and mentors, using the protective case described above secured against movement in the vehicle.

15. CAN-7USAT READY TO LAUNCH AND FINAL COMMENTS

CAN-7USAT balances competition mandatory consistency such as redundant avionics, dual-confirmed deployment triggers, conservative structural margins and meaningful innovation in its atmospheric biosciences payload and NavIC-enabled navigation. With outstanding verification items completed, Team VRITRA is confident in CAN-7USAT's readiness to proceed toward full integrated testing and flight.

16. CONCLUSION

Team VRITRA's CAN-7USAT presents a Critical Design Review that carries forward and matures every design commitment made at PDR into a fabricated, analytically verified, and largely bench-tested flight vehicle. The two-stage descent architecture of a drag-deployed primary parachute followed by a GPS-guided RAM-air parafoil with dual-axis brake-line steering meets all mandatory descent-rate and activation-altitude requirements with demonstrated margin, while the Teensy 4.1 avionics architecture, dual LR1121 telemetry links, and dual-confirmed altitude triggers reflect a design cantered rather than single points of failure.

The communication link comes as **71.6 dB margin** at the full 1 km design range even under the competition-specified 867 MHz/SF10 configuration. Beyond mandatory compliance, CAN-7USAT differentiates itself through its innovation payload of biochemical characterization of atmospheric bacterial isolates collected during descent and also a five-sensor bonus environmental suite and NavIC-enabled dual-band GNSS, reflecting the team's intent to contribute genuine atmospheric science value beyond the competition's baseline mission profile.

A small set of verification items such as full-scale vibration testing, field RF range confirmation, and power budget revalidation remain open ahead of the Flight Readiness Review. Team VRITRA views these not as design uncertainties but as the final confirmatory steps of a design that has already been carried through structural analysis, FEA simulation, link budget calculation, and subsystem-level bench testing.

With eight team members contributing across aerospace, electronics, and robotics disciplines under faculty and mentor guidance from Lovely Professional University, Team VRITRA submits this Critical Design Review confident that CAN-7USAT is on track to meet every mission requirement of the IN-SPACe CAN-7USAT India Student Competition 2026, and looks forward to demonstrating its performance at launch.